

Propulsion Considerations in Tactical Missile Design

3.1 Introduction

Tactical missile propulsion considerations that are addressed in this text emphasize conceptual design methods, design trades, and technologies for rocket, ramjet, and turbojet propulsion. Consideration is given to propulsion system alternatives, turbojet compressor and turbine temperature limits, turbojet specific impulse and thrust prediction, ramjet Mach number and temperature limits, ramjet specific impulse and thrust prediction, ramjet engine/booster integration options, ramjet inlet options, inlet spillage prediction, inlet shock loss, inlet start Mach number, ramjet missile drag from booster integration, fuel alternatives, propellant fraction required for an incremental velocity, solid rocket burn area requirement, solid rocket throat area requirement, solid rocket specific impulse and thrust prediction, solid rocket grain alternatives, solid rocket thrust magnitude control, solid propellant alternatives, motor case material alternatives, and rocket nozzle material alternatives.

Figure 3.1 shows that, after aerodynamic configuration design, the next step in the missile configuration synthesis process is propulsion system design. Propulsion system design is an iterative process requiring consideration of alternative propulsion systems, alternative propulsion subsystems, alternative technologies, and propulsion system resizing. The propulsion sizing output includes the thrust, specific impulse, and the propellant/fuel weight of the propulsion system.

3.2 Propulsion Alternatives Assessment

Figure 3.2 compares the efficiency of tactical missile propulsion alternatives across the Mach number ranges of subsonic through hypersonic. Shown is a typical specific impulse envelope and typical constraints for the alternatives of turbojet, ramjet, ducted rocket, scramjet, and solid rocket propulsion.

Subsonic turbojet/turbofan propulsion is a relatively mature technology. It is most suited for subsonic cruise missiles, providing high efficiency against non-time-critical targets. Beyond Mach 2, there is a technology challenge of the maximum allowable temperature of current turbine materials. Other problems include increasingly complex inlet systems to match the inlet airflow to the

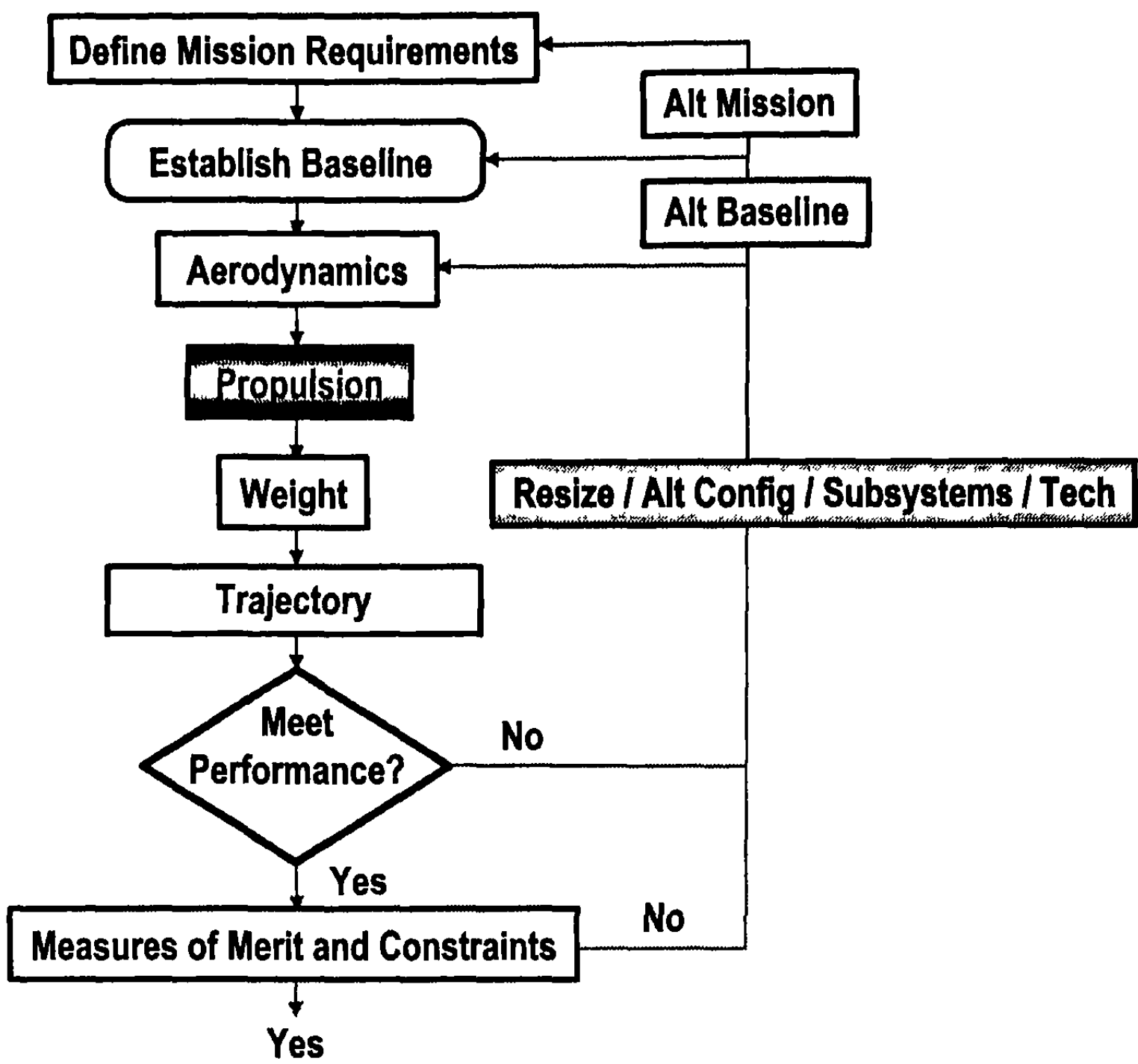


Fig. 3.1 Missile concept synthesis requires evaluation of alternatives and iteration.

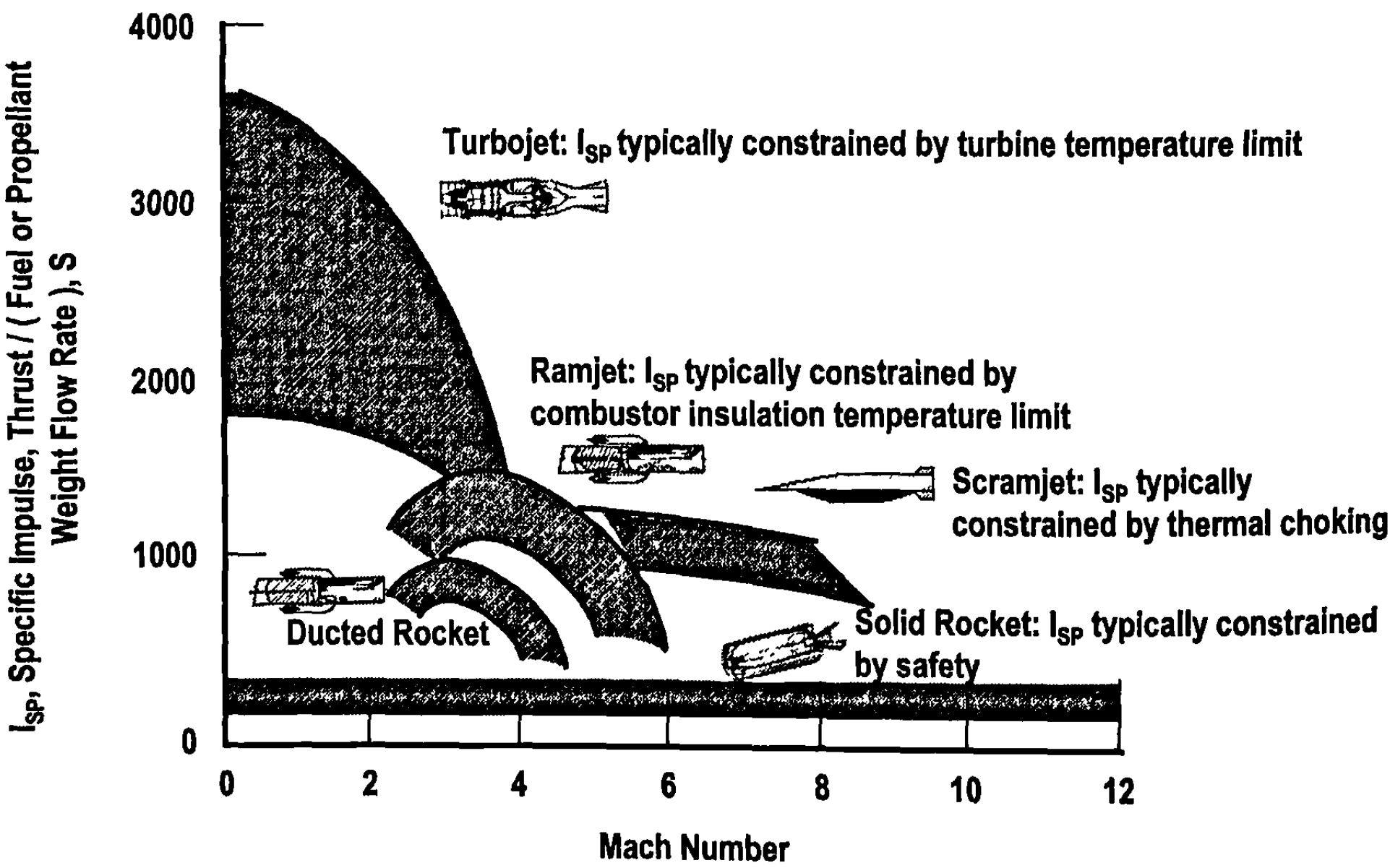


Fig. 3.2 High specific impulse is indicative of lower fuel/propellant consumption.

compressor and expensive cooling to avoid exceeding the current material temperature limit of about 2500° F at the turbine.

A ramjet is efficient from Mach 2.5 to 5. Above Mach 5, the combustor maximum material temperature limits the achievable exit velocity and thrust. Also, deceleration of the inlet airflow to subsonic velocity results in chemical dissociation of the air, which absorbs heat and negates a portion of the energy input of the combustor. For a subsonic launch platform, a rocket boost is required to achieve the ramjet thrust takeover at about Mach 2.5.

The maximum specific impulse of ducted rocket propulsion is about 800 to 1000 s, which is intermediate to that of a solid rocket and a ramjet. Ducted rockets are most efficient for a Mach number range from about 2.5–4. Ducted rockets have higher acceleration capability (higher thrust) than ramjets and generally have longer range capability (higher specific impulse) than solid propellant rockets.

Scramjet propulsion has supersonic flow through the entire flowpath, including supersonic combustion. Efficient cruise is about Mach 6, 100,000 ft altitude. A typical limit in scramjet specific impulse at lower Mach number is thermal choking. A larger inlet is required to avoid thermal choking for Mach numbers less than Mach 6. Scramjet propulsion challenges include fuel/air mixing, efficient combustion, airframe integration, and sufficient thrust to overcome the drag. Compared to a ramjet, a longer combustion chamber is required because of the mixing time for supersonic combustion. An enabling technology to enhance supersonic combustion is endothermic fuels. Endothermic fuels decompose at high temperature into lighter weight molecular products that burn more readily, providing higher specific impulse and permitting shorter combustor length. An endothermic fuel also acts as a heat sink, cooling the adjacent structure. Because scramjet thrust is less than drag for Mach numbers less than about 4, a scramjet requires a large rocket motor to boost it to a high takeover speed. Due to its relatively low thrust, a scramjet may not have sufficient thrust to accelerate or even maintain a cruise condition of thrust equal to drag.

A dual-combustion ramjet/scramjet (DCR) has some of the higher Mach number capability of a scramjet and some of the relatively higher thrust capability and lower takeover Mach number of a ramjet. DCR propulsion separates the inlet airflow into two streams. The main airflow remains supersonic. The smaller airflow is decelerated to subsonic speed for fuel-rich combustion, after which it accelerates through a nozzle. Efficient flight is Mach 3 to Mach 7.

Solid propellant rockets are capable of providing thrust across the entire Mach number range. Safety concerns of explosive sensitivity and toxicity provide constraints on the practical specific impulse of tactical rocket propellants. Although the specific impulse of tactical rockets is relatively low, on the order of 250 s, rockets have an advantage of much higher acceleration capability than air-breathing propulsion. Also, because a rocket can operate at any altitude, a boost-climb-glide trajectory can provide extended range by minimizing drag in high-altitude flight.

Figure 3.3 compares four propulsion alternatives for a long-range precision strike missile. The propulsion alternatives are subsonic cruise turbojet, supersonic cruise liquid hydrocarbon fuel ramjet, hypersonic cruise liquid hydrocarbon fuel scramjet, and supersonic cruise solid propellant rocket. All four propulsion types are held to a missile launch weight of 2000 lb, a representative weight limit

$R = (L / D) I_{sp} V \ln [W_L / (W_L - W_p)]$, Breguet Range Equation

Parameter	Typical Value for 2,000 lb Precision Strike Missile			
	Subsonic Turbojet Missile 	Liquid Fuel Ramjet Missile 	Hydrocarbon Fuel Scramjet Missile 	Solid Rocket 
L / D, Lift / Drag	10	5	3	5
I _{sp} , Specific Impulse	3000 s	1300 s	1000 s	250 s
V _{AVG} , Average Velocity	1000 ft / s	3500 ft / s	6000 ft / s	3000 ft / s
W _p / W _L , Cruise Propellant or Fuel Weight / Launch Weight	0.3	0.2	0.1	0.4
R, Cruise Range	1800 nm	830 nm	310 nm	250 nm
Note: Ramjet and Scramjet missiles booster propellant for Mach 2.5 to 4 take-over speed not included in W _p for cruise. Rockets require thrust magnitude control (e.g., pintle, pulse, or gel motor) for effective cruise. Max range for a rocket is usually a semi-ballistic flight profile, instead of cruise flight. Multiple stages may be required for rocket range greater than 200 nm.				

Fig. 3.3 Cruise range is driven by L/D , I_{sp} , velocity, and propellant or fuel weight fraction.

for carriage on a small fighter aircraft such as the F-18C. The predicted cruise range is based on the Breguet range equation

$$R = (L/D)I_{sp} V \ln[W_L/(W_L - W_P)]$$
(3.1)

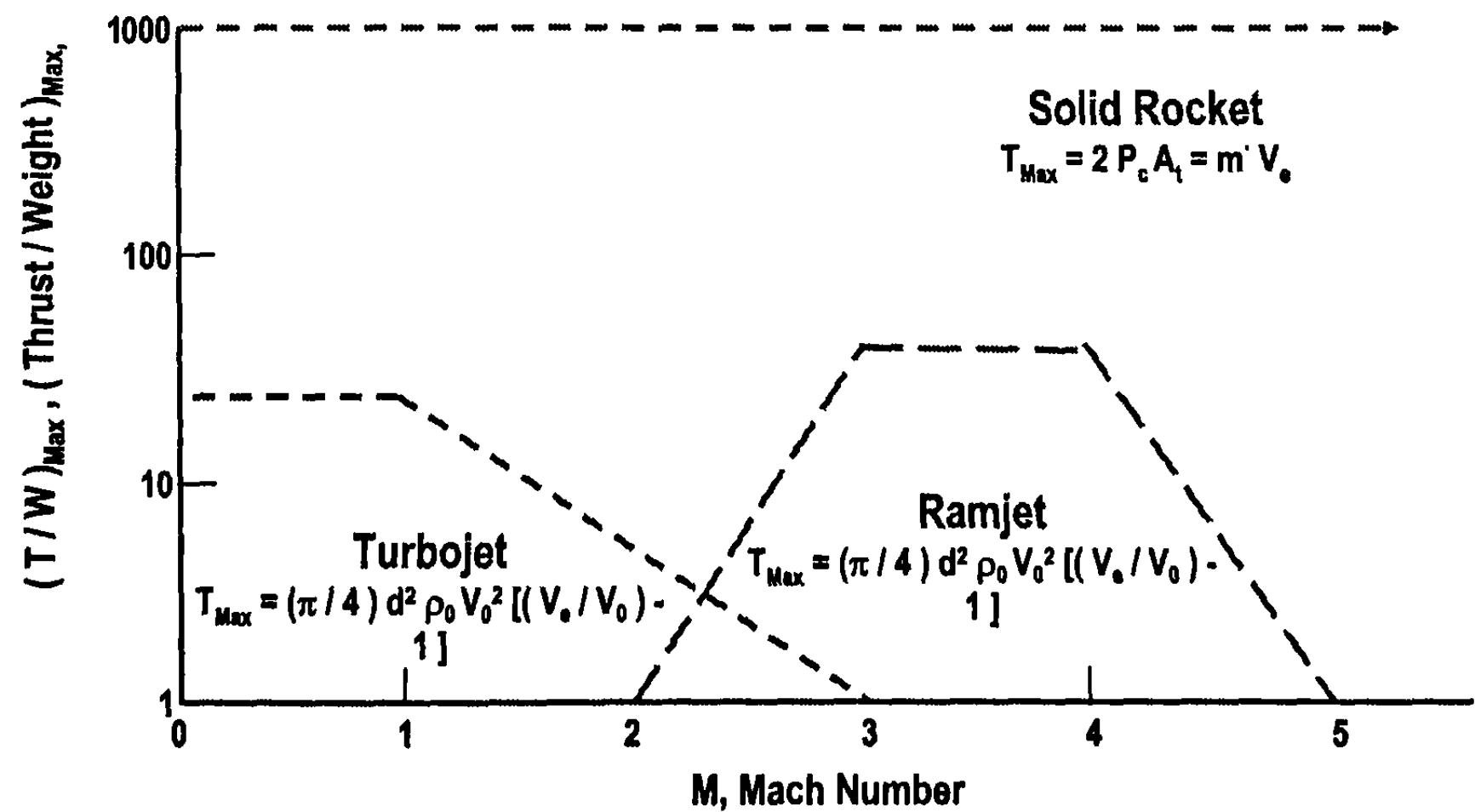
Note from the figure that subsonic cruise turbojet propulsion is the preferred approach for long-range strike against targets that are not time critical. Subsonic cruise turbojet propulsion has 120% greater range than the next best alternative, a supersonic cruise liquid-fuel ramjet (1800 vs 830 n miles). An examination of the Breguet range equation explains the difference in performance. The subsonic cruise turbojet missile is superior to the supersonic cruise ramjet missile in the maximum lift-to-drag ratio ($L/D = 10$ vs 5), specific impulse ($I_{SP} = 3000$ vs 1300 s), and available fuel for a specified launch weight of 2000 lb (600 vs 400 lb of fuel). The ramjet missile has less available weight for fuel because it requires a rocket to boost the missile up to about Mach 2.5 for transition to ramjet propulsion. However, a ramjet missile has an advantage of a shorter response time against time-critical targets. It may also have an advantage in survivability because of the higher flight altitude and higher speed. If time-critical targets are of utmost importance, scramjet propulsion may be preferred. As shown in the figure, the scramjet missile example is 70% faster than the ramjet (6000 vs 3500 ft/s). However, the maximum range of a scramjet missile that is limited to 2000 lb launch weight is only 37% that of a liquid-fuel ramjet (310 vs 830 n miles). Again, examining the Breguet range equation is instructive. The liquid-fuel ramjet missile is superior to the scramjet in its aerodynamic efficiency ($L/D = 5$ vs 3), specific impulse ($I_{SP} = 1300$ vs 1000 s),

and available fuel for a missile limited to 2000 lb launch weight (400 vs 200 lb of fuel). The scramjet missile has less available weight for fuel because it requires a larger rocket booster for a higher takeover Mach number (Mach 4 vs 2.5), requires a longer combustor for efficient combustion, and requires more insulation. Finally, the supersonic cruise rocket has a maximum flight range of 250 n miles. The most efficient cruise condition for the long-range rocket was found to be Mach 3 cruise at high altitude. The solid propellant rocket example uses thrust magnitude control from a pintle motor for more efficient acceleration and cruise. Although it is not shown, a semiballistic flight trajectory (e.g., launch, pitch-up, ballistic climb, glide) would have provided a more efficient flight trajectory for the rocket. For ranges greater than about 200 n miles, a rocket may benefit from multiple stages.

Based on an examination of the Breguet range equation, new technology development has payoff in the areas of high cruise velocity, aerodynamic efficiency (lift/drag), specific impulse, lightweight structure, lightweight/low volume subsystems, and high density fuel/propellant.

Shown in Fig. 3.4 is a comparison of tactical missile propulsion alternatives based on acceleration capability. It is a typical envelope of the maximum thrust-to-weight ratio. The comparison is shown as a function of Mach number. Note that the solid rocket has the highest thrust-to-weight ratio. This is because of high exit velocity, the independence of the exit velocity from that of the freestream velocity, and the capability for high mass flow rate. The exit velocity of a solid rocket is about 6000 ft/s, much higher than that of a typical ramjet (exit velocity about 4500 ft/s) or a typical turbojet (exit velocity about 2000 ft/s). Note from the maximum thrust equation for air-breathing propulsion

$$T_{Max} = (\pi/4)d^2\rho_0V_0^2[(V_e/V_0) - 1] \tag{3.2}$$



Note:
 P_c = Chamber pressure, A_t = Nozzle throat area, m' = Mass flow rate
 d = Diameter, ρ_0 = Free stream density, V_0 = Free stream velocity,
 V_e = Nozzle exit velocity (Turbojet: $V_e \sim 2000$ ft / s, Ramjet: $V_e \sim 4500$ ft / s, Rocket: $V_e \sim 6000$ ft / s)

Fig. 3.4 Solid rockets have high acceleration capability.

Turbojets and ramjets produce thrust only if the exit velocity is greater than the freestream velocity ($V_e > V_0$). The maximum velocity of an air-breathing missile is less than the exit velocity.

3.3 Turbojet Propulsion Temperature Technology Limit

The material temperature limits of the turbojet combustor and turbine usually limit the specific impulse and thrust of a turbojet, particularly for high-speed flight. Figure 3.5 is a schematic of the flow through a turbojet. The flow path consists of the initial free stream (station 0), inlet entrance (station 1), inlet exit/compressor entrance (station 2), compressor exit/combustor entrance (station 3), combustor exit/turbine entrance (station 4), and turbine exit/nozzle entrance (station 5). For a theoretical, ideal turbojet, the thermodynamic cycle consists of an isentropic compression of the flow by the inlet, isentropic compression by the compressor, constant pressure combustion in the combustor, isentropic expansion through the turbine, and isentropic expansion through the nozzle to freestream static pressure. For an ideal turbojet, the work extracted from the flow by the turbine is equal to the work that is required in driving the compressor.

Figure 3.6 shows the impact of the compressor pressure ratio and the flight Mach number on the compressor exit temperature T_3 . Note from the figure that the compressor has high temperatures for high compressor pressure ratios and high freestream Mach Numbers. An assumption is low subsonic flow through the compressor such that the static temperature is nearly equal to the total temperature ($T_3 \approx T_{3t}$). For an ideal compressor, with isentropic flow, the compressor exit temperature is

$$T_3 \approx T_0 \{ 1 + [(\gamma_0 - 1)/2] M_0^2 \} (p_3/p_2)^{(\gamma_3 - 1)/\gamma_3}$$

(3.3)

If the temperature of air is less than 800 R, it behaves as a calorically perfect gas with a constant specific heat ratio of $\gamma = 1.4$. If the temperature of air is

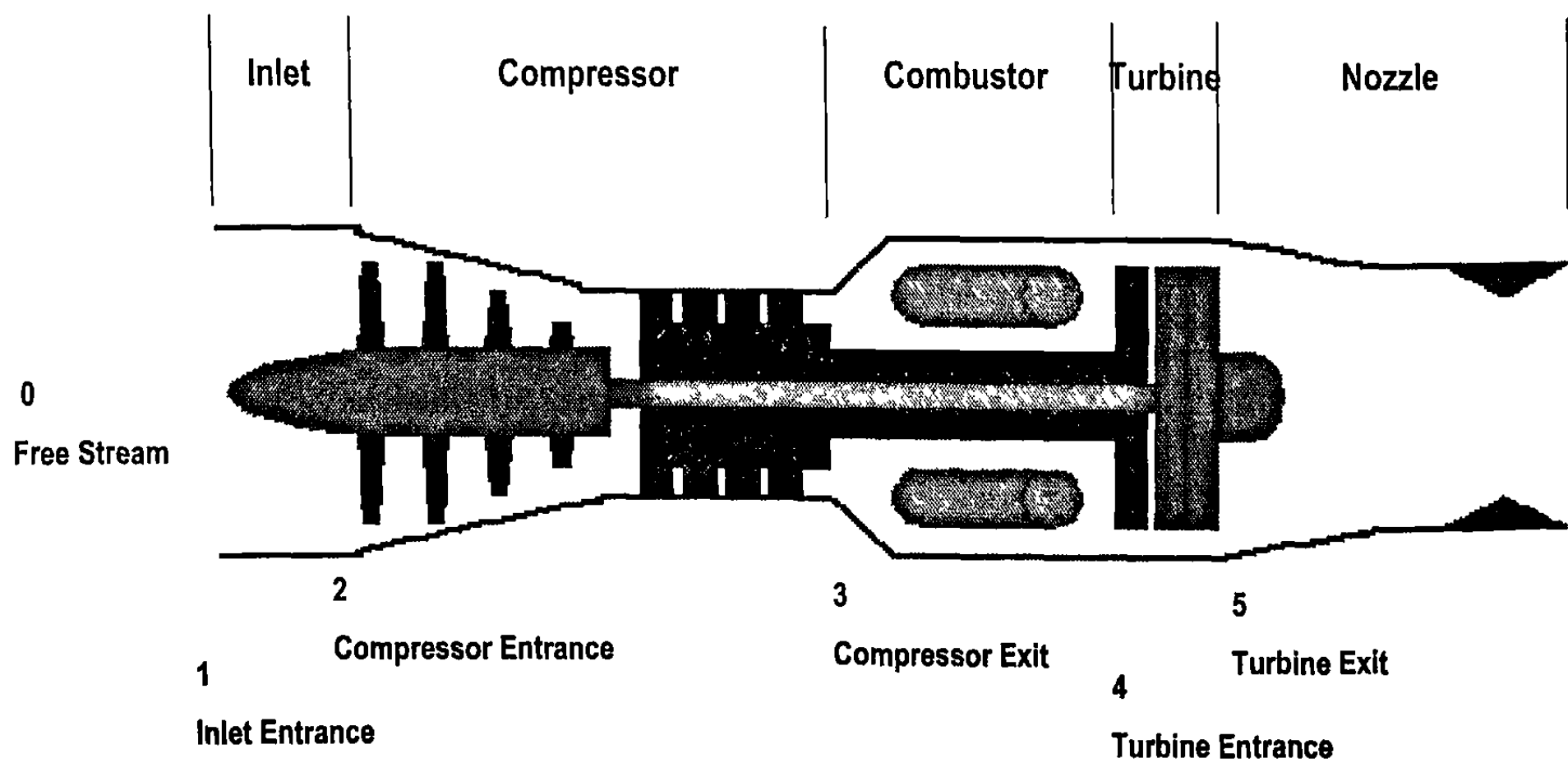
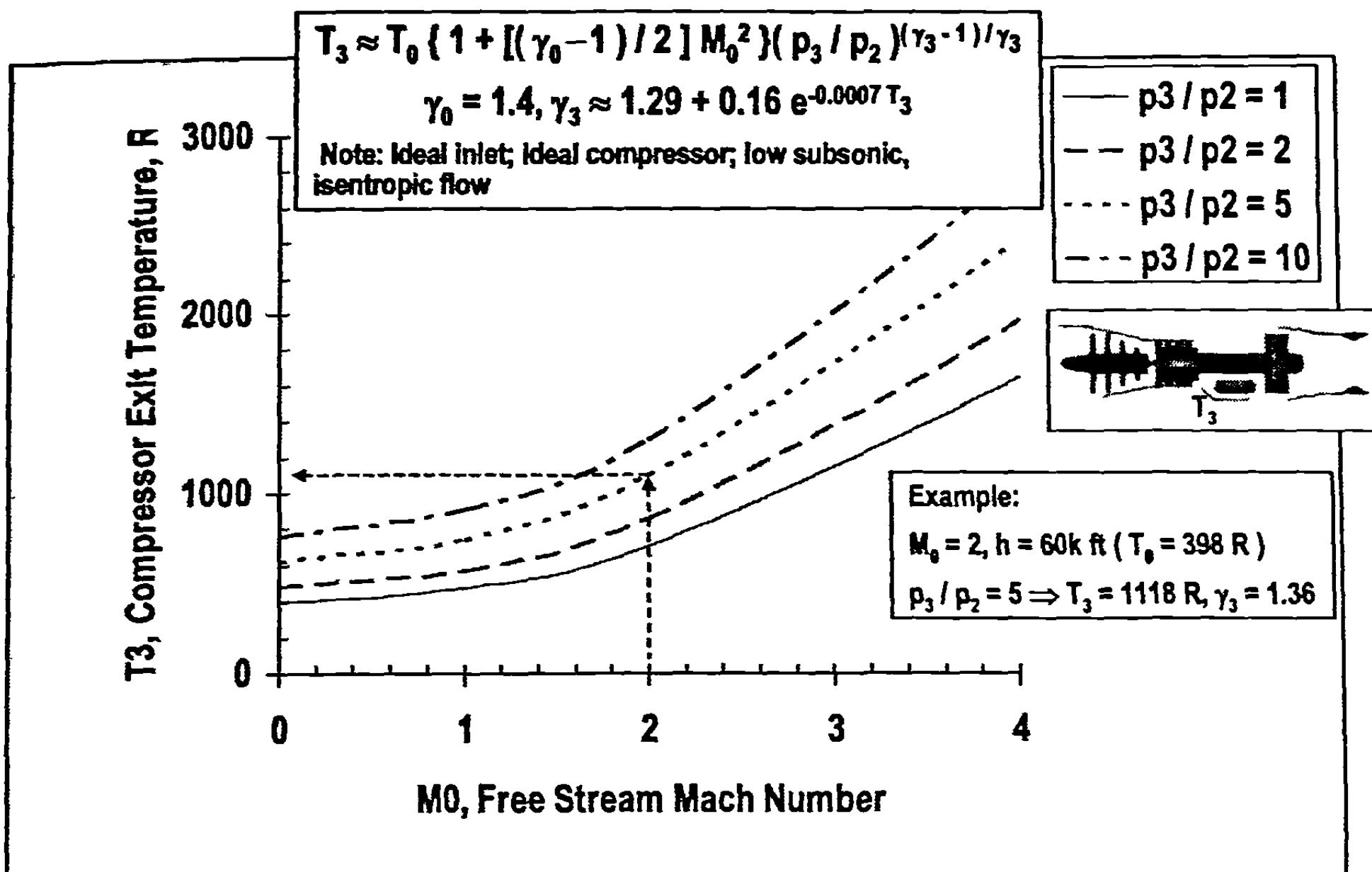


Fig. 3.5 Turbojet nomenclature.



T_3 = Compressor exit temperature in Rankine, T_0 = freestream temperature in Rankine, γ = specific heat ratio, M_0 = free stream Mach number, p_3 = compressor exit pressure, p_2 = compressor entrance pressure

Fig. 3.6 High-temperature compressors are required to achieve high pressure ratio at high speed.

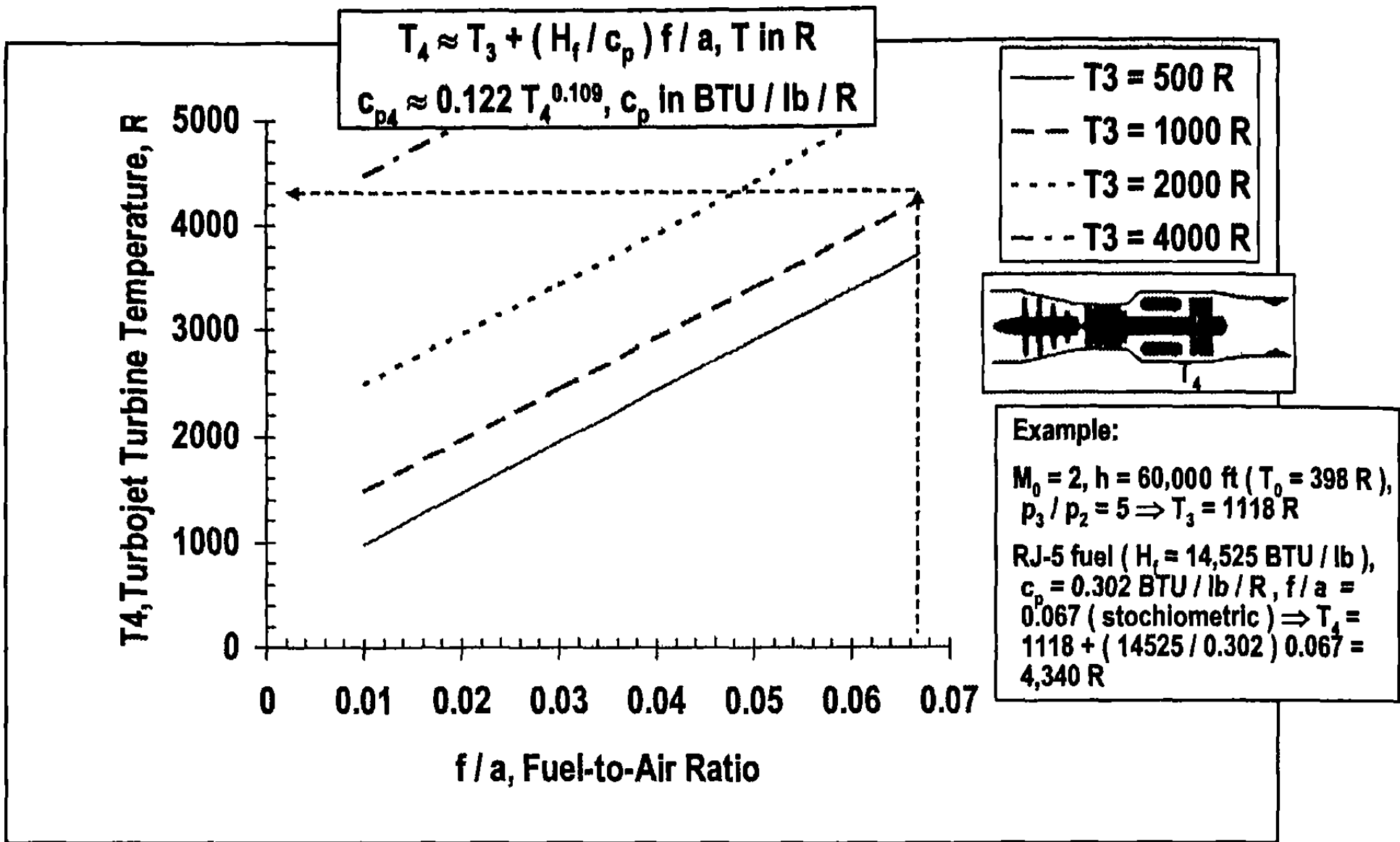
greater than 800 R but less than 5000 R, the vibration states of oxygen and nitrogen molecules affect the specific heat ratio. The air behaves as a thermally perfect gas with the specific heat ratio a function of temperature. If the temperature of air is greater than 5000 R, the oxygen molecules of the air chemically react to form dissociated atomic oxygen. An equation for γ for air temperatures up to 5000 R is

$$\gamma \approx 1.29 + 0.16 e^{-0.0007 T} \quad (3.4)$$

As an example, assume a flight number of $M_0 = 2$, a flight altitude of $h = 60,000 \text{ ft}$, and a compressor pressure ratio of $p_3 / p_2 = 5$. Substituting into the equations for compressor exit temperature and specific heat ratio gives $T_3 = 1118 \text{ R}$ for a value of $\gamma_3 = 1.36$.

Figure 3.7 shows the impact of compressor exit temperature and fuel-to-air ratio on the turbine entrance temperature. It is assumed that the turbine entrance temperature is the same as the combustor exit temperature. Note from the figure that the turbine temperature is high for high values of compressor temperature and fuel-to-air ratio. An assumption is low subsonic flow into the turbine, such that the static temperature is nearly equal to the total temperature ($T_4 \approx T_{4t}$). For an ideal turbine, the turbine entrance temperature T_4 is equal to the sum of the compressor exit temperature T_3 plus the temperature rise from combustion. The equation is

$$T_4 \approx T_3 + (H_f / c_p) f / a \quad (3.5)$$



T_4 = Turbojet turbine entrance temperature in Rankine, T_3 = compressor exit temperature in Rankine, H_f = heating value of fuel, c_p = specific heat at constant pressure, f / a = fuel-to-air ratio

Fig. 3.7 High turbine temperature is required for high-speed turbojet missiles.

In the above equation H_f = heating value of fuel, c_p = specific heat at constant pressure, and f/a = fuel-to-air ratio. The specific heat at constant pressure is a function of temperature, given by the equation

$$c_p \approx 0.122T^{0.109} \tag{3.6}$$

The units of c_p are Btu/lb/R.

The range of fuel-to-air ratio shown in the figure is from $f/a = 0.01$, which is a lean typical lower limit of combustor blowout, to $f/a = 0.067$, which is stoichiometric combustion.

As an example, let us calculate the turbine entrance temperature for flight at Mach 2 and $h = 60,000$ ft altitude. Assume a compressor pressure ratio $p_3/p_2 = 5$; RJ-5 hydrocarbon fuel, which has a heating value of $H_f = 14525$ Btu/lb; and stoichiometric combustion, which has a fuel-to-air ratio $f/a = 0.067$. Substituting into the previous equation for compressor exit temperature gives $T_3 = 1118$ R. Substituting into the equations for turbine entrance temperature and specific heat at constant pressure gives $T_4 = 4340$ R for a value of $c_p = 0.302$ Btu/lb/R.

Hypersonic turbojet propulsion is not currently practical because of the turbine material temperature limits. An assessment is given in Table 3.1 of the achievable specific impulse at Mach 4 cruise for current and advanced materials. Nickel-based super alloys such as Inconel, Rene, Hastelloy, and Haynes have a maximum temperature limit of about 3000 R for short-duration turbines. For Mach 4 cruise, limiting the turbine temperature to 3000 R has a large negative impact on the specific impulse I_{SP} . Limiting the compressor ratio to less than

Table 3.1 Turbine material temperature limit is a constraint for a high-speed turbojet missile

Max Short Duration Temp	Turbine Material	Temperature Constrained Turbines for Mach 4 Cruise	I_{SP} for Mach 4 Cruise
$\approx 3000R$	Nickel Super Alloys	Very Highly Constrained Turbojet, Air Turbo Rocket, Turbo Ramjet	$\approx 1000s$
$\approx 3000R$	Titanium Aluminides (lighter weight than nickel super alloys)	Very Highly Constrained Turbojet, Air Turbo Rocket, Turbo Ramjet	$\approx 1000s$
$\approx 3500R$	Single Crystal Nickel Aluminides	Highly Constrained Turbojet	$\approx 1200s$
$\approx 4000R$	Ceramic Matrix Composites	Moderately Constrained Turbojet	$\approx 1500s$
$\approx 4500R$	Rhenium Alloys	Moderately Constrained Turbojet	$\approx 2000s$
$\approx 5000R$	Tungsten Alloys	Slightly Constrained Turbojet	$\approx 2500s$

Note: Constrained turbojet for Mach 4 cruise imposes a limit on turbine temperature that is less than ideal. Constraints could consist of a combination of:

- Constraint on compressor pressure ratio to limit turbine temperature
- Constraint on fuel-to-air ratio to limit turbine temperature
- Use of afterburner to limit turbine temperature

optimum, limiting the fuel-to-air ratio to less than optimum, or using an afterburner can constrain the turbine temperature, but with reduced I_{SP} . For a turbine temperature limit of 3000 R, a very highly constrained turbojet has an I_{SP} of about 1000 s for Mach 4 flight, about 33% of the subsonic I_{SP} . Other potential turbine-based propulsion systems, such as an air turbo rocket or a turbo ramjet, would also have a Mach 4 I_{SP} of about 1000 s. Another advanced high-temperature material is titanium aluminide. Titanium aluminide is a more expensive but lighter weight alternative to nickel super alloys. The temperature limit of a titanium aluminide turbine is also about 3000 R, giving a Mach 4 I_{SP} of about 1000 s. As before, the turbine-based Mach 4 propulsion alternatives would be a very highly constrained turbojet, an air turbo rocket, or a turbo ramjet. Another advanced material, with a higher temperature capability, is single crystal nickel aluminide. The maximum short-duration turbine temperature is about 3500 R, which would provide a Mach 4 I_{SP} of about 1200 s for a highly constrained turbojet, with about 40% of the subsonic I_{SP} . For a turbine temperature of 3500 R, a turbojet is preferred over the alternatives of an air turbo rocket or a turbo ramjet because of its higher I_{SP} . A higher temperature but higher risk turbine material is ceramic matrix composite. A concern is fracture sensitivity. Short-duration maximum turbine temperature is about 4000 R, which would provide a Mach 4 I_{SP} of about 1500 s for a moderately constrained turbojet, about 50% of the subsonic I_{SP} . As before, the turbojet would be preferred over either an air turbo rocket or a turbo ramjet. Rhenium would allow a short-duration turbine temperature of about 4500 R, allowing a Mach 4 turbojet

I_{SP} of about 2000 s, about 67% of the subsonic I_{SP} . Disadvantages of rhenium are high cost and weight. However, rhenium has good potential for high-temperature applications, and its cost should decrease in the future. The SM-3 Standard Missile Third Stage Rocket Motor (TSRM) is an example of an application of rhenium. Finally, the highest temperature turbine material is tungsten, which has a maximum temperature of about 5000 R for a short-duration application. The I_{sp} of a Mach 4 turbojet using a tungsten-based turbine is about 2500 s, which is 83% of the subsonic I_{SP} . Although a tungsten-based turbine would only slightly constrain a Mach 4 turbojet, tungsten has a risk of fracture sensitivity, is difficult to fabricate to precision tolerance, and is very heavy. However, the relatively small constraint of limiting a tungsten turbine to a temperature of 5000 R, a value that is somewhat lower than ideal temperature for Mach 4 cruise, would result in a compressor pressure ratio slightly less than ideal and a fuel-to-air ratio slightly less than stoichiometric, or it would require a small amount of afterburning.

It was previously shown that a turbine-based missile has longer range than a ramjet. Another advantage is a capability to loiter over the target area. Although a turbine-based missile is capable of subsonic through supersonic cruise, to date there are few examples. These are shown in Fig. 3.8. Examples of supersonic turbojets are the low supersonic Firebee II, the low supersonic Regulus II, and the higher supersonic SS-N-19 Shipwreck. There are no examples of a turbo ramjet missile. Turbo ramjets are complicated, requiring variable bypass air, and may be too expensive for missile applications. The high supersonic SR-71 has turbo ramjet propulsion. An air turbo rocket is a relatively new technology. The relatively low temperature gas from a fuel-rich gas generator drives the turbine, which drives the compressor. Freestream air is compressed by the compressor and then flows into the combustor, where it mixes with the fuel-rich gas from the gas generator. Secondary combustion occurs, followed by

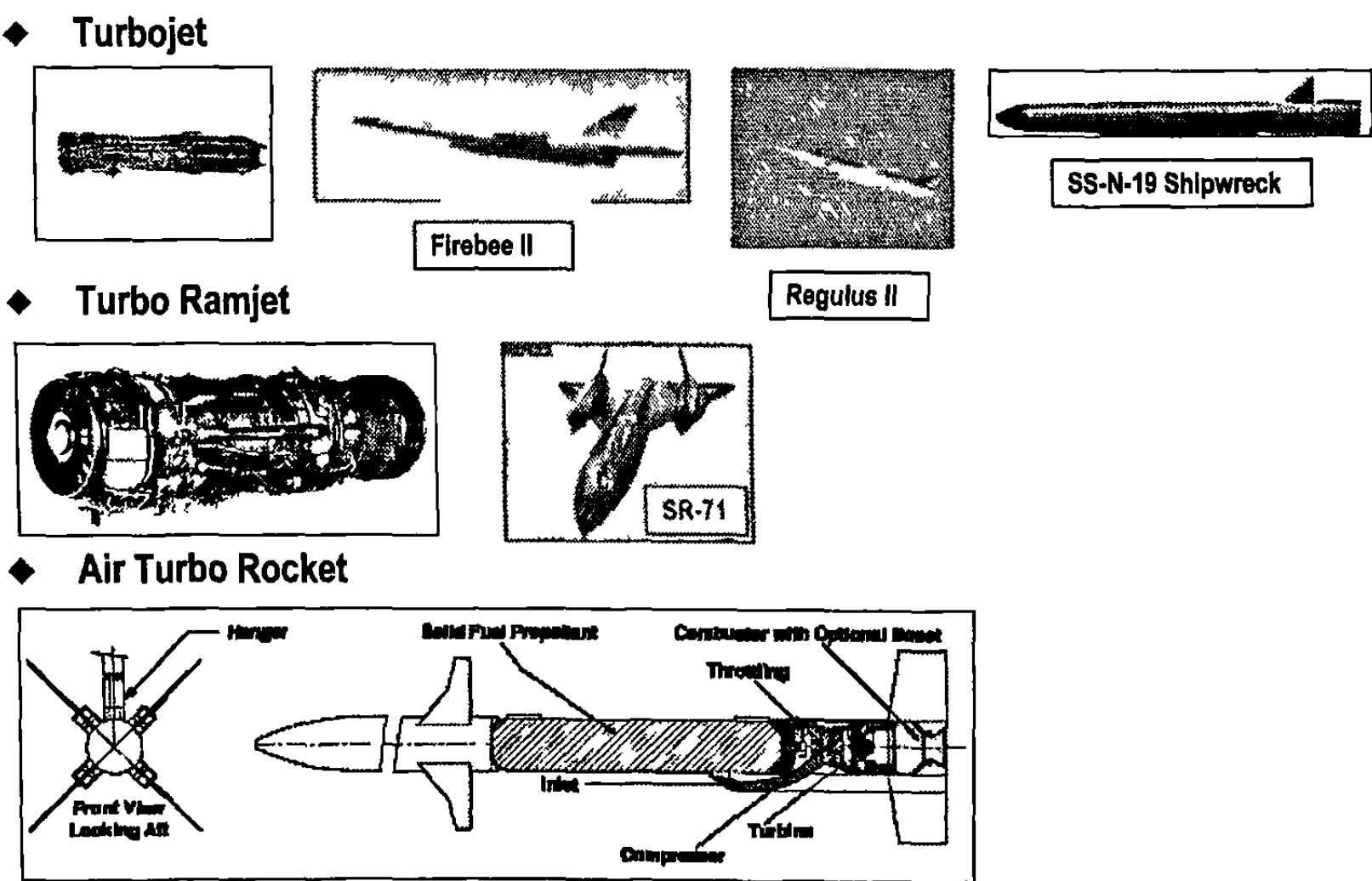


Fig. 3.8 Turbine-based missiles are capable of subsonic to supersonic cruise. Figure also in color section.

expansion through the nozzle. An advantage of an air turbo rocket is the turbine is not exposed to the high temperature of secondary combustion in the combustion chamber. An air turbo rocket has a higher I_{SP} than a ducted rocket but a lower I_{SP} than an unconstrained turbojet.

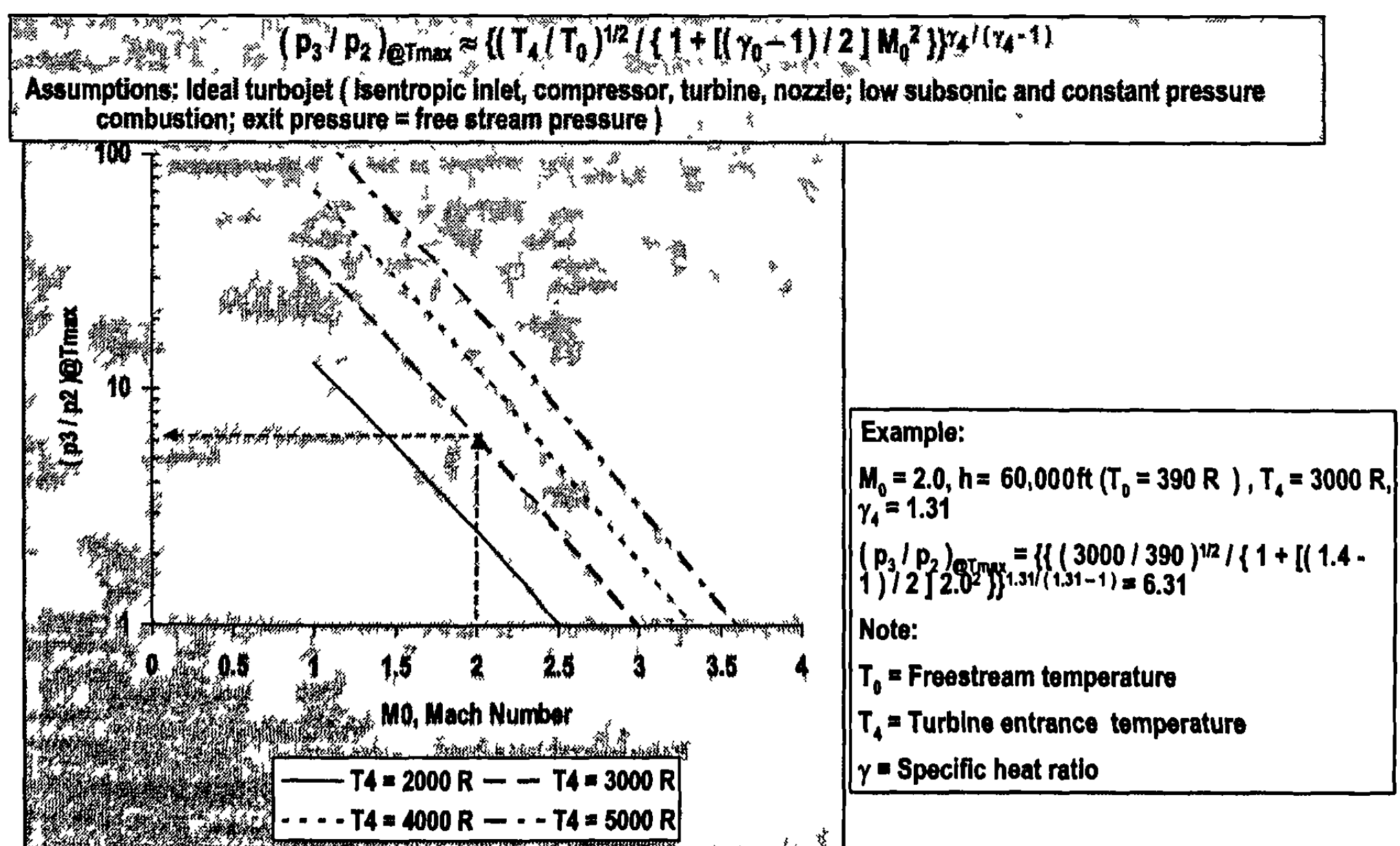
A multimission strike missile using turbine-based propulsion has high potential. It could counter both conventional targets and time-critical targets. Benefits of a turbine-based propulsion system could include 1) subsonic cruise efficiency against long-range targets that are not time critical; 2) fast response (e.g., Mach 4 cruise) against time-critical targets; 3) subsonic loiter capability; and 4) reduced logistics/cost of a multimission missile.

Turbojet thrust is usually limited by the turbine temperature. For a specified turbine temperature, there is an optimum value of the compressor ratio that gives maximum thrust. For an ideal turbojet, from Ashley¹¹

$$(p_3/p_2)_{@T_{max}} \approx \{(T_4/T_0)^{1/2} / \{1 + [(\gamma_0 - 1)/2] M_0^2\}\}^{\gamma_4/(\gamma_4 - 1)} \quad (3.7)$$

The assumptions in the above equation are isentropic inlet, compressor, turbine, and nozzle; low subsonic flow through the compressor ($p_3 \approx p_{3i}$, $p_2 \approx p_{2i}$); low subsonic flow and constant pressure through the combustor; and the nozzle exit pressure equal to the freestream pressure.

Figure 3.9 shows that at subsonic Mach number the ideal pressure ratio for maximum thrust is very high [e.g., $(p_3/p_2)_{@T_{max}} \gg 10$]. A technology limit at low Mach number is achieving high compressor ratios. At the other extreme, the figure shows that at high supersonic Mach numbers, low values of p_3/p_2 are optimum for maximum thrust. Considering the limit for the practical use of a



Source: Ashley, H., *Engineering Analysis of Flight Vehicles*, Dover Publications, Inc., New York, 1974

Fig. 3.9 Compressor pressure ratio for a maximum thrust turbojet is limited by turbine temperature.

turbojet, at very high supersonic Mach numbers the ideal pressure ratio approaches 1 (ramjet operation). Turbojet operation at very high supersonic Mach number, with $p_3/p_2 > 1$, requires high-temperature combustion/high-temperature turbine.

As an example calculation of $(p_3/p_2)_{@T_{max}}$, assume $M_0 = 2$, $h = 60,000$ ft altitude ($T_0 = 390$ R), $T_4 = 3000$ R, and $\gamma_4 = 1.31$. Calculate

$$(p_3/p_2)_{@T_{max}} = \{ \{ (3000/390)^{1/2} / \{ 1 + [(1.4 - 1)/2] 2^2 \} \}^{1.31/(1.31-1)} = 6.31$$

(3.8)

3.4 Turbojet Thrust Prediction

Shown in Fig. 3.10 is the nondimensional thrust of an ideal turbojet. The equation is

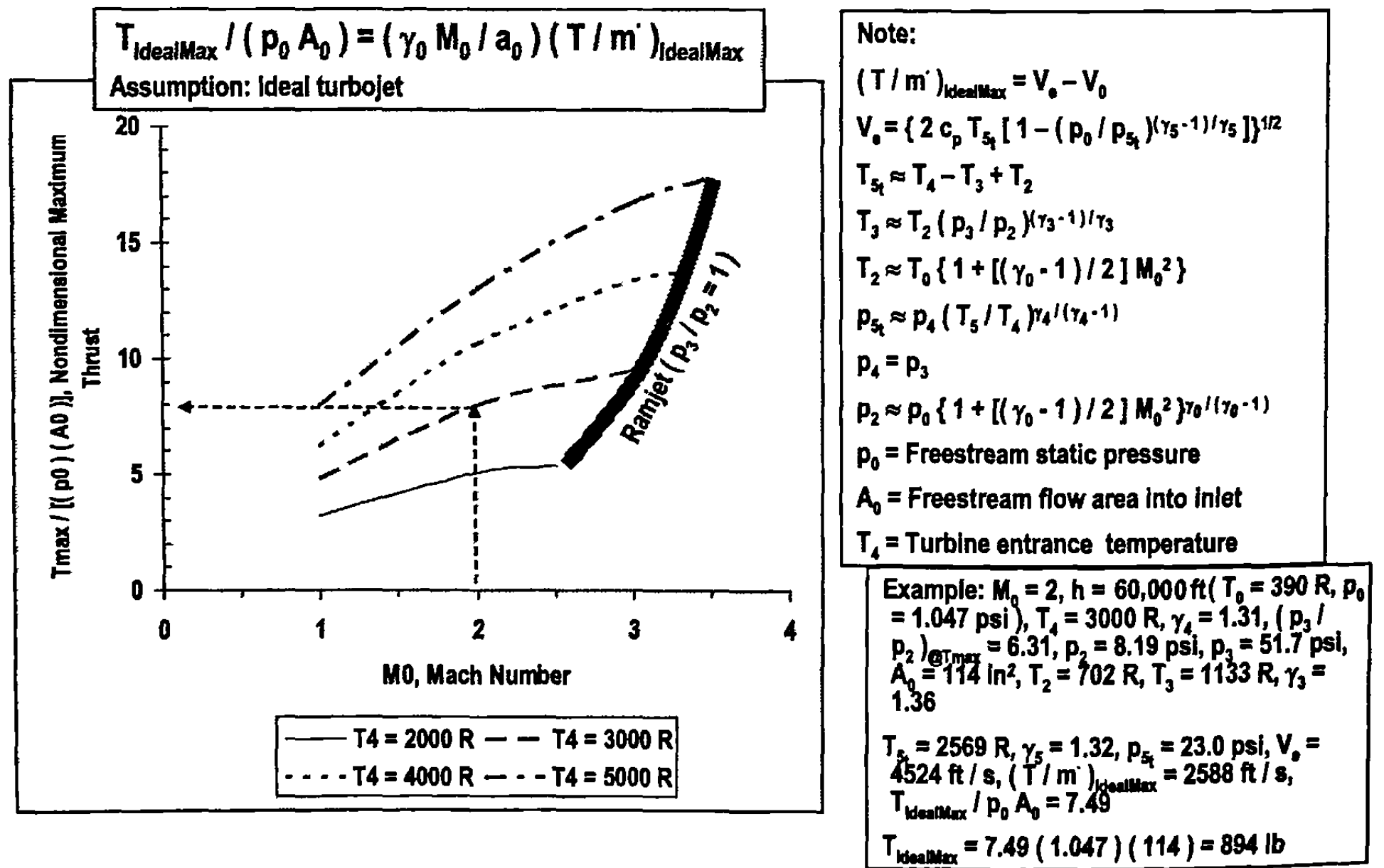
$$T_{IdealMax}/(p_0 A_0) = (\gamma_0 M_0 / a_0) (T/\dot{m})_{IdealMax}$$

(3.9)

Assumptions are isentropic flow through the inlet, compressor, turbine, and nozzle; low subsonic flow through the compressor; low subsonic flow and constant pressure through the combustor; and nozzle exit pressure equal to the freestream pressure. The thrust equation is derived from the momentum equation

$$T = \dot{m}(V_e - V_0)$$

(3.10)



The equation for the exit velocity of an ideal turbojet is

$$V_e = \{2c_p T_{5_t} [1 - (p_0/p_{5_t})^{(\gamma_5-1)/\gamma_5}]\}^{1/2} \quad (3.11)$$

The primary terms to be determined in the above equation are the turbine exit total temperature T_{5_t} and the turbine exit pressure p_{5_t} . The equation for the turbine exit total temperature of an ideal turbojet is $T_{5_t} \approx T_4 - T_3 + T_2$. The temperatures in this equation are the combustor temperature T_4 , compressor exit temperature T_3 , and the compressor entrance/inlet exit temperature T_2 . In this analysis the combustor temperature T_4 is specified. For the ideal turbojet, the compressor exit temperature is

$$T_3 \approx T_2 (p_3/p_2)^{(\gamma_3-1)/\gamma_3} \quad (3.12)$$

The compressor entrance temperature is

$$T_2 \approx T_0 \{1 + [(\gamma_0 - 1)/2] M_0^2\} \quad (3.13)$$

Next, the term p_{5_t} in the exit velocity equation can be solved from

$$p_{5_t} \approx p_4 (T_{5_t}/T_4)^{\gamma_4/(\gamma_4-1)} \quad (3.14)$$

We will assume constant pressure combustion ($p_4 = p_3$). In this analysis, the compressor pressure ratio p_3/p_2 is specified. Compressor entrance/inlet exit pressure p_2 , for adiabatic flow through the inlet, is

$$p_2 \approx p_0 \{1 + [(\gamma_0 - 1)/2] M_0^2\}^{\gamma_0/(\gamma_0-1)} \quad (3.15)$$

Note from the figure that high thrust for an ideal turbojet occurs for high atmospheric pressure p_0 (low altitude), large inlet capture area A_0 , high freestream Mach number M_0 , and high combustor exit/turbine entrance temperature T_4 . Also note the boundary at the right side of the figure (ramjet operation, with a compressor pressure ratio $p_3/p_2 = 1$).

As an example calculation of the maximum thrust of an ideal turbojet, assume $A_0 = 114 \text{ in}^2$, $T_4 = 3000 \text{ R}$, $M_0 = 2$, and $h = 60,000 \text{ ft}$ altitude ($T_0 = 390 \text{ R}$, $p_0 = 1.047 \text{ psi}$, $a = 968 \text{ ft/s}$). From the previous example, $\gamma_4 = 1.31$ and $(p_3/p_2)_{@T_{\max}} = 6.31$. Calculate

$$p_2 = 1.047 \{1 + [(1.4 - 1)/2] 2^2\}^{1.4/(1.4-1)} = 8.19 \text{ psi} \quad (3.16)$$

$$T_2 = 390 \{1 + [(1.4 - 1)/2] 2^2\} = 702 \text{ R} \quad (3.17)$$

Solve for

$$p_3 = p_4 = 6.31(8.19) = 51.7 \text{ psi} \quad (3.18)$$

$$T_3 = 702(6.31)^{(1.36-1)/1.36} = 1133 \text{ R} \quad (3.19)$$

$$\gamma_3 = 1.29 + 0.16 e^{-0.0007(1133)} = 1.36 \quad (3.20)$$

Substituting the values of the T_4 , T_3 , and T_2 temperatures gives

$$T_{5_t} = 3000 - 1133 + 702 = 2569 \text{ R} \quad (3.21)$$

$$\gamma_5 = 1.32 \quad (3.22)$$

Substituting the values of p_4 , T_{5_t} , T_4 , and γ_4 into the equation for the turbine exit total pressure gives

$$p_{5_t} = 51.7(2569/3000)^{1.31/(1.31-1)} = 23.0 \text{ psi} \quad (3.23)$$

Also, substituting the values of c_{p_5} , T_{5_t} , p_{5_t} , p_0 , and γ_5 in into the exit velocity equation gives

$$V_e = 4524 \text{ ft/s} \quad (3.24)$$

Thrust per unit mass flow rate (also the incremental velocity), is then

$$T/\dot{m} = 4524 - 1936 = 2588 \text{ ft/s} \quad (3.25)$$

All of the required terms are now available to substitute in the nondimensional thrust equation, giving

$$T_{\text{IdealMax}}/(p_0 A_0) = [1.4(2)/968](2588) = 7.49 \quad (3.26)$$

Substituting the freestream static pressure p_0 and inlet capture area A_0 into the above equation gives the thrust

$$T_{\text{IdealMax}} = 7.49(1.047)(114) = 894 \text{ lb} \quad (3.27)$$

3.5 Turbojet Specific Impulse Prediction

Figure 3.11 shows the nondimensional specific impulse of an ideal turbojet. It is based on the equation

$$(I_{SP})_{\text{Ideal@}T_{\text{max}}} g_c c_p T_0 / (a_0 H_f) = T_{\text{IdealMax}} T_0 / [(p_0 A_0 \gamma_0 M_0)(T_4 - T_3)] \quad (3.28)$$

Assumptions are maximum thrust; isentropic flow through the inlet, compressor, turbine, and nozzle; low subsonic flow through the compressor ($T_3 \approx T_{3_t}$);

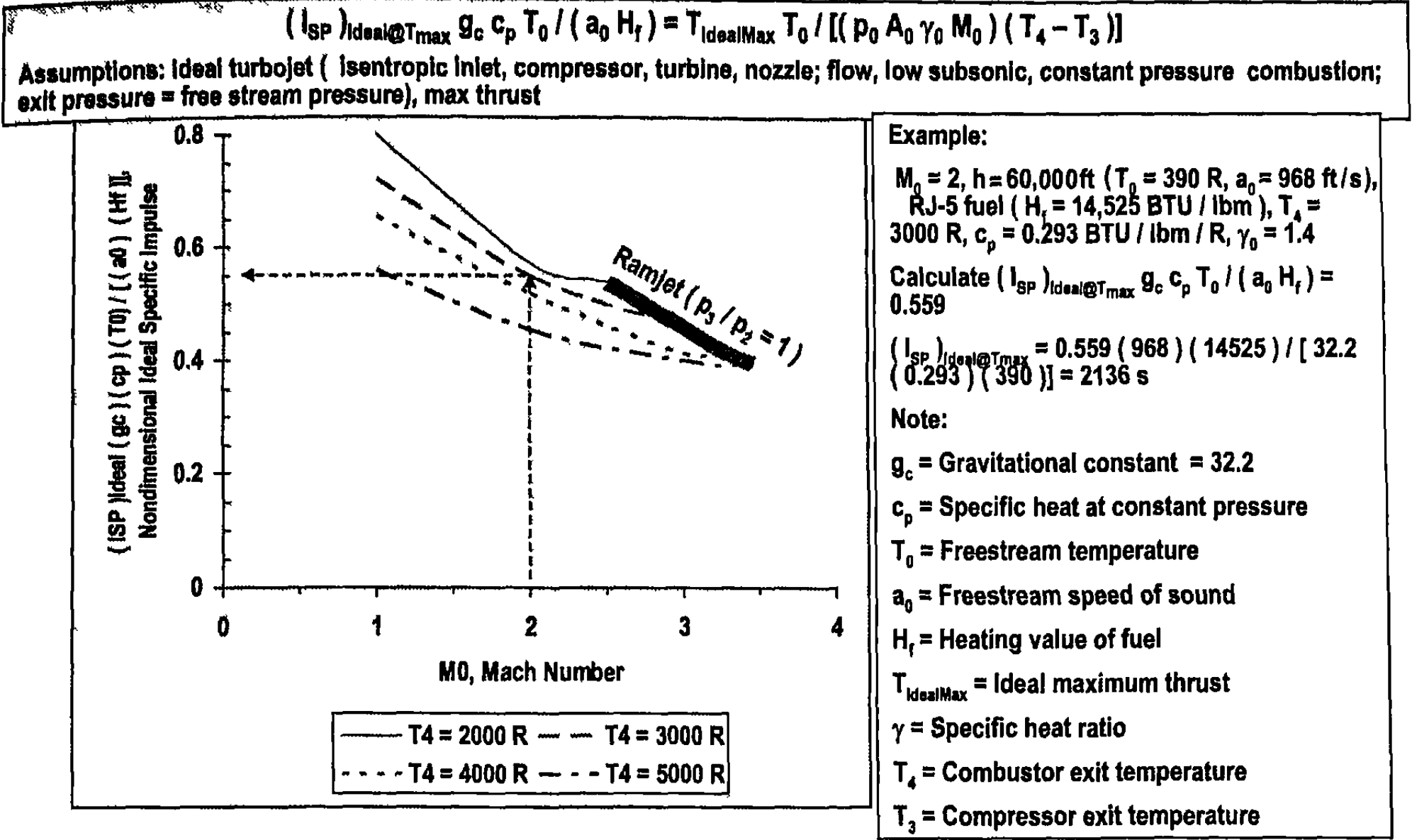


Fig. 3.11 Turbojet specific impulse decreases with supersonic Mach number.

combustor has low subsonic flow ($T_4 \approx T_{4_i}$) and constant pressure ($p_4 = p_3$); and nozzle exit pressure equal to the freestream pressure.

The example calculation of turbojet specific impulse will be based on the same conditions as the previous example calculation of turbojet thrust, which were $A_0 = 114$ in², $T_4 = 3000$ R, $M_0 = 2$, and $h = 60,000$ ft altitude ($T_0 = 390$ R, $p_0 = 1.047$ psi, $a = 968$ ft/s). Also, from the previous example, $T_3 = 1133$ R, $c_p = 0.293$ Btu/lbm/R, and $T_{IdealMax} = 894$ lb. We shall assume the fuel is RJ-5, which has a heating value $H_f = 14,525$ Btu/lbm/R. Substituting, the non-dimensional specific impulse is

$$(I_{SP})_{Ideal@T_{max}} g_c c_p T_0 / (a_0 H_f) = 894(390) / [1.047(114)(1.4)(2)(3000 - 1133)]$$
$$= 0.559$$

(3.29)

Substituting the values of the freestream speed of sound a_0 , heating value of RJ-5 fuel H_f , gravitational constant g_c , specific heat at constant pressure c_p , and freestream temperature T_0 gives the specific impulse

$$(I_{SP})_{Ideal@T_{max}} = 0.559(968)(14525) / [32.2(0.293)(390)] = 2136$$

s (3.30)

3.6 Ramjet Propulsion Alternatives

Figure 3.12 illustrates tactical missile ramjet propulsion alternatives of liquid-fuel ramjet, solid-fuel ramjet, and ducted rocket. The examples are based on an integral rocket ramjet, with the rocket booster occupying the same volume as the ramjet engine combustor. Shown in the upper portion of each example is the

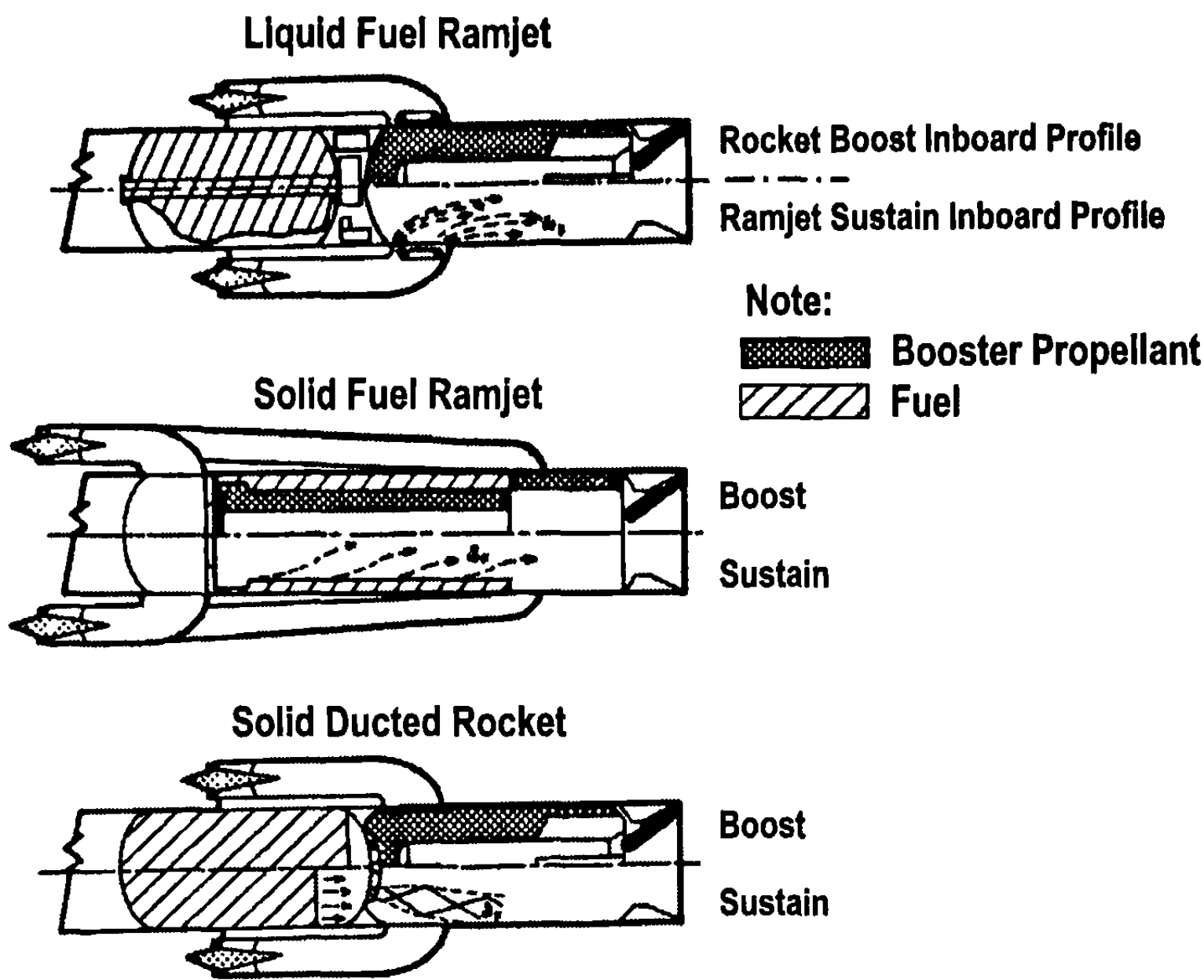


Fig. 3.12 Tactical missile ramjet propulsion alternatives.

rocket booster configuration during boost. The bottom portion of each example shows the ramjet engine configuration during sustain. Ramjet propulsion is relatively simple compared to turbojet propulsion; it has no moving parts. Air enters the inlet at supersonic velocity and is decelerated to low subsonic speed (e.g., Mach 0.2) at the combustor entrance. Downstream of the combustor is a convergent-divergent nozzle, which accelerates the airflow back to supersonic velocity. The maximum specific impulse of a hydrocarbon liquid-fuel ramjet is about 1500 s, much higher than the specific impulse of a solid rocket. An efficient cruise condition for a liquid-fuel ramjet is about Mach 4, 80,000 ft altitude. As mentioned previously, the maximum Mach number is about 5. Because a ramjet is inefficient at low supersonic Mach number, a rocket is required for subsonic launch to boost the missile to the ramjet thrust takeover at about Mach 2.5. Liquid-fuel ramjets are synergistic with noncircular, lifting body airframes because fuel can be stored in noncircular tanks. Liquid-fuel ramjets can be throttled for efficient matching of the fuel with the inlet airflow over a broad flight envelope of Mach number and altitude.

A variation on the traditional liquid-fuel ramjet is the solid-fuel ramjet. The solid-fuel ramjet has a fuel grain with a hollow center core for the inlet airflow. An advantage of the higher density solid fuel is its volumetric performance. Disadvantages are limited throttle capability, reduced volumetric loading of the fuel because of the hollow center grain, and lower combustion efficiency. Some passive fuel flow compensation is achieved from changing flight conditions. If a larger variation in fuel flow rate is required, shaping of the fuel grain is needed, at a loss in volumetric loading. Other means of thrust management, such as variable geometry inlets or nozzles, are not cost effective for tactical missiles.

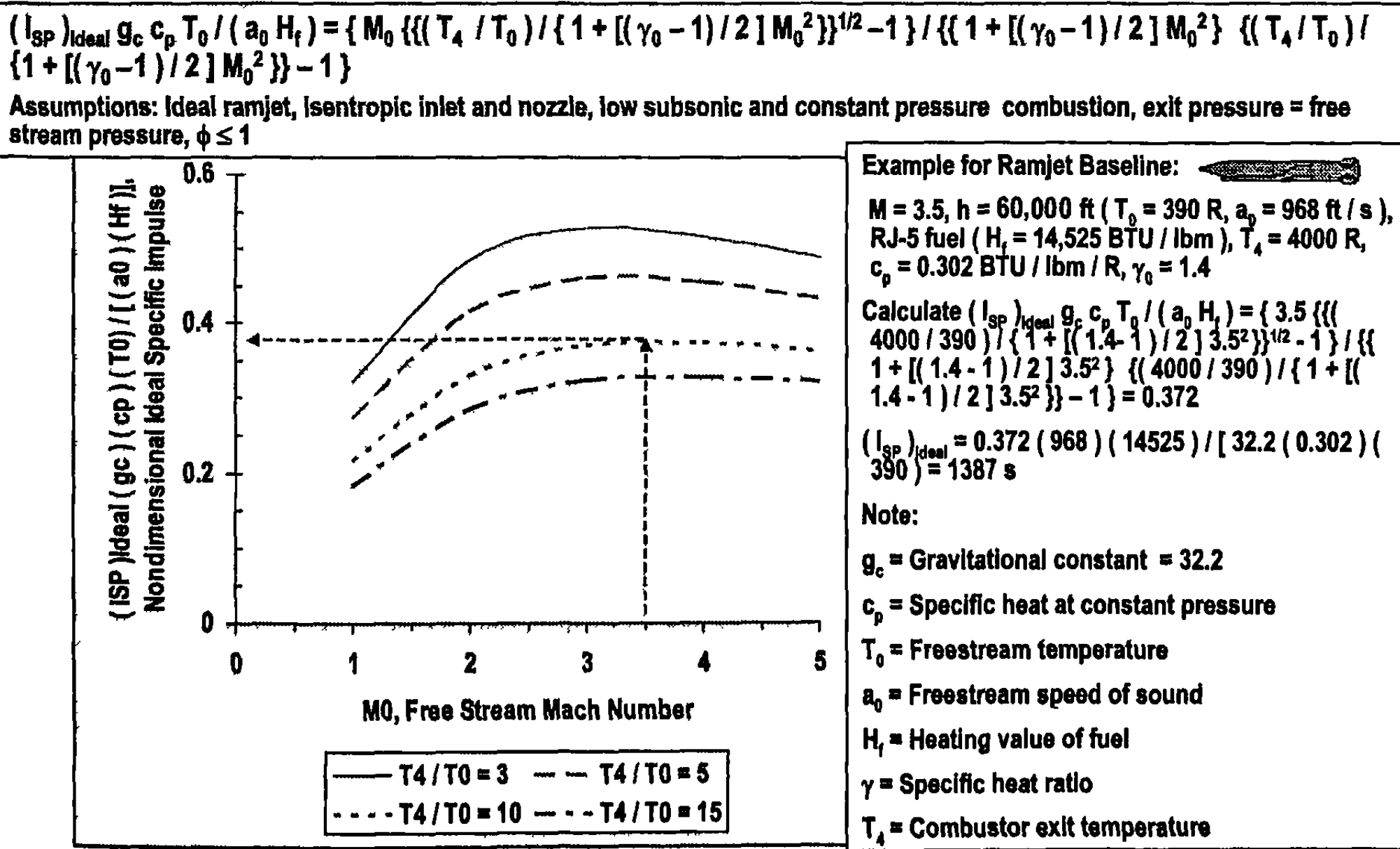
As mentioned previously, a ducted rocket has a specific impulse of about 500–800 s, intermediate to that of a solid rocket and a ramjet. Ducted rockets are most efficient for Mach 2.5 to 4. Ducted rockets have higher acceleration (higher thrust) than ramjets and longer range (higher specific impulse) than solid rockets. A gas generator provides fuel-rich products to the combustor. The gas generator flow rate can be controlled, providing a throttle capability for thrust magnitude control.

The integration of the ramjet booster impacts the missile compatibility for launch platform length and diameter constraints. Integral boosters that occupy the combustor case improve volumetric efficiency and reduce weight, enhancing integration with the launch platform. Another integration challenge for the booster is to minimize fratricide by having no ejecta. A nozzleless booster with a common booster case/ramjet combustor avoids nozzle and booster ejecta. A nozzleless booster has a cast propellant grain that maintains the throat and divergence geometry during motor burn.

3.7 Ramjet Specific Impulse Prediction

Shown in Fig. 3.13 is the nondimensional specific impulse of an ideal ramjet.¹¹ The equation is

$$(I_{SP})_{Ideal} g_c c_p T_0 / (a_0 H_f) = \{ M_0 \{ \{ (T_4 / T_0) / \{ 1 + [(\gamma_0 - 1) / 2] M_0^2 \} \}^{1/2} - 1 \} / \{ \{ 1 + [(\gamma_0 - 1) / 2] M_0^2 \} \{ (T_4 / T_0) / \{ 1 + [(\gamma_0 - 1) / 2] M_0^2 \} \} - 1 \} \} \quad (3.31)$$



Source: Ashley, H., *Engineering Analysis of Flight Vehicles*, Dover Publications, Inc., New York, 1974

Fig. 3.13 High specific impulse for a ramjet occurs when using high heating value fuel at Mach 3 to 4.

Note that ideal specific impulse is a function of freestream Mach number M_0 , heating value of fuel H_f , combustion temperature T_4 , freestream temperature T_0 , specific heat at constant pressure c_p , freestream speed of sound a_0 , and the freestream specific heat ratio γ_0 . Results for the ideal ramjet are based on assumptions of isentropic inlet and nozzle, low subsonic combustion with the combustor static temperature approximately equal to the combustor total temperature, constant pressure combustion, ideal expansion to freestream pressure at the nozzle exit, and an equivalence ratio less than 1. It is also assumed that the ramjet combustor is sufficiently long to provide complete combustion. This may be a problem for flight at a dynamic pressure less than 500 psf and an altitude above 60,000 ft. A design guideline for a typical length-limited combustor with subsonic combustion is that the pressure inside the combustor should be greater than 5 psi to ensure complete combustion.

The ideal specific impulse from the above equation may be corrected to account for losses in efficiency. A typical ramjet has an overall specific impulse efficiency of about 81%. The typical efficiencies of the combustor, nozzle, and inlet are about 0.86, 0.96, and 0.83 respectively.

An example will be calculated for the ramjet baseline at Mach 3.5 and an altitude $h = 60,000$ ft ($T_0 = 390$ R, $a_0 = 968$ ft/s). The assumed fuel is RJ-5 fuel (heating value $H_f = 14,525$ Btu per lbm of fuel). The assumed combustion temperature is $T_4 = 4000$ R. From the equation, the nondimensional specific impulse is calculated to be

$$\begin{aligned} (I_{SP})_{Ideal} g_c c_p T_0 / (a_0 H_f) &= \{3.5 \{[(4000/390) / \{1 + [(1.4 - 1)/2]3.5^2\}]^{1/2} - 1\} / \\ &\quad \{1 + [(1.4 - 1)/2]3.5^2\} \{[(4000/390) / \\ &\quad \{1 + [(1.4 - 1)/2]3.5^2\}] - 1\} = 0.372 \end{aligned} \quad (3.32)$$

The specific impulse, after substituting the values of freestream speed of sound $a_0 = 968$ ft/s, heating value of RJ-5 fuel $H_f = 14525$ Btu/lbm, gravitational constant $g_c = 32.2$, specific heat at constant pressure $c_p = 0.302$ Btu/lbm/R, and freestream temperature $T_0 = 390$ R, is

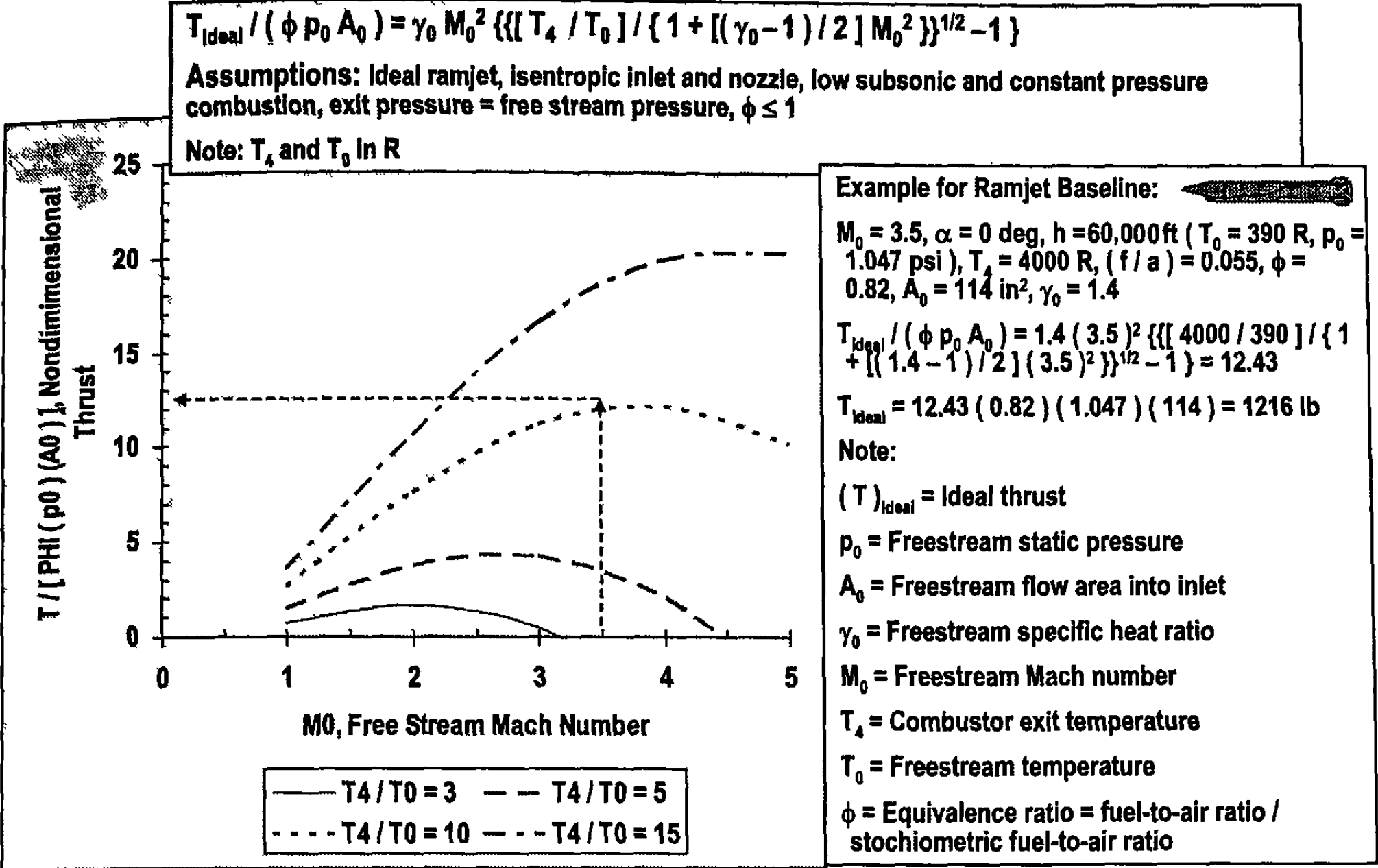
$$(I_{SP})_{Ideal} = 0.372(968)(14525)/[32.2(0.302)(390)] = 1387 \text{ s} \quad (3.33)$$

3.8 Ramjet Thrust Prediction

Shown Fig. 3.14 is the thrust of an ideal ramjet.¹¹ The nondimensional thrust equation is

$$T_{Ideal} / \phi p_0 A_0 = \gamma_0 M_0^2 \{ [T_4/T_0] / \{1 + [(\gamma_0 - 1)/2] M_0^2\} \}^{1/2} - 1 \} \quad (3.34)$$

Note that the ideal thrust is a function of the equivalence ratio ϕ , combustion temperature T_4 , freestream flow area into the inlet A_0 , freestream Mach number M_0 , freestream static pressure p_0 , freestream temperature T_0 , and ratio of specific heat γ_0 . For typical values of the parameters, maximum thrust



Source: Ashley, H., *Engineering Analysis of Flight Vehicles*, Dover Publications, Inc., New York, 1974

Fig. 3.14 High thrust for a ramjet occurs from Mach 3 to 5 with high combustion temperature.

occurs at a Mach number between Mach 3 to 5. Results are based on an ideal ramjet with isentropic inlet and nozzle, low subsonic and constant pressure combustion, fuel-to-air equivalence ratio ϕ less than stoichiometric combustion, and ideal expansion of the nozzle exit pressure to the freestream static pressure.

As an example, assume that the ramjet baseline is at a flight condition of Mach 3.5/60,000 ft altitude ($T_0 = 390 \text{ R}, p_0 = 1.047 \text{ psi}$). Also assume a combustion temperature $T_4 = 4000 \text{ R}$ and a fuel-to-air ratio $f/a = 0.055$ (which gives an equivalence ratio of $\phi = 0.82$ and provides a combustion temperature $T_4 = 4000 \text{ R}$). The freestream flow area is assumed to be equal to the inlet capture area, $A_0 = 114 \text{ in}^2$. Substituting in the nondimensional thrust equation

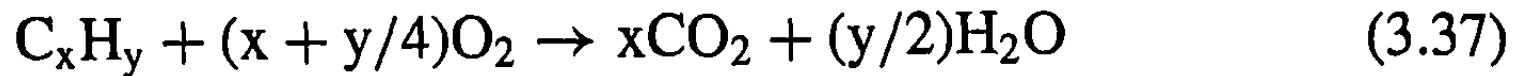
$$\begin{aligned} T_{ideal} / (\phi p_0 A_0) &= 1.4(3.5)^2 \{ [4000/390] / \{ 1 + [(1.4 - 1) / 2] (3.5)^2 \} \}^{1/2} - 1 \} \\ &= 12.43 \end{aligned} \tag{3.35}$$

Substituting equivalence ratio $\phi = 0.82$, freestream pressure $p_0 = 1.047$, and freestream flow area $A_0 = 114 \text{ in}^2$ in the thrust equation gives

$$T_{ideal} = 12.43(0.82)(1.047)(114) = 1216 \text{ lb} \tag{3.36}$$

Note that the actual thrust is lower because of combustor, nozzle, and inlet efficiency losses and because of the inlet capture efficiency.

The fuel-to-air ratio for stoichiometric combustion is a function of the type of fuel. For typical liquid hydrocarbon fuels such as RJ-5, JP-4, and kerosene the fuel-to-air ratio for stoichiometric combustion is $f/a \approx 0.067$. For lighter weight hydrocarbon fuels, stoichiometric f/a values are somewhat lower. The general equation for combustion of a hydrocarbon fuel of composition C_xH_y with oxygen (O_2) is



Because air is approximately 21% oxygen and 79% nitrogen, the equation for combustion of a hydrocarbon fuel in air is $C_xH_y + (x + y/4)(0.21O_2 + 0.79 N_2) \rightarrow xCO_2 + (y/2)H_2O + [(x + y)/4](0.79/0.21)N_2$. Substituting the molecular weight of hydrogen (1), carbon (12), oxygen (16), and nitrogen (14) gives a fuel-to-air ratio equation for stoichiometric combustion of a hydrocarbon fuel as

$$f/a = (12x + y)/\{(x + y/4)[32 + (0.79/0.21)28]\} \quad (3.38)$$

As an example, octane (C_8H_{18}) has a stoichiometric fuel-to-air ratio

$$f/a = (96 + 18)/\{(8 + 18/4)[32 + (0.79/0.21)28]\} = 0.066 \quad (3.39)$$

For methane (CH_4), which is a lightweight gas, the stoichiometric fuel-to-air ratio is

$$f/a = (12 + 4)/\{(1 + 4/4)[32 + (0.79/0.21)28]\} = 0.058 \quad (3.40)$$

3.9 Ramjet Temperature Technology Limit

Figure 3.15 shows ramjet combustor exit temperature T_4 as a function of free-stream Mach number M_0 and fuel-to-air ratio f/a . The equation used to develop the figure is

$$T_4 \approx T_0\{1 + [(\gamma_0 - 1)/2]M_0^2\} + (H_f/c_p)(f/a) \quad (3.41)$$

In the equation, T_0 is the freestream temperature, γ_0 is the freestream specific heat ratio, H_f is the fuel heating value, and c_p is the specific heat of air at constant pressure in the combustor. Assumptions are low subsonic combustor, no heat transfer through the inlet (isentropic flow), and fuel-to-air ratio less than stoichiometric. The figure is based on the heating value of RJ-5 fuel ($H_f = 14,525$ Btu/lbm/R).

Note that the ramjet combustor temperature exceeds 4000 R for flight conditions of high Mach number and fuel-to-air ratios approaching stoichiometric flow $[(f/a)_{\phi=1} = 0.067$ for RJ-5 hydrocarbon fuel]. Operation at less than stoichiometric ($\phi < 1$), with excess inlet air, may be required to avoid exceeding the temperature limit of the combustor and its insulation. The state-of-the art of ramjet combustor insulation technology is often a limiting factor in achieving the thrust and specific impulse that are required for efficient flight at hypersonic Mach number.

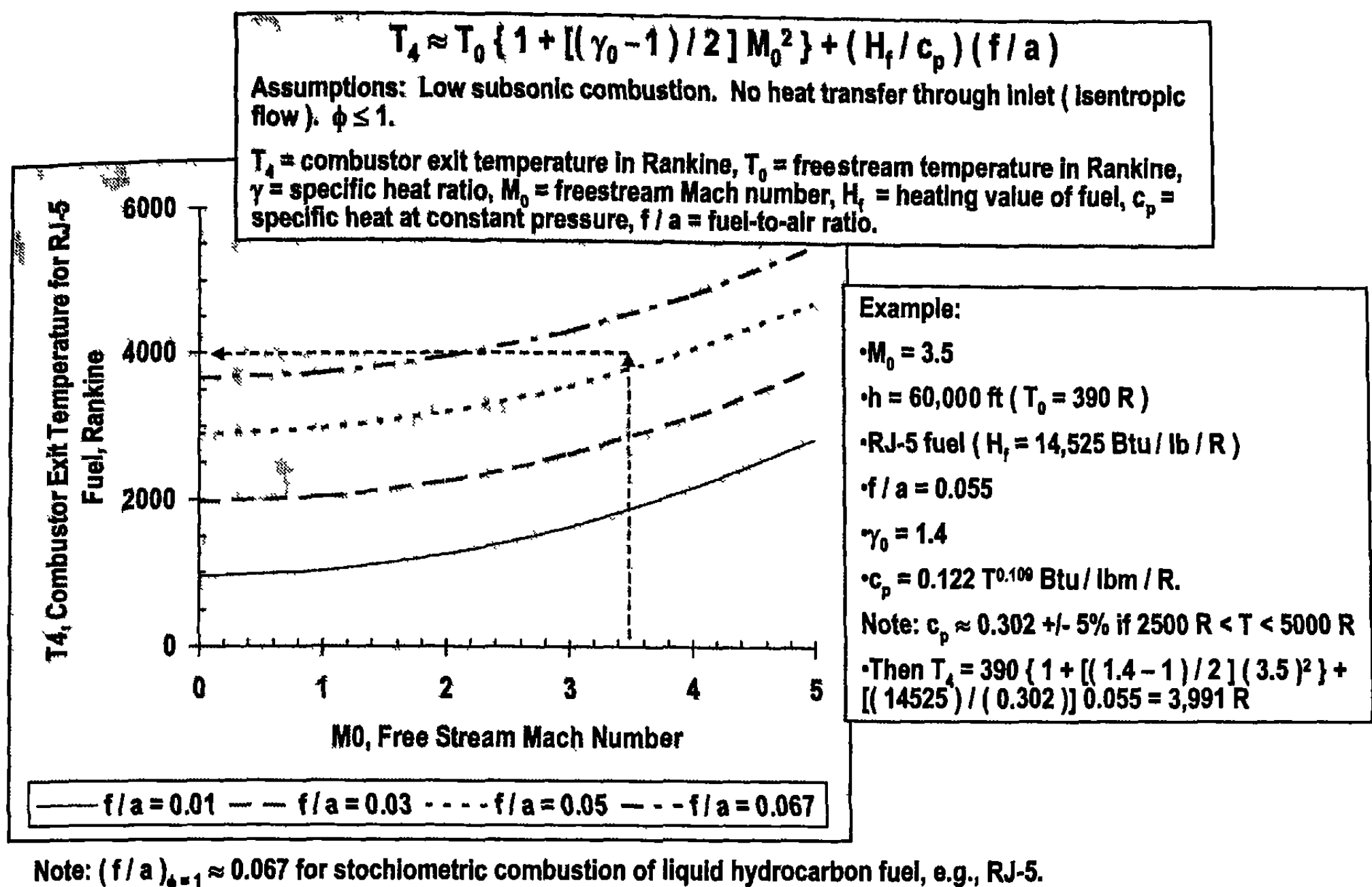


Fig. 3.15 Ramjet combustor temperature increases with Mach number and fuel flow.

As an example, assume a subsonic combustion ramjet flying at a flight Mach number $M_0 = 3.5$ and an altitude $h = 60,000 \text{ ft}$ ($T_0 = 390 \text{ R}$). The assumed fuel is RJ-5 fuel ($H_f = 14,525 \text{ Btu/lb}$) and the assumed fuel-to-air ratio $f/a = 0.055$. It is also assumed that $\gamma_0 = 1.4$. The specific heat at constant pressure is approximately $c_p \approx 0.302 \text{ Btu/lbm/R}$ for the flow through the combustor at typical temperatures. A previous equation showed that c_p is a weak function of temperature, $c_p = 0.122 T^{0.109} \text{ Btu/lbm/R}$. For the typical temperature range,

$$c_p = 0.302 \pm 5\% \quad \text{if } 2500 \text{ R} < T < 5000 \text{ R} \quad (3.42)$$

Substituting into the equation for combustion temperature gives

$$T_4 \approx T_0 \left\{ 1 + \left[\frac{\gamma_0 - 1}{2} \right] M_0^2 \right\} + (H_f / c_p) (f / a) = 390 \left\{ 1 + \left[\frac{1.4 - 1}{2} \right] \times (3.5)^2 \right\} + (14525 / 0.302) (0.055) = 3991 \text{ R} \quad (3.43)$$

The assumption of constant pressure combustion in a constant area combustor can have significant error if there is a high combustor entrance Mach number or if there is high heating. A limiting condition occurs when the local Mach number in the combustor is increased from the subsonic entrance Mach number M_3 to a thermally choked combustion Mach number $M_4 = 1$. Shown in Fig. 3.16 is the thermally choked combustor entrance Mach number $(M_3)_{TC}$ versus freestream Mach number M_0 and the ratio of the combustor total temperature to freestream static temperature (T_4/T_0). The figure is based on

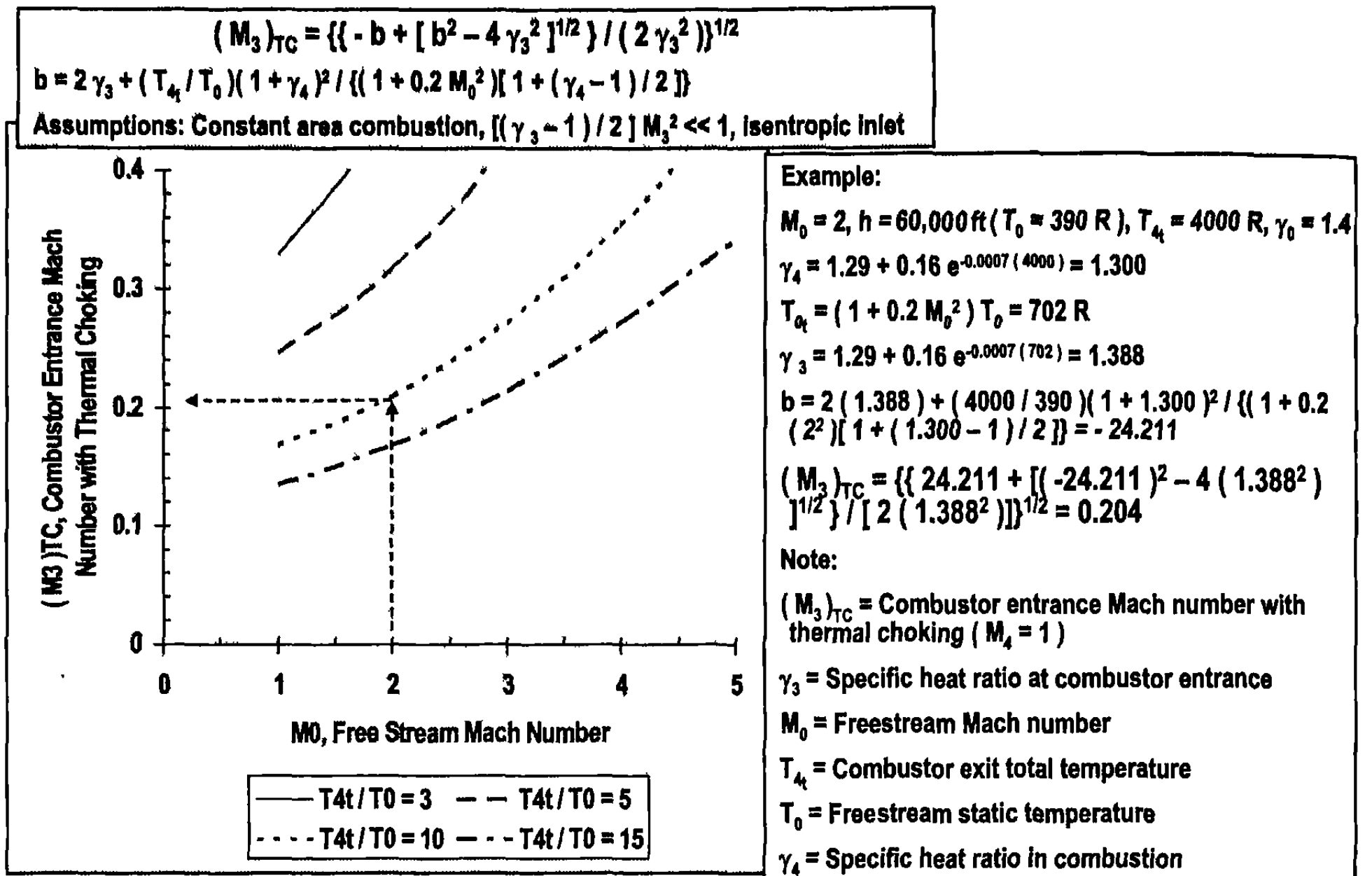


Fig. 3.16 Ramjet combustor entrance Mach number should be low to avoid thermal choking.

the equation

$$(M_3)_{TC} = \{ \{-b + [b^2 - 4\gamma_3^2]^{1/2}\} / (2\gamma_3^2) \}^{1/2} \quad (3.44)$$

where $b = 2\gamma_3 + (T_{4t}/T_0)(1 + \gamma_4)^2 / \{ (1 + 0.2M_0^2)[1 + (\gamma_4 - 1)/2] \}$.

Terms in the above equations are $(M_3)_{TC}$ = combustor entrance Mach number with thermal choking ($M_4 = 1$), γ_3 = specific heat ratio at the combustor entrance, M_0 = freestream Mach number, T_{4t} = combustor exit total temperature, T_0 = freestream static temperature, and γ_4 = specific heat ratio in combustion. Assumptions are constant area combustion with negligible skin friction (Rayleigh flow), $[(\gamma_3 - 1)/2]M_3^2 \ll 1$, and isentropic flow from the freestream to the combustor entrance (i.e., isentropic inlet). A source for the generalized Rayleigh flow equation is Ref. 12. Note from the figure that a low subsonic combustor entrance Mach number is required to avoid thermal choking, particularly for high values of combustor temperature ratio T_{4t}/T_0 and low supersonic free-stream Mach number (e.g., booster-to-ramjet transition).

As an example, assume $M_0 = 2$, $h = 60,000 \text{ ft}$ ($T_0 = 390 \text{ R}$), and $T_{4t} = 4000 \text{ R}$. For this case, the freestream specific heat ratio $\gamma_0 = 1.4$. Calculate

$$\gamma_4 = 1.29 + 0.16 e^{-0.0007(4000)} = 1.300 \quad (3.45)$$

$$T_{0t} = (1 + 0.2M_0^2)T_0 = 702 \text{ R} \quad (3.46)$$

$$\gamma_3 = 1.29 + 0.16 e^{-0.0007(702)} = 1.388 \quad (3.47)$$

Solving for the term b in the $(M_3)_{TC}$ equation,

$$b = 2(1.388) + (4000/390)(1 + 1.300)^2 / \{[1 + 0.2(2^2)][1 + (1.300 - 1)/2]\} \\ = -24.211 \quad (3.48)$$

Then for this case, the maximum combustor entrance Mach number to avoid thermal choking is

$$(M_3)_{TC} = \{[24.211 + [(-24.211)^2 - 4(1.388^2)]^{1/2}] / [2(1.388^2)]\}^{1/2} \\ = 0.204 \quad (3.49)$$

Reference 12 also gives the static pressure at thermal choking relative to the static pressure at the combustor entrance as

$$(p_4/p_3)_{M_4=1} = [1 + \gamma_3(M_3)^2] / (1 + \gamma_4) \quad (3.50)$$

Substituting into the above equation, the decrease in static pressure is

$$(p_4/p_3)_{M_4=1} = [1 + 1.388(0.204)^2] / (1 + 1.300) = 0.46 \quad (3.51)$$

For this case, thermal choking would result in a large decrease (54%) in combustor pressure. A design consideration to minimize the pressure drop and avoid thermal choking during constant area combustion is to have a relatively small inlet throat area (low value of A_{IT}/A_3). This results in a low subsonic Mach number at the combustor entrance ($M_3 \ll 1$).

The inlet entrance Mach number M_3 for a supersonic ramjet can be obtained comparing the cross-sectional area at the combustor entrance to the cross-sectional area at the inlet throat.

Downstream of the normal shock, the inlet subsonic Mach number is a function of the ratio of the local cross-sectional area to the inlet sonic throat area A_{IT} . From Ref. 12, the combustor entrance Mach number M_3 can be obtained from the equation

$$A_{IT}/A_3 = [(\gamma + 1)/2]^{(\gamma+1)/[2(\gamma-1)]} M_3 [1 + (\gamma - 1)/2] M_3^2]^{-(\gamma+1)/[2(\gamma-1)]} \quad (3.52)$$

For $\gamma = 1.4$,

$$A_{IT}/A_3 = (216/215)M_3(1 + M_3^2/5)^{-3} \quad (3.53)$$

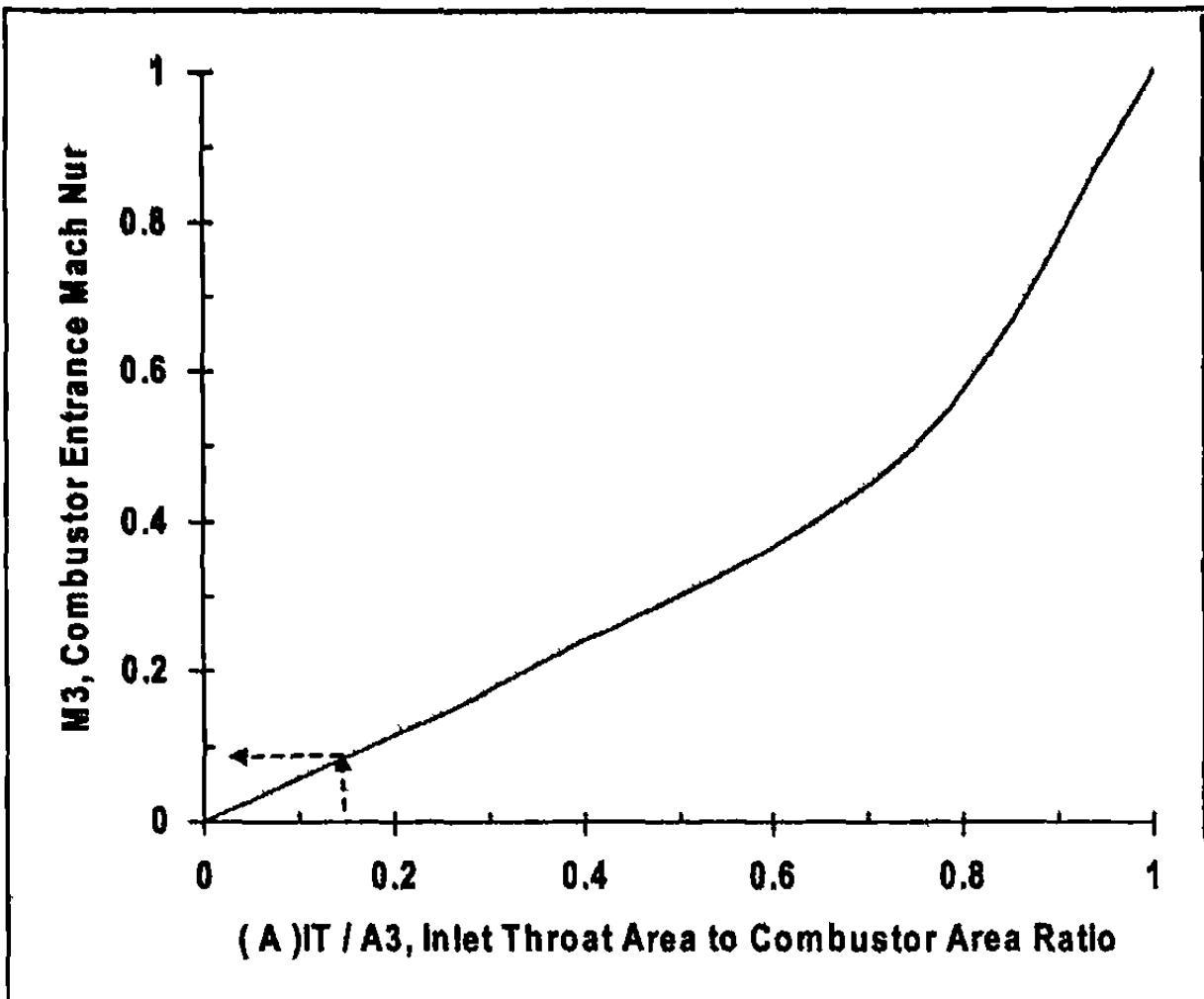
Assumptions are an isentropic inlet, $M_{IT} = 1$, and $\gamma = 1.4$. Figure 3.17 shows the combustor entrance Mach number M_3 versus the ratio of inlet throat area to combustor area (A_{IT}/A_3).

As an example, for the ramjet baseline $A_{IT} = 41.9 \text{ in}^2$ and $A_3 = 287 \text{ in}^2$. Assume sonic flow ($M = 1$) at the inlet throat. Calculate

$$A_{IT}/A_3 = 41.9/287 = 0.1459 \quad (3.54)$$

$$A_{IT} / A_3 = [(\gamma + 1) / 2]^{(\gamma + 1) / [2(\gamma - 1)]} M_3 [1 + (\gamma - 1) / 2] M_3^2]^{-(\gamma + 1) / [2(\gamma - 1)]} = (216 / 215) M_3 (1 + M_3^2 / 5)^{-3}$$

Assumptions: Isentropic Inlet, $M_{IT} = 1$, $\gamma = 1.4$



Note:

A_{IT} = Inlet throat area

A_3 = Combustor entrance area

M_3 = Combustor entrance Mach number

γ = Specific heat ratio

Example:

Ramjet Baseline

$A_{IT} = 41.9 \text{ in}^2$

$A_3 = 287 \text{ in}^2$

$A_{IT} / A_3 = 41.9 / 287 = 0.1459$

Assume sonic flow ($M = 1$) at A_{IT}

$M_3 = 0.085$

$M_3 = 0.085 < (M_3)_{TC} = 0.204$

Fig. 3.17 A ramjet combustor with a low entrance Mach number requires a small inlet throat area.

Substituting into the above equation gives

$$M_3 = 0.085 \quad (3.55)$$

Comparing to the prior example, note that $M_3 = 0.085 < (M_3)_{TC} = 0.204$. Since the ramjet baseline has sonic flow at its throat and $M_3 < (M_3)_{TC}$, the flow is not thermally choked.

From Ref. 12, the pressure loss in the combustor, assuming Rayleigh flow, is given by the equation

$$p_4 / p_3 = (1 + \gamma_3 M_3^2) / (1 + \gamma_4 M_4^2) \quad (3.56)$$

The Rayleigh flow temperature equation is

$$T_{4t} / T_0 = [(1 + \gamma_3 M_3^2) / (1 + \gamma_4 M_4^2)]^2 (M_4 / M_3)^2 \{1 + [(\gamma_4 - 1) / 2] M_4^2\} / \{1 + [(\gamma_3 - 1) / 2] M_3^2\} \quad (3.57)$$

Assumptions for Rayleigh flow are heat addition at a constant combustor cross-sectional area and negligible friction. An approximate equation for the Rayleigh flow combustion Mach number is

$$M_4 \approx \{[-b - (b^2 - 4ac)^{1/2}] / (2a)\}^{1/2} \quad (3.58)$$

The terms in the above equation are

$$a = (T_{4t}/T_0)\gamma_4^2 M_3^2 \quad (3.59)$$

$$b = 2(T_{4t}/T_0)\gamma_4 M_3^2 - \gamma_3 M_3^2 - 1 \quad (3.60)$$

$$c = (T_{4t}/T_0)M_3^2 \quad (3.61)$$

Assume the conditions from the prior example, $M_0 = 2$, $h = 60,000$ ft ($T_0 = 390$ R), $T_{4t} = 4000$ R, $\gamma_0 = 1.4$, $\gamma_4 = 1.300$, and $\gamma_3 = 1.388$. Also assume the ramjet baseline with a sonic inlet throat flow condition. Then for

$$A_{IT}/A_3 = 41.9/287 = 0.1459, M_3 = 0.085 \quad (3.62)$$

Solving for the terms in the combustion Mach number equation,

$$a = 10.256(1.300)^2(0.085)^2 = 0.1252 \quad (3.63)$$

$$b = 2(10.256)(1.300)(0.085)^2 - 1.388(0.085)^2 - 1 = -0.8174 \quad (3.64)$$

$$c = 10.256(0.085)^2 = 0.0741 \quad (3.65)$$

Substituting into the equation for combustion Mach number gives

$$M_4 \approx 0.304 \quad (3.66)$$

Solving for the pressure ratio loss in the combustor,

$$p_4/p_3 = [1 + 1.388(0.085)^2]/[1 + 1.300 M_4^2] = 0.902 \quad (3.67)$$

Note that there is a 10% reduction in combustion pressure for this example, indicating that the assumption of constant pressure combustion is reasonably accurate for conceptual design (10% error).

3.10 Ramjet Engine/Booster Integration

The integral rocket ramjet (IRR) engine is the most popular type of ramjet. After rocket booster burnout, the rocket booster motor case is used as the ramjet engine. Typically the booster case length is sufficiently long to provide efficient combustion as a ramjet engine. Shown in Fig. 3.18 is the minimum length of the combustor for complete combustion as a function of time required for combustion and the velocity of the airflow through the combustor. The equation is

$$(l_{\text{comb}})_{\text{min}} = t_{\text{comb}} V_{\text{comb}} \quad (3.68)$$

A subsonic combustion ramjet has a typical combustion time $t_{\text{comb}} \approx 0.002$ s and a combustion velocity $V_{\text{comb}} \approx 200$ ft/s. The minimum length of the combustor for efficient combustion is $(l_{\text{comb}})_{\text{min}} \approx 0.4$ ft. For a typical ramjet with

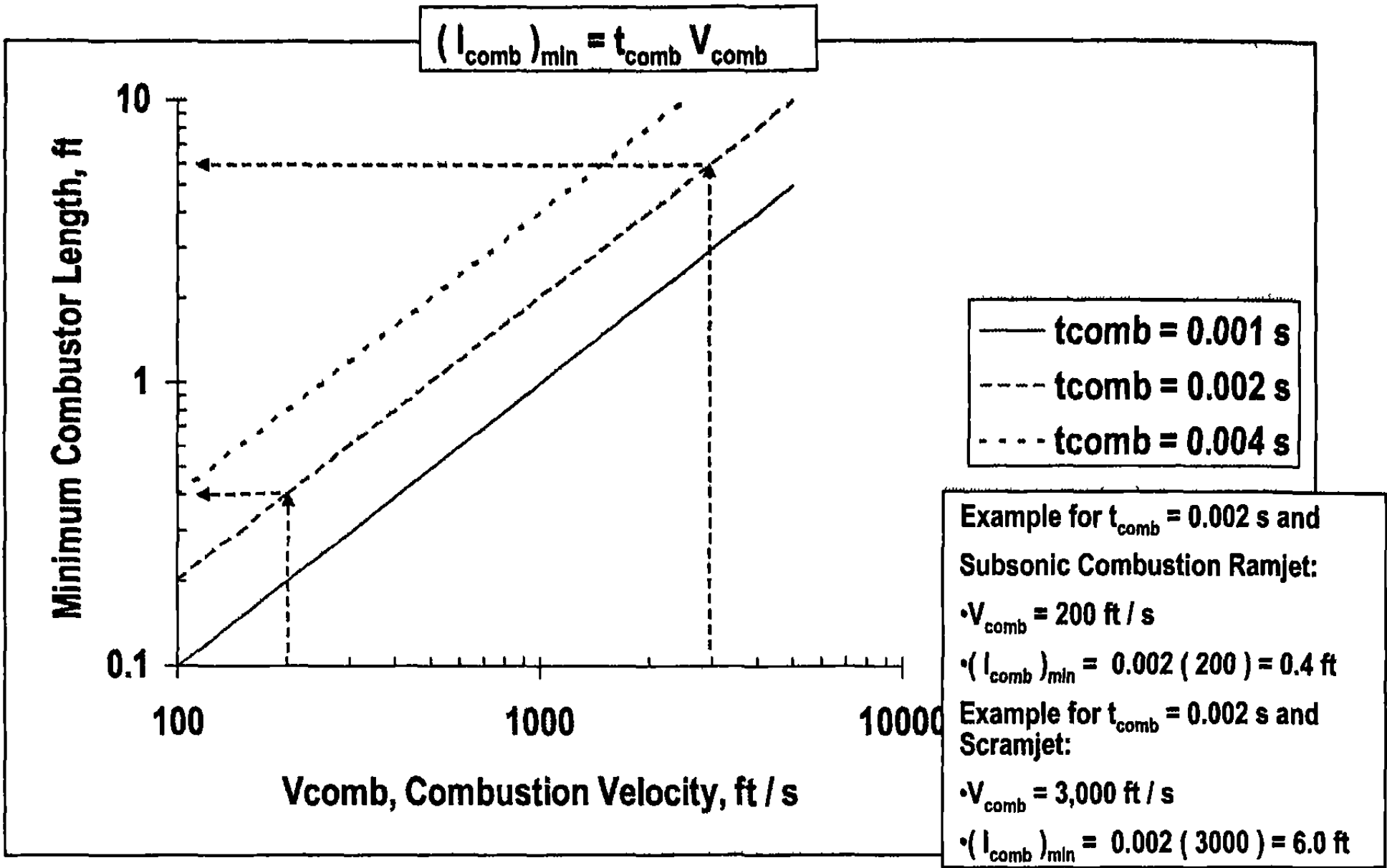


Fig. 3.18 Minimum length for the combustor is a function of combustion velocity.

a diameter greater than 1 ft and a total fineness ratio greater than 10, the impact of the minimum combustor length is relatively small, less than 4% of the length of the missile. For an IRR, the typical booster length is much longer than the minimum combustor length, easily accommodating the combustor.

A supersonic combustion ramjet also has a typical combustion time $t_{\text{comb}} \approx 0.002 \text{ s}$. However, a typical combustion velocity is $V_{\text{comb}} \approx 3000 \text{ ft/s}$. The minimum length of the combustor for efficient combustion is $(l_{\text{comb}})_{\text{min}} \approx 6 \text{ ft}$. In the case of the scramjet, the impact of the minimum combustor length is large, requiring a much longer and heavier missile.

Figure 3.19 from Ref. 13 characterizes ramjet engine/booster integration options according to their cruise drag. Low cruise drag ramjets are tandem IRR, aft drop-off booster, forward-located booster, and podded drop-off booster. The high cruise drag ramjets shown in the figure are the podded ramjet, podded IRR, and the podded ramjet with a drop-off booster. Most modern ramjets have low cruise drag. The IRR is particularly attractive. As noted previously, after booster burnout, an IRR uses the booster case as the combustion chamber for the ramjet. The IRR is often preferred because of advantages of low cruise drag, low volume, light weight, and small diameter.

A further comparison of ramjet engine/booster integration options is given in Fig. 3.20 based on Ref. 13. Note that the integral rocket-ramjet engine has above-average-to-superior characteristics for the selection factors of length, diameter, weight, no ejecta, cruise drag, launch platform carriage drag, cost, compatibility of the ramjet engine with the booster, and compatibility of the ramjet engine with the inlet. The other candidate engines are below average in one or more of the selection factors. However, depending upon the weighting of the selection factors and the mission, one or more of the other candidate engines may be

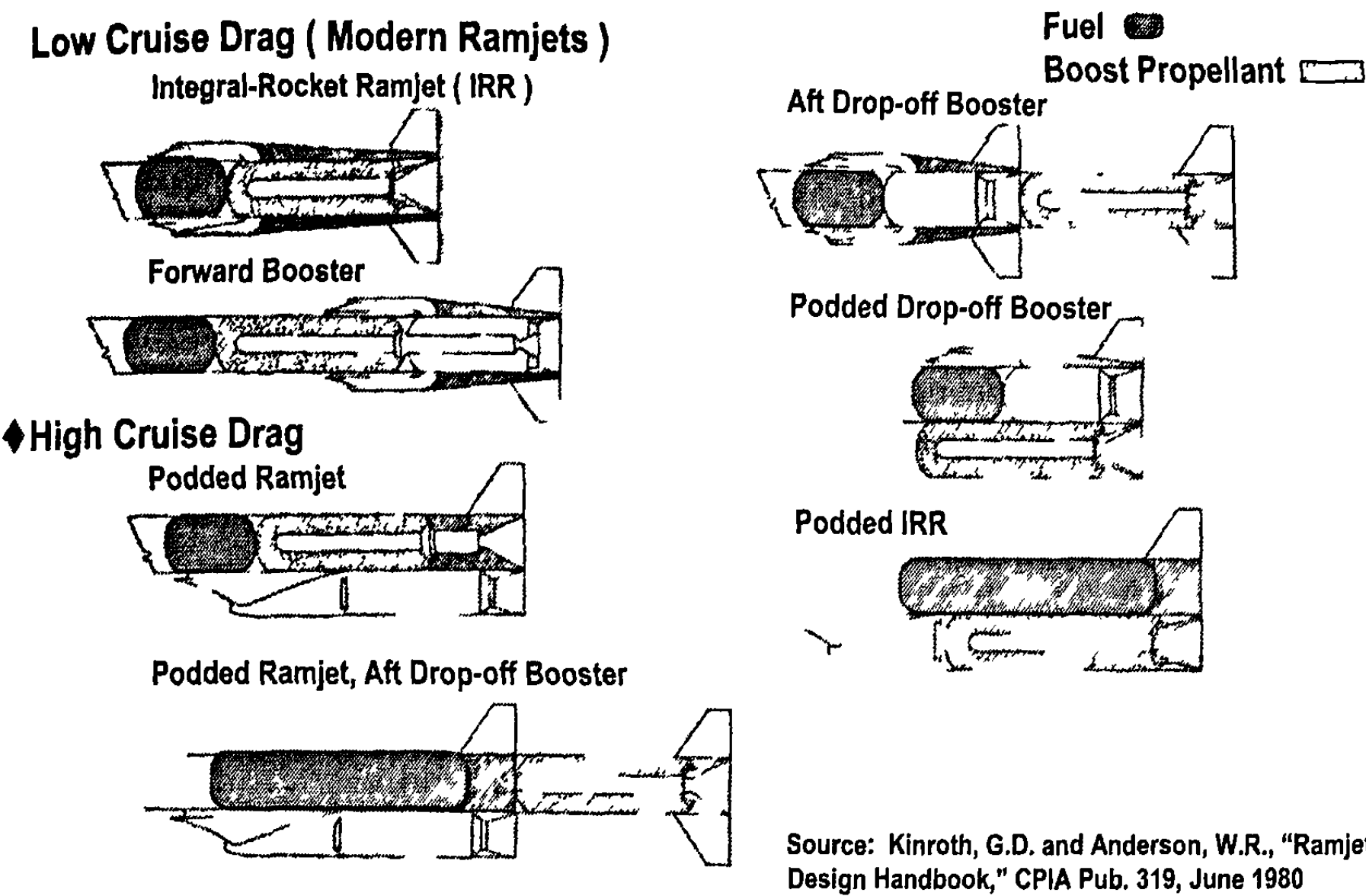


Fig. 3.19 Ramjet engine/booster integration options.

acceptable, or even preferable. The strengths and potential weaknesses of the other candidate ramjet engine alternatives to an IRR include the following:

- 1) Aft drop-off booster attractive features include small diameter, low cruise drag, low cost, and compatibility of the engine with the inlet. Disadvantages of an aft drop-off booster include long length and ejecta.

		Selection Factors								
		Length	Diameter	Weight	Ejectables	Cruise Drag	Carry Drag	Cost	Cycle Compatibility	Inlet Compatibility
Integral Rocket – Ramjet (IRR)		●	●	●	●	●	●	●	●	●
Aft Booster (Drop-off)		—	●	○	—	●	●	●	●	●
Forward Booster		—	●	○	●	●	●	○	○	○
Podded Booster (Drop-off)		●	—	○	—	●	○	●	●	●
Podded Ramjet		●	—	○	●	—	○	●	○	—
Podded IRR		●	—	●	●	—	○	○	○	—
Podded Ramjet Aft Booster (Drop-off)		○	—	—	—	—	—	●	○	—

● Superior ● Above Average ○ Average — Below average

Source: Kinroth, G.D. and Anderson, W.R., "Ramjet Design Handbook," CPIA Pub. 319, June 1980

Fig. 3.20 Ramjet engine/booster integration trades.

2) Forward booster has attractive features of small diameter, no ejecta, and low launch platform carriage drag. A disadvantage is long length.

3) Podded drop-off booster has attractive features of short length, low cruise drag, and low cost. Disadvantages include a large diameter and ejecta.

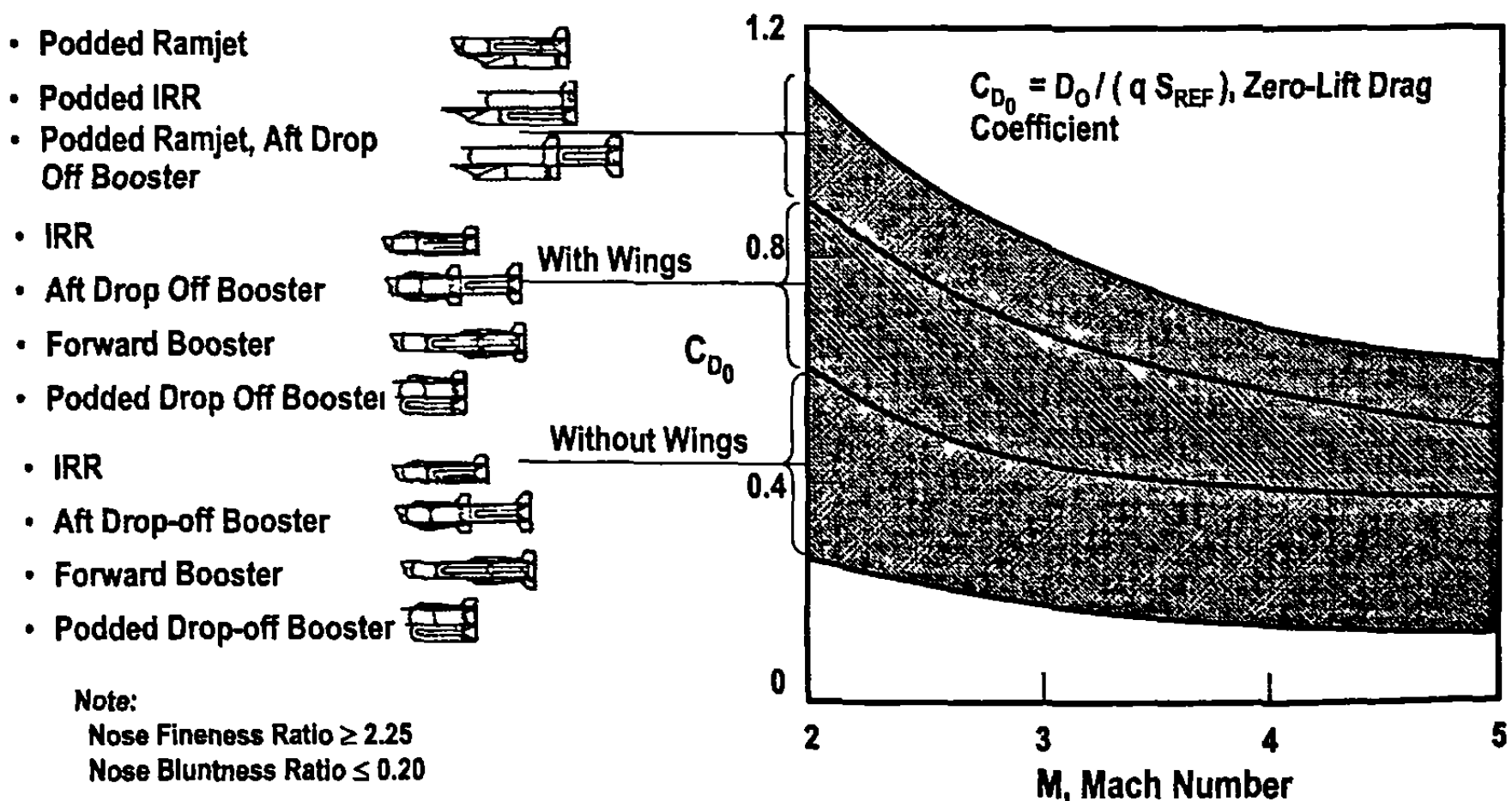
4) Podded ramjet has an attractive feature of no ejecta. Disadvantages of a podded ramjet include a large diameter, high cruise drag, and a potential problem of inlet compatibility. The disadvantages make it unsuitable for most modern applications.

5) Podded IRR has attractive features of short length and lighter weight. Disadvantages include large diameter, high cruise drag and a potential problem of inlet compatibility. The disadvantages make it unsuitable for most modern applications.

6) Podded ramjet with a drop-off booster has disadvantages of large diameter, heavy weight, ejecta, high cruise drag, high carriage drag, and a potential problem of inlet compatibility. The disadvantages make it unsuitable for most modern applications.

3.11 Ramjet Missile Drag

A first-order estimate of ramjet zero-lift drag, based on Ref. 13, is given in Fig. 3.21. Results are applicable to ramjets that have a high fineness nose ($l_N/d \geq 2.25$) with small bluntness ($d_{\text{NoseTip}}/d \leq 0.2$). The examples of ramjets that have low cruise drag are wingless ramjets based on one of the following: an IRR, an aft drop-off booster, a forward-located booster, or a podded drop-off booster. As an example, the wingless low-drag ramjet missiles have a zero-lift drag coefficient at Mach 2 of $C_{D_0} \approx 0.3$ to 0.6. Adding wings to the low drag ramjets increases the drag. At Mach 2, the value of the zero-lift drag coefficient is $C_{D_0} \approx 0.6$ to 0.9. Examples of the older high drag configurations are the



Source: Kinroth, G.D. and Anderson, W.R., "Ramjet Design Handbook," CPIA Pub. 319, June 1980

Fig. 3.21 Ramjets with internal boosters and no wings have low drag.

podded ramjets. The figure shows that at Mach 2, the zero-lift drag coefficient for podded ramjets is the highest, $C_{D_0} \approx 0.9$ to 1.1. As stated previously, the high drag podded ramjets are unsuitable for most modern applications.

3.12 Ramjet Inlet Options

Ramjet inlet design considerations include the type of opening, the location of the inlet on the body, and the number of inlets. Figure 3.22 from Ref. 13 shows the inlet placement and geometry of the following nine ramjet inlet options:

1) Annular nose inlet. The inlet surrounds the body. It may rely only on the cowl lip oblique shock for the initial compression, or it may also have a center spike for compression.

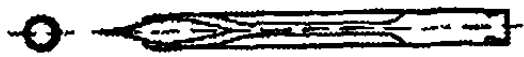
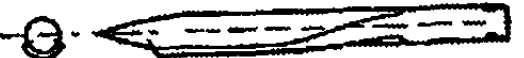
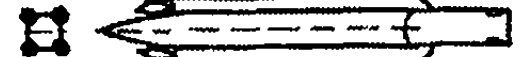
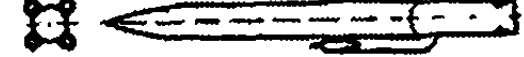
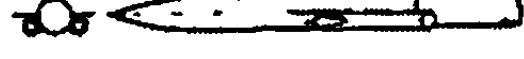
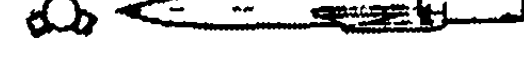
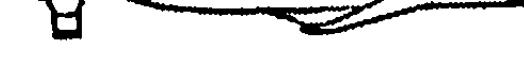
2) Chin inlet located forward underneath the nose. The nose compression angle provides an oblique shock in front of the inlet. The ramjet missile baseline of this textbook has a chin inlet.

3) Forward cruciform axisymmetric inlet. The nose compression angle provides an oblique shock in front of each inlet. The inlets are usually in tandem with cruciform fins.

4) Aft cruciform axisymmetric inlet. There are four inlets located near the center of the missile. The inlets are usually in tandem with cruciform fins.

5) Underwing axisymmetric inlet. There are two inlets located near the center of the missile, under a planar wing. The wing provides an initial compression.

6) Twin two-dimensional inlet. This concept has two inlets near the center of the missile. They are usually located on the lower portion of the body, separated by a radial angle of about 90 deg. The inlets are usually in tandem with the bottom cruciform tails.

Type Inlet	Sketch	Placement
Nose		Nose-full axisymmetric
Chin		Forward underside in nose compression field-partial axisymmetric
Forward Cruciform Axisymmetric		Forward in nose compression field-cruciform (four) axisymmetric
Aft Cruciform Axisymmetric		Aft-cruciform (four) axisymmetric
Under Wing Axisymmetric		In planar wing compression field-twin axisymmetric
Twin Two-dimensional		Aft-twin cheek-mounted two dimensional
Underslung Axisymmetric		Aft underside-full axisymmetric
Underslung Two-dimensional		Aft underside-belly mounted two dimensional
Cruciform Two-dimensional		Aft-cruciform (four) two dimensional

Source: Kinroth, G.D. and Anderson, W.R., "Ramjet Design Handbook," CPIA Pub. 319, June 1980.

Fig. 3.22 Ramjet inlet options.

7) Underslung axisymmetric inlet. The underslung inlet is located near the center of the missile.

8) Underslung two-dimensional inlet. Similar to the above.

9) Cruciform two-dimensional inlet. Similar to the aft cruciform axisymmetric inlet.

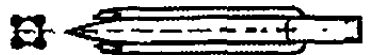
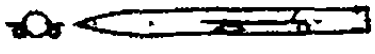
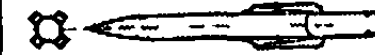
Tradeoff considerations of inlet alternatives are shown in Fig. 3.23 from Ref. 13. The selection factors are:

1) High pressure recovery. The chin inlet and the forward cruciform axisymmetric inlet benefit from relatively high recovery pressure provided by an oblique shock from the forebody in front of the inlet. Another benefit of the oblique shock is a lower Mach number at the inlet entrance, which allows the inlet to swallow the shock at lower flight Mach number. Other inlets that also have relatively good pressure recovery are the nose inlet, underwing axisymmetric inlet, and the twin two-dimensional inlet. The underwing axisymmetric inlet benefits from the wing compression, whereas the twin two-dimensional inlet benefits from the underbody compression.

2) Small carriage envelope on the launch platform. The annular inlet and the chin inlet require only a small inlet height to capture airflow because of the large wraparound angle.

3) High angle of attack capability. The chin inlet and the underwing axisymmetric inlet allow high positive angles of attack, providing a relatively high recovery pressure for bank-to-turn maneuvering.

4) Lightweight. The underwing axisymmetric inlet and the underslung axisymmetric inlet, located aft on the missile body, require less material for structure.

Type Inlet	Selection Factors									
	Pressure Recovery	Carriage Envelope	Alpha Capability	Weight	Drag	Warhead Shrouding	Inlet Cost	Preferred Steering	Preferred Control	Prime Mission Suitability
	●	●	○	●	●	—	●	STT	W, C	ATS, STA
	●	●	●	○	●	○	○	BTT	T	ATS, ATA, STA
	●	○	●	—	○	—	—	STT	T	ATS, ATA, STA
	○	●	○	●	○	●	○	STT	T	ATS
	●	—	●	●	○	●	●	BTT	T	ATS, ATA, STA
	●	○	●	●	●	●	○	BTT	T	ATS, ATA, STA
	○	—	○	●	○	●	●	BTT	T	ATS
	—	—	●	●	●	●	●	BTT	T	ATS, ATA, STA
	—	○	○	●	●	●	○	STT	T	ATS

Note:
BTT = Bank to Turn
STT = Skid to Turn
W = Wing C = Canard
T = Tail
● Superior ● Above Average ○ Average — Below average

Source: Kinroth, G.D. and Anderson, W.R., "Ramjet Design Handbook," CPIA Pub. 319, June 1980

Fig. 3.23 Ramjet inlet concept trades.

5) Low drag. The annular nose inlet and the chin inlet have reduced drag on the nose.

6) Does not shroud the warhead. The aft cruciform axisymmetric inlet, aft twin axisymmetric inlet, and the aft cruciform two-dimensional inlet, as shown in the figure, are located aft of the warhead section of the missile.

7) Low cost. The lowest cost, simplest fabrication inlet is probably an underslung axisymmetric inlet.

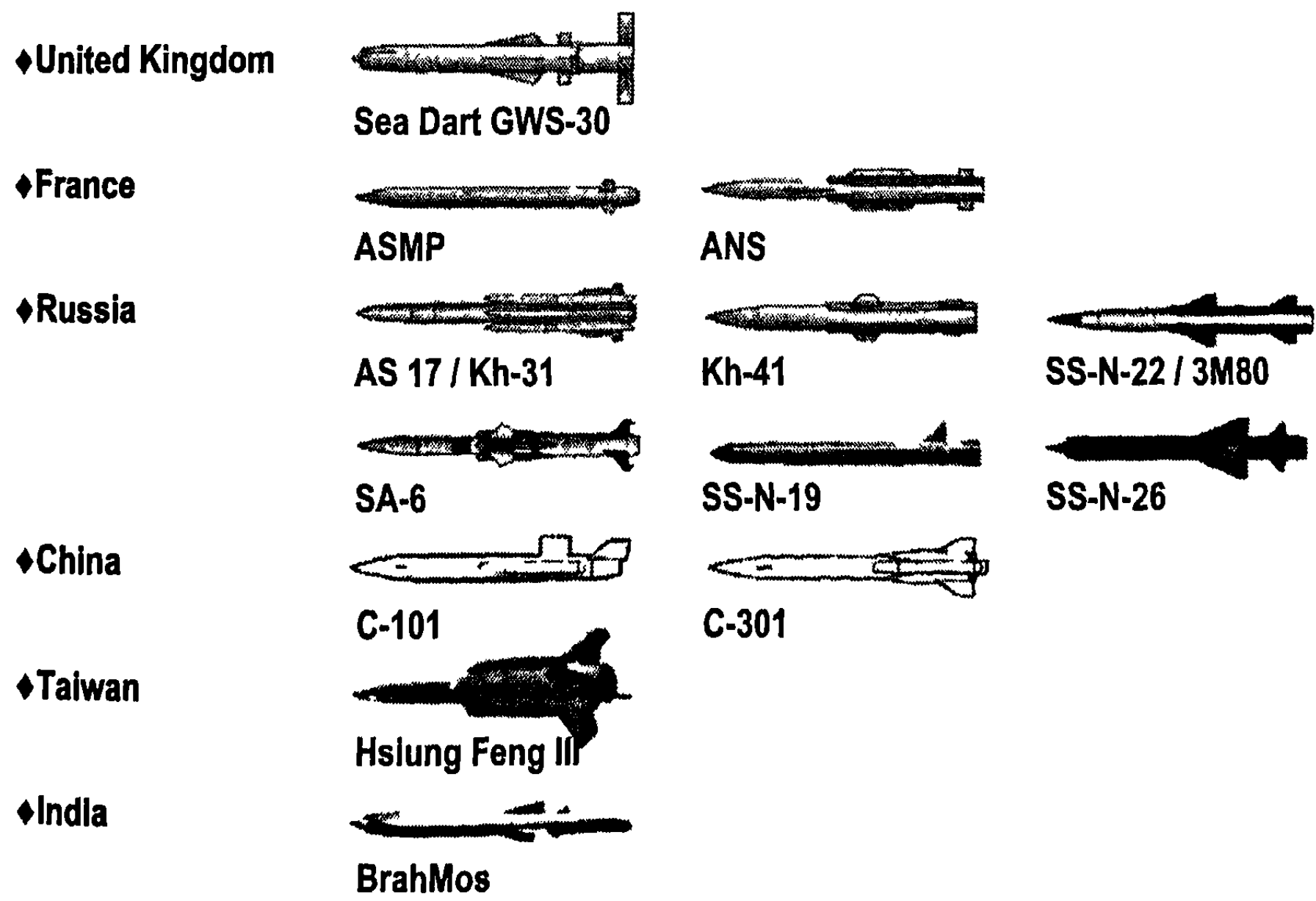
8) Preferred steering approach (skid to turn, bank to turn). Cruciform inlets are consistent with skid-to-turn maneuvering, whereas chin, twin, or underslung inlets are consistent with bank-to-turn maneuvering.

9) Preferred flight control (tail, canard, wing). An annular inlet has volume available for the forward packaging of the flight control actuators for canard or wing control. The other inlets are more compatible with tail control.

10) Preferred mission application (air to surface, air to air, or surface to air). The air intercept mission requires high maneuverability and high angle of attack, which is especially consistent with the capability of the chin inlet and underwing axisymmetric inlets. These inlets require bank-to-turn maneuvering.

There is no single inlet that is best for all missions. The selection factors must be weighted for the mission as part of the inlet selection.

The current supersonic/hypersonic airbreathing missiles are shown in Fig. 3.24. Except for the SS-N-19 Mach 2.5 turbojet, the missiles use ramjet propulsion. These missiles have either a nose inlet (U.K. Sea Dart, Russian



• Aft inlets have lower inlet volume and do not degrade lethality of forward located warhead.
• Nose inlet may have higher flow capture, pressure recovery, smaller carriage envelope, and lower drag.

Fig. 3.24 Current supersonic airbreathing missiles have either a nose inlet or have axisymmetric aft inlets.

SS-N-19 and SS-N-26, and Indian BrahMos) or aft axisymmetric inlets (France ANS and ASMP, Russian AS-17/Kh-31, Kh-41, SS-N-22/3M80, and SA-6, China C-101 and C-301, and Taiwan Hsiung Feng III). A nose inlet has advantages of low drag and small carriage envelope and may have higher flow capture and pressure recovery at low supersonic Mach number. Aft axisymmetric inlets have advantages of light weight and low volume, and they do not shroud/degrade the warhead. Note that the flight control for all of these missiles is based on tail control surfaces. The tail flight control actuators are packaged around the throat of the nozzle.

3.13 Supersonic Inlet/Airframe Integration

Inlet design and integration for a supersonic or hypersonic tactical missile involve a host of considerations, including matching the flow delivered to the ramjet nozzle, swallowing capacity, and inlet start. There may not be sufficient time to consider all of these in the initial conceptual design. The inlet design considerations that are addressed in this text are those that are considered to have a large effect on the aerodynamic configuration.

Figure 3.25 shows the shock wave location for an inlet without external compression compared to the shock wave location for an external compression inlet. The external compression inlet has a body deflection angle in front of the inlet. The inlet without external compression, shown at the top of the figure, has an advantage in that it swallows 100% of the oncoming air, with no spillage. A disadvantage is that it may have lower pressure recovery, particularly at high

◆ Inlet w/o External Compression

◆ Inlet Swallows 100% of the Free Stream Flow

◆ External Compression Required for Efficient Pressure Recovery if Mach Number > 2 and Inlet Start at Low Supersonic Mach number

◆ External Compression Inlet (with Spillage)

◆ Shocks Converge Outside Inlet Lip (Results in Spillage Air)

◆ External Compression Inlet (w/o Spillage)

◆ Inlet Swallows 100% of the Free Stream Flow

◆ Shocks Converge at Inlet Lip (Inlet Captures Maximum Free Stream Flow)

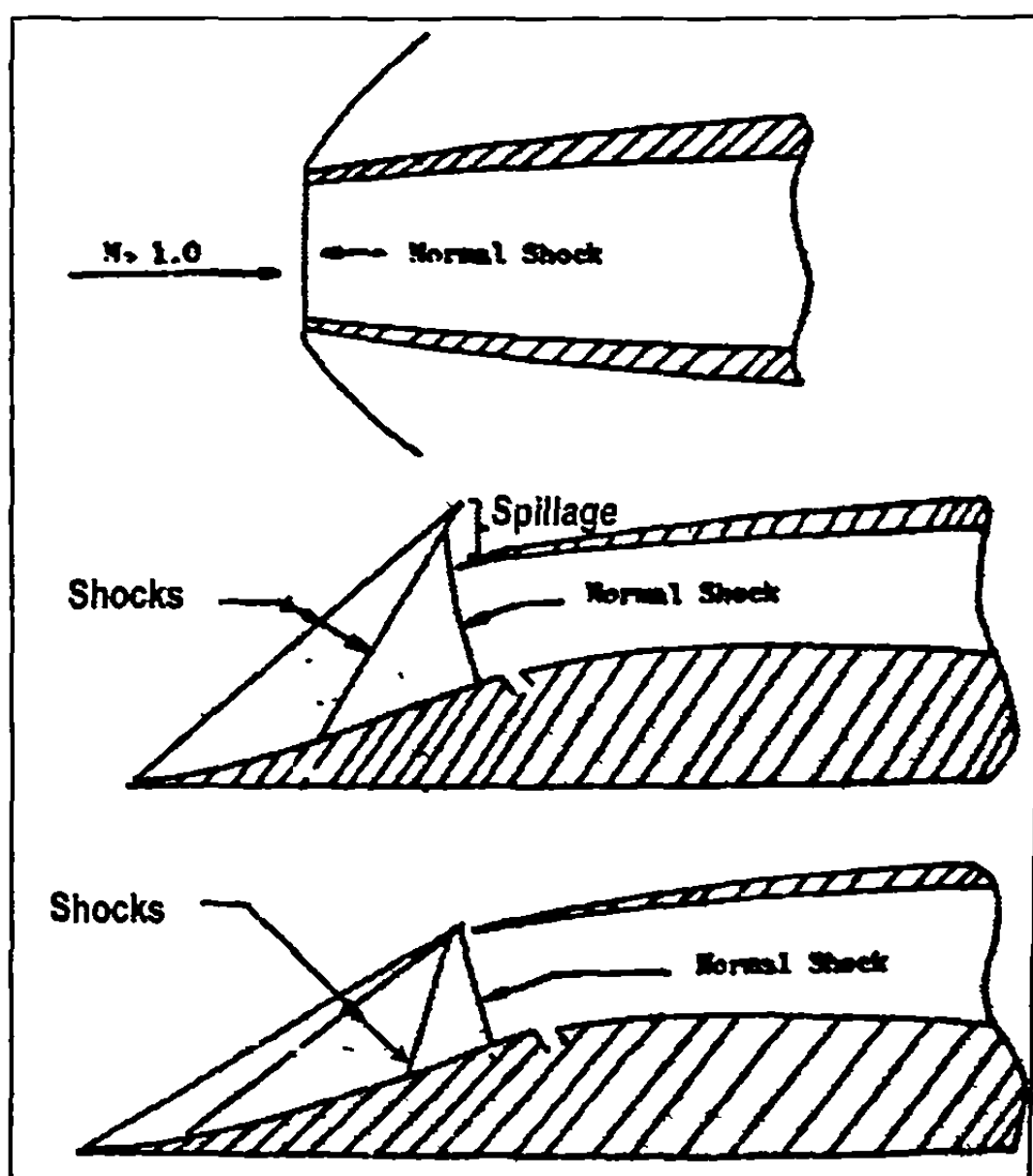


Fig. 3.25 Shock on inlet cowl lip prevents spillage.

Mach number, because of a strong normal shock inside the inlet. External compression (e.g., external compression spike) is usually required for efficient pressure recovery at a Mach number greater than 2. External compression is also usually required to swallow the shock and have an inlet start at low supersonic Mach number. External compression reduces the problem of a strong normal shock but adds an additional concern of matching the airflow of the inlet with the requirement of the engine. An external compression inlet is illustrated at the middle and bottom of the figure, operating in two conditions. The first condition, shown in the middle of the figure, is a lower Mach number where the external shocks have high shock wave angles, passing over the inlet. The inlet captures less of the freestream air and has spillage, which results in lower thrust and tends to increase the drag. The second condition for the external compression inlet is shown at the bottom of the figure. It is a higher Mach number condition with the shocks converging on the inlet lip. In this condition the external compression inlet swallows 100% of the oncoming air, providing greater thrust.

The forebody oblique shock wave angle provides a guideline for the inlet height that is required to achieve a shock-on-lip condition. Shown in Fig. 3.26 is the relationship of the oblique shock wave angle θ to the flow deflection angle $(\alpha + \delta)$ and Mach number M . The equation is

$$\tan(\alpha + \delta) = 2 \cot \theta_{2D} (M^2 \sin^2 \theta_{2D} - 1) / [2 + M^2 (\gamma + 1 - 2 \sin^2 \theta_{2D})] \quad (3.69)$$

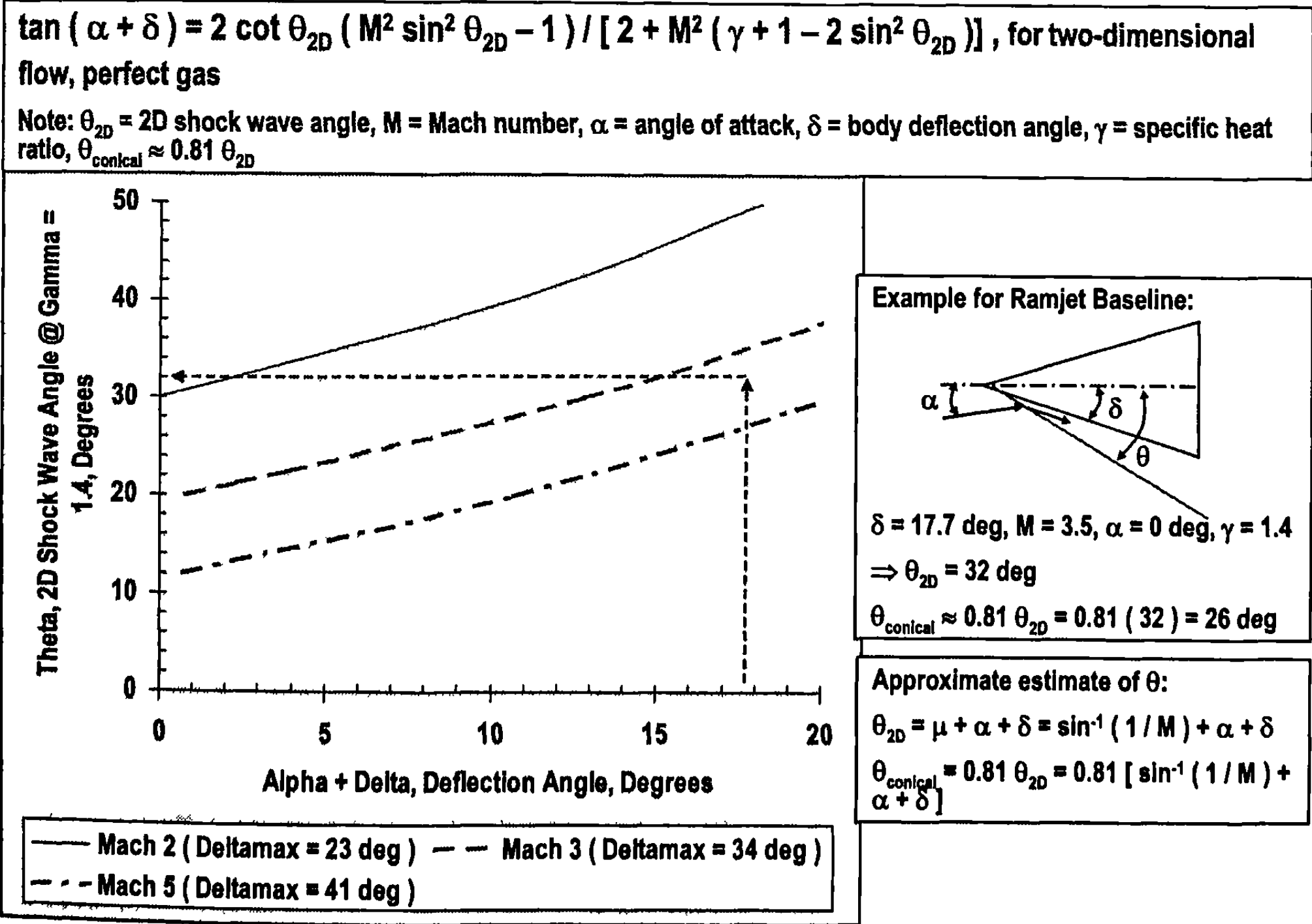


Fig. 3.26 Shock wave angle increases with deflection angle and decreases with Mach number.

The prediction is based on assumptions of two-dimensional flow of a perfect gas. Note that the shock wave angle increases with body deflection angle and decreases with Mach number. As a result, the inlet height to avoid air spillage must be larger for a missile with a low fineness nose at a low supersonic Mach number. Also, the maximum allowable flow deflection increases with Mach number. For example, at Mach 2 the maximum allowable flow deflection angle for an attached two-dimensional oblique shock with $\gamma = 1.4$ is $(\alpha + \delta)_{\max} = 23$ deg. At Mach 5, the maximum allowable flow deflection angle is greater, $(\alpha + \delta)_{\max} = 41$ deg.

A first-order estimate of the two-dimensional shock wave angle, suitable for conceptual design, is to assume that the shock wave angle θ_{2D} is the sum of the Mach angle μ plus the flow deflection angle $\alpha + \delta$. The approximate equation is

$$\theta_{2D} \approx \mu + \alpha + \delta = \sin^{-1}(1/M) + \alpha + \delta \quad (3.70)$$

The approximate equation for θ_{2D} becomes less accurate for lower supersonic Mach number and body deflection angle $\delta > 20$ deg.

For conical flow around a conical nose, the shock wave angle is

$$\theta_{\text{conical}} \approx 0.81 \theta_{2D} \quad (3.71)$$

As an example for the ramjet baseline, which has a body deflection angle $\delta = 17.7$ deg (from its conical nose), assume Mach 3.5 and an angle of attack $\alpha = 0$ deg. The exact equation with $\gamma = 1.4$ gives a two-dimensional shock wave angle $\theta_{2D} = 32$ deg. The approximate equation gives $\theta_{2D} = 34.3$ deg, an error of 7%. The conical shock wave angle for the ramjet baseline is $\theta_{\text{conical}} = 26$ deg (81% of a two-dimensional shock wave angle).

Supersonic inlets are designed for shock-on-lip operation at the highest Mach number cruise condition. For a shock-on-lip design, a higher flight Mach number should be avoided. A higher Mach number would result in the shock wave entering the inlet, causing shock-wave boundary-layer interaction, flow separation, a large loss in total pressure, and a large loss in thrust. For a shock-on-lip design, operating at a lower Mach number results in a larger shock wave angle, streamlines spilling outside the inlet, a reduction in the captured air, a reduction in thrust, and spillage drag.

Inlet capture efficiency can be determined by tracing the shockwave and the streamlines back to the inlet lip. For a two-dimensional (wedge) nose, tracing the shock wave and streamlines shown in the right side of Fig. 3.27 results in the following two-dimensional inlet capture efficiency equation

$$(A_0/A_c)_{2D} = (h/l)(1 + \delta M + \alpha M)/[(1 + \alpha M)(\delta + h/l)] \quad (3.72)$$

For a conical nose, a similar tracing gives the following conical inlet capture efficiency equation

$$(A_0/A_c)_{\text{conical}} = (h/l)(1 + \delta M + \alpha M)/[(1 - 0.23\delta M + \alpha M)(\delta + h/l)] \quad (3.73)$$

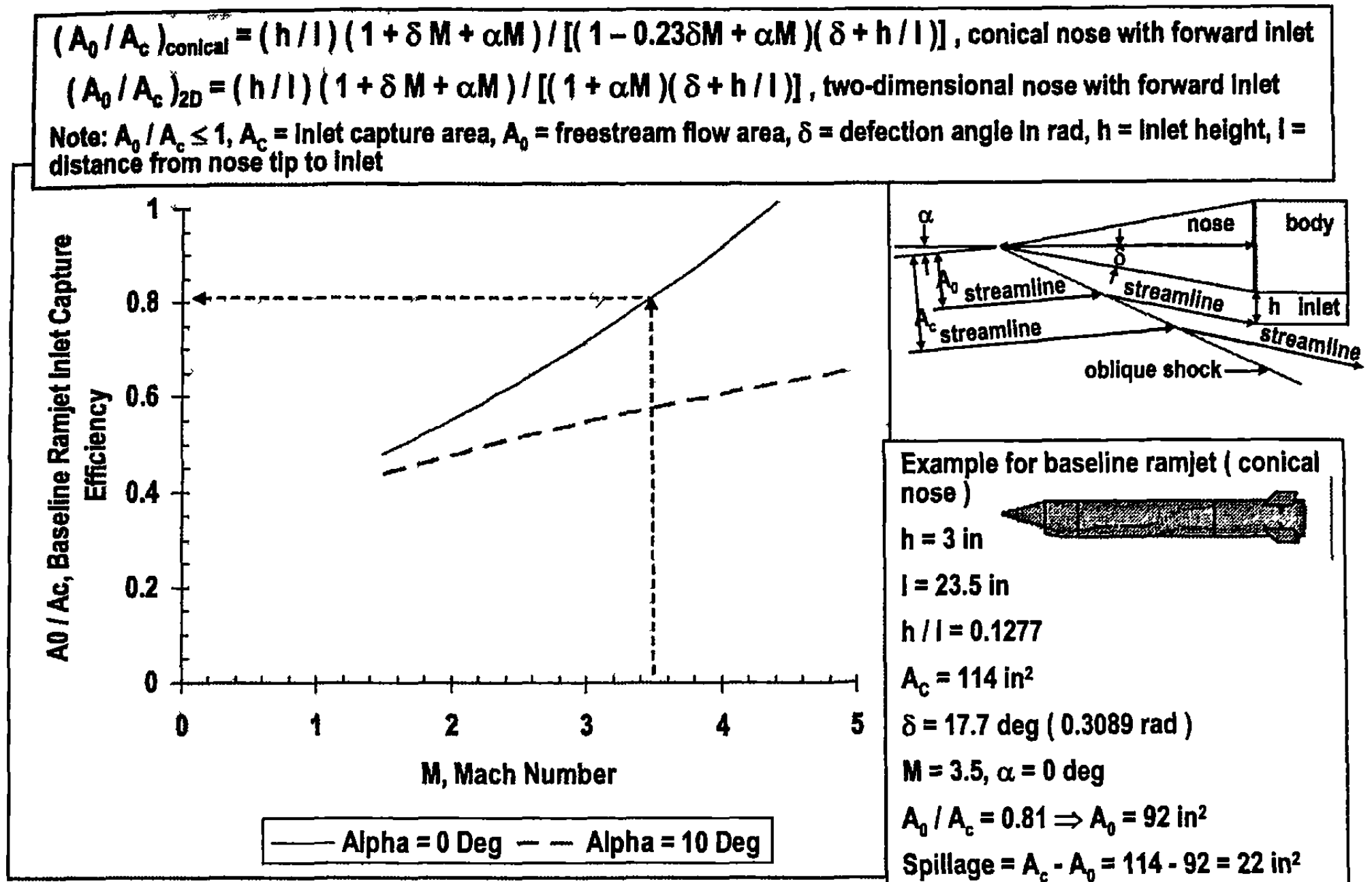


Fig. 3.27 Capture efficiency of an inlet increases with Mach number.

At a given Mach number, a conical shock wave has a smaller shock angle ($\theta_{\text{conical}} = 0.81\theta_{2D}$), resulting in higher capture efficiency for a given body deflection angle δ .

The ramjet baseline capture efficiency is shown in the figure as a function of Mach number ($2 < M < 5$) and angle of attack ($\alpha = 0, 10$ deg). Note from the figure that the ramjet baseline has 100% capture efficiency (shock-on-lip operation) at the cruise design maximum Mach number $M = 4$. Lower capture efficiency and spillage occur at lower Mach number because of the larger value of the shock wave angle θ .

The ramjet baseline will be used as an example calculation of inlet capture efficiency. The chin inlet ramjet baseline has an inlet height $h = 3$ in, a distance from the nose tip to the inlet of $l = 23.5$ in, capture area $A_c = 114$ in², and a conical nose half angle $\delta = 17.7$ deg = 0.309 rad. For an assumed flight condition of $M = 3.5$ and an angle of attack $\alpha = 0$ deg, the ramjet baseline inlet capture efficiency is computed

$$\begin{aligned} (A_0/A_c) &= (3/23.5)[1 + 0.309(3.5) + 0(3.5)] / \\ &\quad \{[1 - 0.23(0.309)(3.5) + 0(3.5)](0.309 + 3/23.5)\} \\ &= 0.128(2.08)/[0.751(0.437)] = 0.81 \end{aligned} \quad (3.74)$$

The freestream flow area is computed $A_0 = 0.81(114) = 92$ in² and the air spillage is given by $A_c - A_0 = 114 - 92 = 22$ in². For the ramjet baseline, a greater value of inlet height would be required for shock-on-lip operation at Mach 3.5.

The equation for the nondimensional inlet throat area that is required to start a two-dimensional supersonic inlet with ideal (isentropic) compression is

$$A_{IT}/A_0 = (M_{IE})_{start} [(\gamma + 1)/2]^{(\gamma+1)/[2(\gamma-1)]}$$
$$\{[1 + (\gamma - 1)/2](M_{IE})_{start}^2\}^{-(\gamma+1)/[2(\gamma-1)]}$$

(3.75)

The terms in the equation are A_{IT} = inlet throat area, A_0 = freestream flow area, $(M_{IE})_{start}$ = inlet start Mach number, and γ = specific heat ratio. Substituting $\gamma = 1.4$ for air gives

$$A_{IT}/A_0 = 1.728(M_{IE})_{start}(1 + 0.2(M_{IE})_{start}^2)^{-3}$$

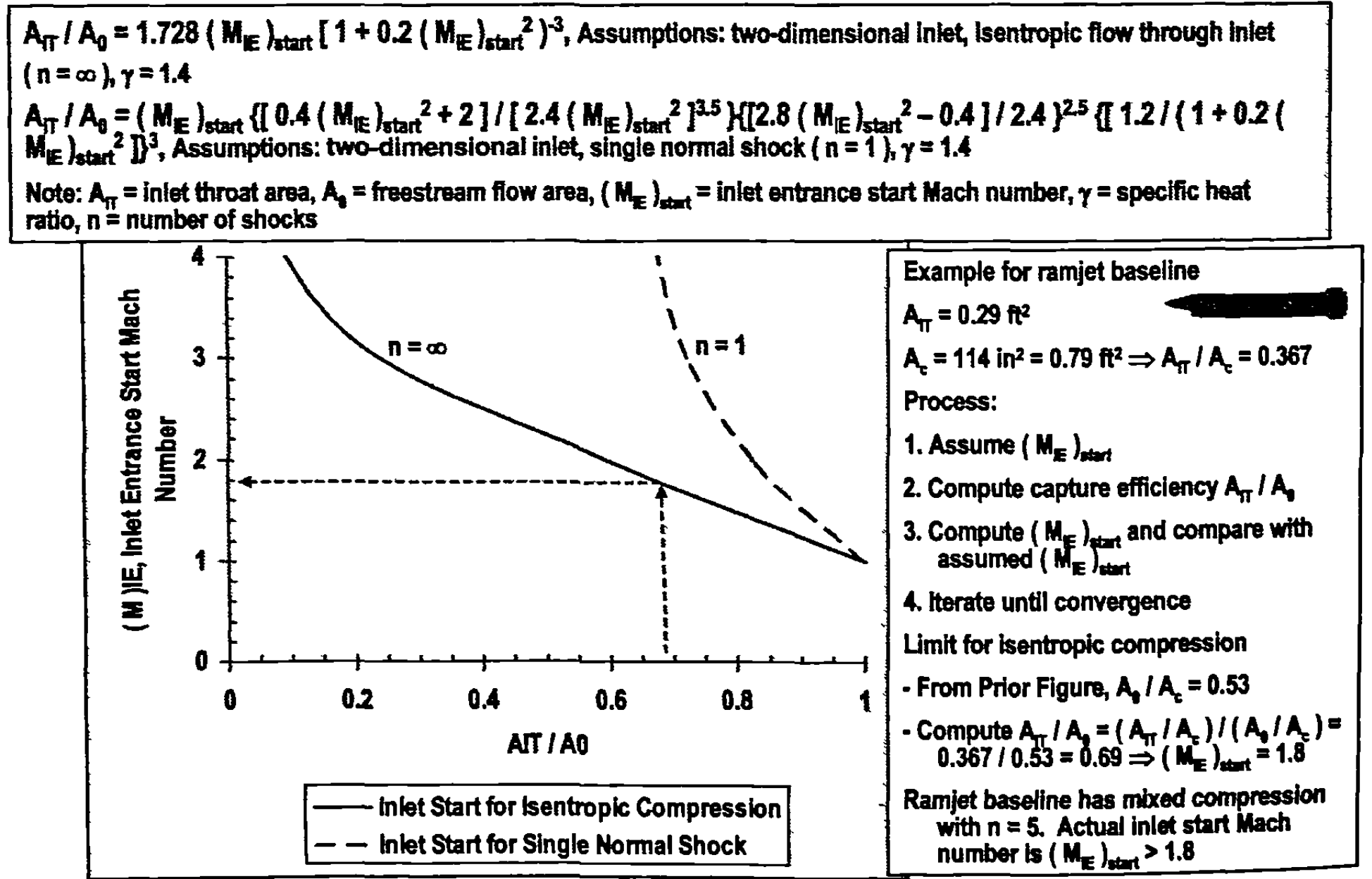
(3.76)

Another extreme is a two-dimensional supersonic inlet with a single, normal shock compression ($n = 1$). The equation for the nondimensional inlet throat area that is required to start the inlet is

$$A_{IT}/A_0 = (M_{IE})_{start} \{ [0.4(M_{IE})_{start}^2 + 2] / [2.4(M_{IE})_{start}^2] \}^{3.5}$$
$$\{ [2.8(M_{IE})_{start}^2 - 0.4] / 2.4 \}^{2.5} \{ [1.2 / (1 + 0.2(M_{IE})_{start}^2)] \}^3$$

(3.77)

Figure 3.28 shows $(M_{IE})_{start}$ versus A_{IT}/A_0 . Note that $A_{IT} = A_0$ for $(M_{IE})_{start} = 1$ and that for values of $A_{IT} < A_0$, $(M_{IE})_{start} > 1$. Also note that an inlet with a single, normal shock requires a larger Mach number to start the inlet.



As an example calculation, assume the ramjet baseline is at an angle of attack $\alpha = 0$ deg. The ramjet baseline inlet throat area is $A_{IT} = 0.29 \text{ ft}^2$ and its inlet capture area is $A_c = 114 \text{ in}^2 = 0.79 \text{ ft}^2$. Compute $A_{IT}/A_c = 0.29/0.79 = 0.367$. The inlet start Mach number is obtained by iteration. The process is to 1) assume a value of $(M_{IE})_{start}$; 2) compute the capture efficiency A_{IT}/A_0 ; 3) compute $(M_{IE})_{start}$ and compare with the assumed $(M_{IE})_{start}$; and 4) iterate until convergence. For this example, assume an ideal isentropic inlet with $(M_{IE})_{start} = 1.8$. From the previous figure, for $(M_{IE})_{start} = 1.8$ and $\alpha = 0$ deg, the inlet capture efficiency is $A_0/A_c = 0.53$. Calculate

$$A_{IT}/A_0 = (A_{IT}/A_c)/(A_0/A_c) = 0.367/0.53 = 0.69 \quad (3.78)$$

Substituting into the inlet start equation $A_{IT}/A_0 = 0.69 = 1.728(1.8)[1 + 0.2(1.8)^2]^{-3}$ verifies $(M_{IE})_{start} = 1.8$.

The ramjet baseline has a mixed compression inlet, with a total of $n = 5$ shocks (four compressions prior to the normal shock inside the inlet). The actual inlet entrance start Mach number is greater than the isentropic inlet start Mach number of $(M_{IE})_{start} = 1.8$. Also, a typical design criteria for a supersonic air-breathing missile is a minimum cruise Mach number of $(M_0)_{MinCruise} \approx 1.2 \times (M_0)_{InletStart}$, to overcome inlet buzz and to have sufficient design margin for the inlet to swallow the shock. Inlet buzz is caused by a transient start and unstart of the inlet. Also, a ramjet needs a thrust/drag design margin following inlet start to accelerate under ramjet thrust to the cruise Mach number. The minimum design Mach number for transition from booster to ramjet flight is $(M_0)_{RamjetTransition} \approx 1.2 \times (M_0)_{InletStart}$.

As discussed previously, it is usually desirable for a ramjet to have a low value of inlet entrance Mach number M_{IE} , which reduces the required start Mach number, which reduces the required size/weight of the booster. A mixed compression inlet provides forebody compression, and the shock wave(s) reduce M_{IE} . The equation for the inlet entrance Mach number for a single oblique shock wave is

$$M_{IE} = \{ \{ 36M_0^4 \sin^2 \theta_{2D} - 5[M_0^2 \sin^2 \theta_{2D} - 1][7M_0^2 \sin^2 \theta_{2D} + 5] \} / \{ [7M_0^2 \sin^2 \theta_{2D} - 1][M_0^2 \sin^2 \theta_{2D} + 5] \} \}^{1/2} \quad (3.79)$$

The oblique shock wave angle θ_{2D} can be determined from the equation

$$\tan(\alpha + \delta) = 2 \cot \theta_{2D} (M_0^2 \sin^2 \theta_{2D} - 1) / [2 + M_0^2 (2.4 - 2 \sin^2 \theta_{2D})] \quad (3.80)$$

Terms in the above equations are M_0 = freestream Mach number, α = angle of attack, and δ = body deflection angle. Assumptions are a single shock, two-dimensional flow, perfect gas, and specific heat ratio $\gamma = 1.4$. Figure 3.29 shows M_{IE} for a flow deflection angle $\alpha + \delta$ up to the maximum value for attached flow. Freestream Mach number range shown in the figure is $2 < M_0 < 5$. Note the large effect of flow deflection on reducing M_{IE} .

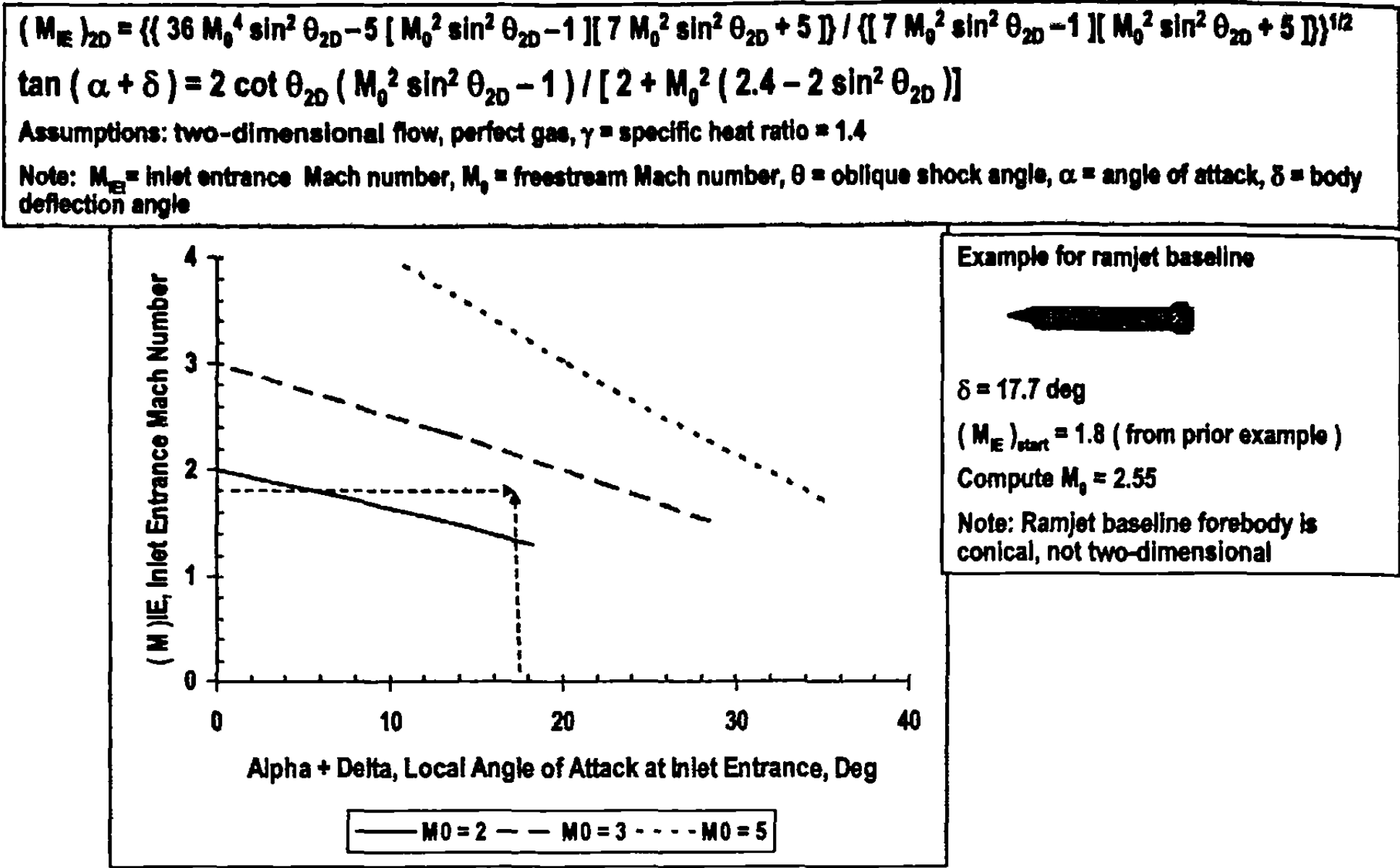
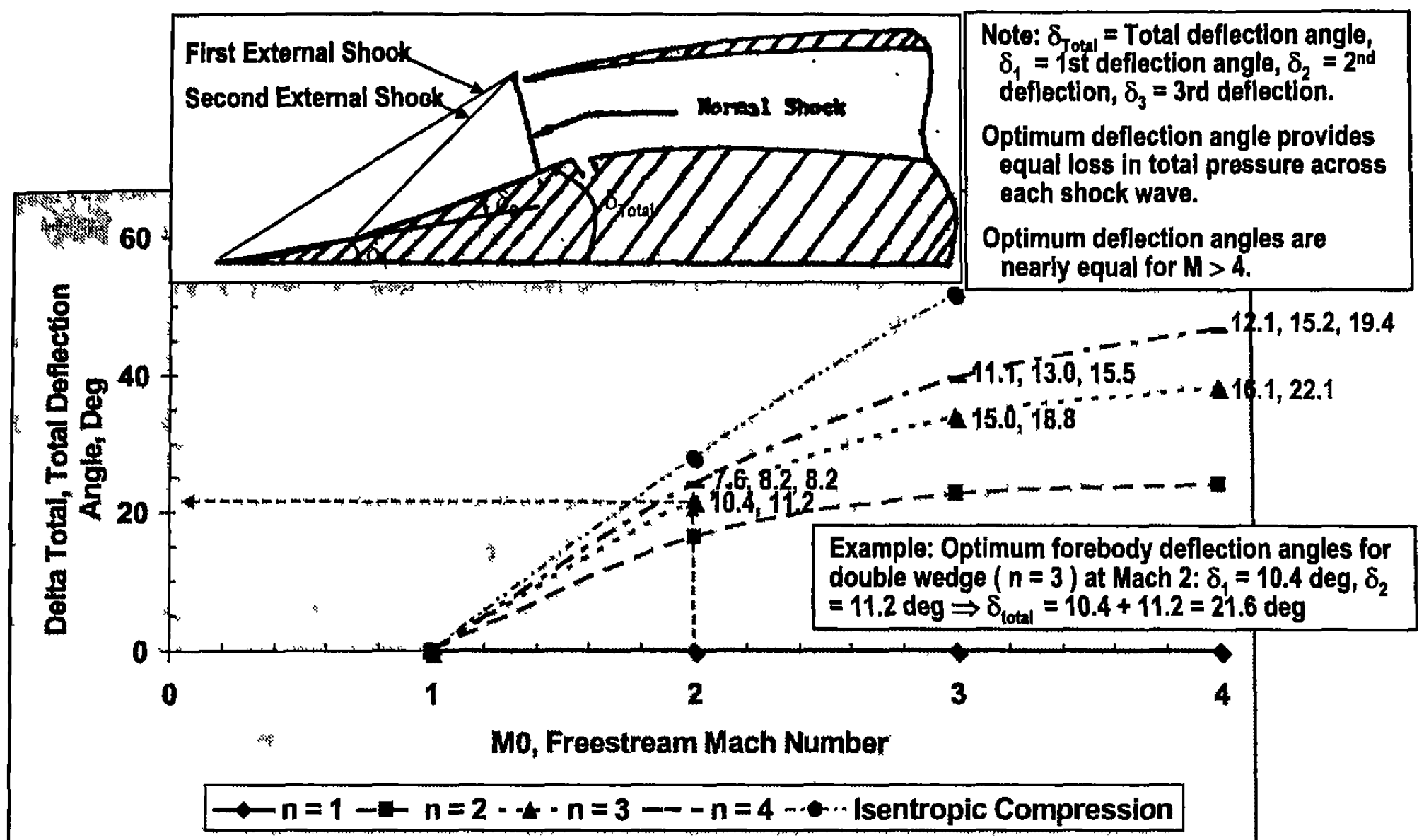


Fig. 3.29 Forebody shock compression reduces the inlet entrance Mach number.

As an example, assume that the ramjet baseline conical forebody can be approximated as a two-dimensional deflection. Then for $\alpha = 0 \text{ deg}$, $\alpha + \delta = 17.7 \text{ deg}$. From the prior example, the inlet start Mach number is $(M_{IE})_{Inletstart} \approx 1.8$. Substituting into the above equations and neglecting the difference between a conical deflection versus a two-dimensional deflection, the ramjet baseline booster must be sized for a ramjet transition Mach number of approximately $(M_0)_{RamjetTransition} \approx 2.55$.

Figure 3.30, derived from Ref. 14 presents the optimum deflection angles of wedge forebodies to give the highest value of total pressure ratio following a normal shock. Oswatitsch¹⁵ showed that the optimum forebody deflection angles for maximum total pressure behind a normal shock provide equal loss in total pressure across each shock wave. Note from the figure that the required deflection angle(s) for optimum pressure recovery increases with Mach number. For freestream Mach numbers greater than 4, the optimum deflection angle(s) increases slowly with increasing Mach number. The figure also shows that two external shocks have smaller optimum deflection angles than the optimum deflection angle of a single external shock. Also note that the sum of the deflections (total turning angle) for two optimum external shocks is greater than the deflection angle of a single optimum external shock. Similarly, three optimum external shocks have smaller deflection angles than two optimum external shocks and the total turning angle of three optimum external shocks is greater than the total turning angle of two optimum external shocks. The advantage of multiple shocks over one shock is higher total pressure recovery. A disadvantage of multiple external shocks prior to the inlet normal shock is the flow is deflected more away from the centerline. As a rule of thumb, the total turning angle of the flow should be less than about 45 deg to avoid flow separation inside the inlet.



Reference: "Technical Aerodynamics Manual," North American Rockwell Corporation, DTIC AD 723823, June 1970.

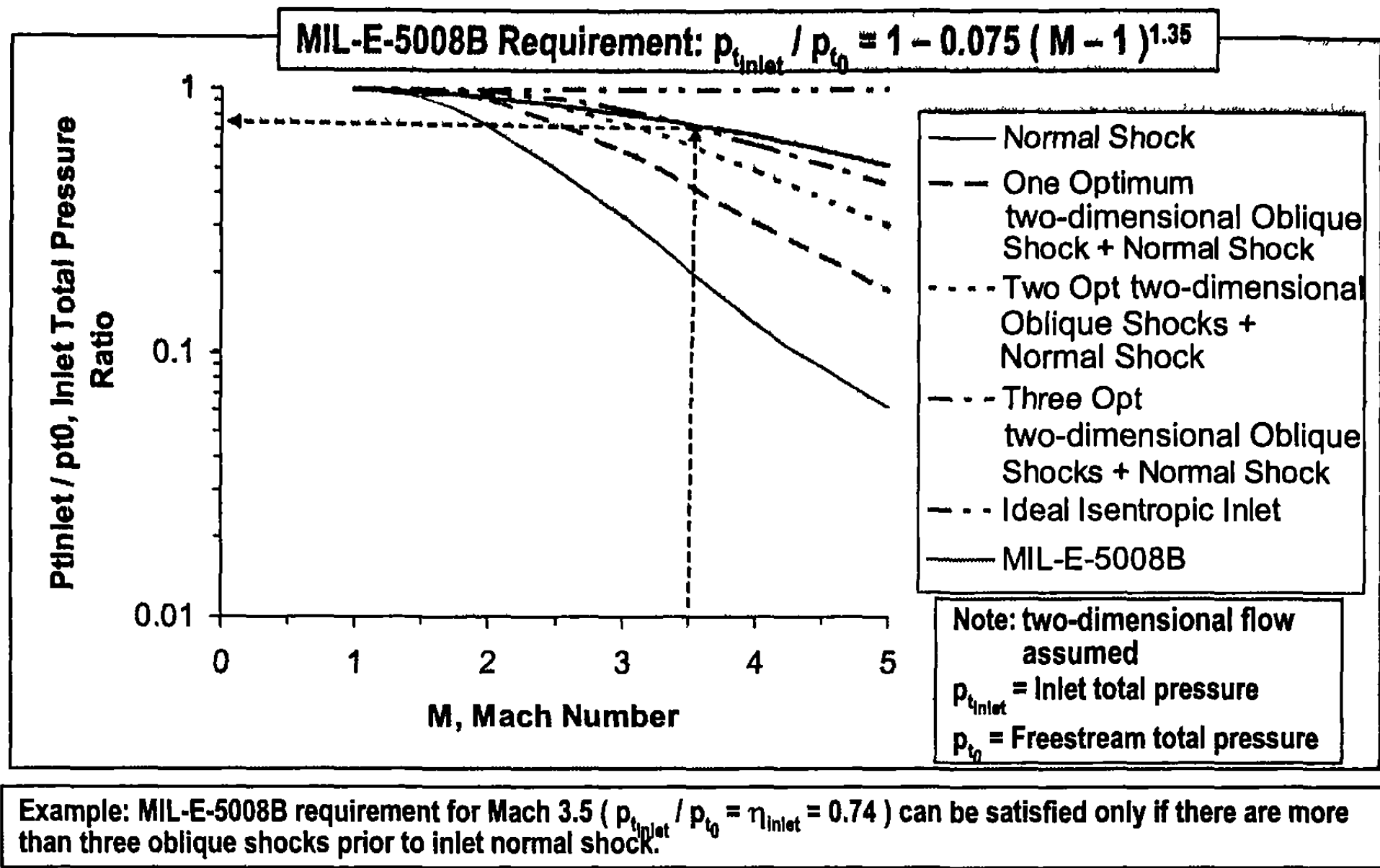
Fig. 3.30 Optimum forebody deflection angle(s) for best pressure recovery increases with Mach number.

As an example, at Mach 2 the forebody optimum total deflection angle for a double-wedge forebody is 21.6 deg, with two oblique shocks prior to a normal shock of 10.4 deg half-angle for the first deflection and 11.2 deg half-angle for the second deflection.

Because ramjet combustion is subsonic, there must be an inlet normal shock to provide subsonic flow into the combustor. Small oblique shocks prior to the normal shock alleviate the problem of total (stagnation) pressure loss across the inlet normal shock. Figure 3.31, based on Ref. 14, compares the military specification MIL-E-5008B total pressure recovery requirement with that of a single normal shock as well as one, two, and three optimum two-dimensional oblique shocks prior to the normal shock. The requirement of MIL-E-5008B, which is a reasonable pressure recovery for a well designed inlet, is given by the equation

$$p_{t_{inlet}}/p_{t_0} = 1 - 0.075(M - 1)^{1.35} \quad (3.81)$$

As stated previously, from Oswatitsch¹⁵ we know that the maximum pressure recovery occurs for an equal loss in total pressure across each shock wave. Note from the figure that for hypersonic missiles, a mixed compression inlet with three or more oblique shocks is required to satisfy MIL-E-5008B. For the ramjet baseline, inlet/airframe integration consists of 1) an external forebody compression (conical nose); 2) a ramp leading up to the cowl; 3) an inlet cowl lip angle; 4) internal turning; and 5) an internal normal shock. A mixed compression inlet reduces excessive flow turning away from the axial direction.



Source for Optimum 2D Shocks: Oswatitsch, K.L., "Pressure Recovery for Missiles with Reaction Propulsion at High Supersonic Speeds", NACA TM - 1140, 1947.

Fig. 3.31 Oblique shocks prior to the inlet normal shock are required to satisfy MIL-E-5008B.

An example is shown in the figure. The MIL-E-5008B Mach 3.5 requirement is compared with the Mach 3.5 total pressure recovery for three optimum oblique shocks plus a normal shock. From Ref. 13, at Mach 3.5 the optimum deflection angles are $\delta_1 = 11.9$ deg, $\delta_2 = 14.1$ deg, and $\delta_3 = 17.2$ deg. As shown in the figure, the stagnation pressure recovery from three optimum oblique shocks at Mach 3.5 is $\eta_{inlet} = 0.72$. The requirement from MIL-E-7008B for Mach 3.5 is

$$\eta_{inlet} = P_{tInlet}/p_{t0} = 1 - 0.075(3.5 - 1)^{1.35} = 0.74 \tag{3.82}$$

Therefore, more than three oblique shocks are required to satisfy the military standard for Mach 3.5.

3.14 Fuel Alternatives

An area of high payoff for airbreathing missiles is that of high-density fuels. As shown in Fig. 3.32, the high-density fuels provide high volumetric performance, which is especially beneficial for volume-limited missiles. Current turbojet fuels such as JP-4, JP-5, JP-7, and JP-10 have relatively low density, on the order of 0.028 lb/in³, and low volumetric performance, on the order of 559 BTu/in³. Liquid-fuel ramjet hydrocarbon fuels such as RJ-4, RJ-5, RJ-6, and RJ-7 have somewhat higher density (0.040 lb/in³) and higher volumetric performance (581 BTu/in³). Binder fuels, such as hydroxyl-terminated polybutadiene (HTPB), have an advantage of slightly higher volumetric performance (606 BTu/in³) with

Type Fuel	Density, lb / in ³	Volumetric Performance, BTU / in ³	Low Observables
Turbine (JP-4, JP-5, JP-7, JP-8, JP-10)	~ 0.028	○ 559	●
Liquid Ramjet (RJ-4, RJ-5, RJ-6, RJ-7)	~ 0.040	○ 581	●
HTPB	~ 0.034	○ 606	○
Slurry (40% JP-10 / 60% carbon)	~ 0.049	◐ 801	○
Solid Carbon (graphite)	~ 0.075	● 1132	—
Slurry (40% JP-10 / 60% aluminum)	~ 0.072	◐ 866	—
Slurry (40% JP-10 / 60% boron carbide)	~ 0.050	● 1191	—
Solid Mg	~ 0.068	● 1200	—
Solid Al	~ 0.101	● 1300	—
Solid Boron	~ 0.082	● 2040	—

● Superior ◐ Above average ○ Average — Below average

Fig. 3.32 High-density fuels provide higher volumetric performance but have higher observables.

a disadvantage of increased visual observables. The density of HTPB is 0.034 lb/in³. Slurry fuels, such as JP-10 with carbon slurry, and solid carbon fuels, such as graphite, have much higher volumetric performance. High volumetric performance is also achievable with high-density solid metal fuels such as magnesium, aluminum, and boron. For example, solid boron fuel, if it could be packaged with a theoretical solids loading of 100%, would provide more than three times the volumetric performance of a liquid hydrocarbon fuel. Disadvantages of solid metal fuels are high visual observables from their plumes and reduced volumetric loading/packaging efficiency.

3.15 Ducted Rocket Design Considerations

Ducted rockets combine the higher specific impulse of a ramjet with some of the simplicity and acceleration capability of a solid rocket. If there is a relatively small amount of excess fuel (e.g., 30% of the propellant weight) discharged by the gas generator, the ducted rocket behaves more like a rocket, with a higher burn rate and a higher temperature but a lower specific impulse. Conversely, if the gas generator discharges a relatively large amount of excess fuel (e.g., 70% of the propellant weight), the ducted rocket behaves more like a ramjet, with a higher specific impulse and a lower burn rate/temperature. The fuel fraction of a ducted rocket varies from about 30 to 80% of the propellant weight. Fuel fractions greater than 80% are not practical for ducted rocket propulsion because the reduced combustion temperature for a high fuel fraction severely reduces the fuel flow rate, resulting in an unacceptably low thrust.

Ducted rocket fuels are either a metal fuel such as boron, aluminum, and magnesium or a carbon-based fuel such as graphite carbon or hydrocarbon HTPB binder. Metal fuels have an advantage of higher specific impulse. Also, the higher density of metal fuels is beneficial for a volume-limited missile. Disadvantages of metal fuels include deposits and higher observables. Metal fuel deposits downstream of the gas generator are a particular problem for a high fuel fraction ducted rocket because of the reduced combustion temperature. A carbon-based fuel has an advantage of lower observables and a higher reliability from reduced deposits. A disadvantage is lower maximum specific impulse compared to a metal fuel (e.g., 800 vs 1,000 s).

The choice of oxidizer is a primary trade of performance versus reduced observables. Ammonium perchlorate (AP) has advantages of higher burn rate and lower hazards classification. However, AP provides an HCl contrail, particularly at high altitude. Minimum smoke oxidizers, such as Her Majesty's Explosive (HMX) and Royal Demolition Explosive (RDX) have reduced observables but have disadvantages of lower performance and are more hazardous.

The most popular approach for thrust magnitude control is to partially block the gas generator throat with a pintle or valve. Partially blocking the gas generator throat reduces the throat area, resulting in an increase in the chamber pressure. The increased chamber pressure increases the gas flow rate, which provides increased thrust. The thrust variation using this approach is about 10:1. A concern is deposits on the pintle or valve that may clog the gas generator throat. Another approach incorporates retractable wires in the gas generator propellant grain. The burn area of the gas generator propellant grain is a function of the retraction speed of the wires. A disadvantage of this approach is the maximum variation in thrust is only about 4:1.

3.16 Rocket Motor Performance

Figure 3.33 shows the relationship of the maximum incremental velocity of a rocket as a function of the propellant weight fraction and the specific impulse. The equation is

$$\Delta V = -g_c I_{sp} \ln(1 - W_p/W_i) \quad (3.83)$$

An assumption for the equation is a high-thrust motor, with thrust much larger than drag. It is also assumed that the thrust is much greater than the component of the missile weight in the thrust direction, which is applicable for either high thrust-to-weight ratio or horizontal flight. Also, the flight path angle is assumed to be constant. Note from the figure that hypersonic Mach number can be achieved for a low-speed launch only if the propellant weight is more than one-half of the missile initial weight. Also note that higher specific impulse also gives higher burnout velocity.

For the rocket baseline missile, the initial boost (launch) weight is $W_{i,boost} = W_L = 500$ lb, the boost propellant weight is $W_p = 84.8$ lb, and the specific impulse during boost is $I_{SP} = 250$ s. Computing the maximum theoretical incremental velocity from boost

$$\Delta V = -32.2(250)\ln(1 - 84.8/500) = 1496 \text{ ft/s} \quad (3.84)$$

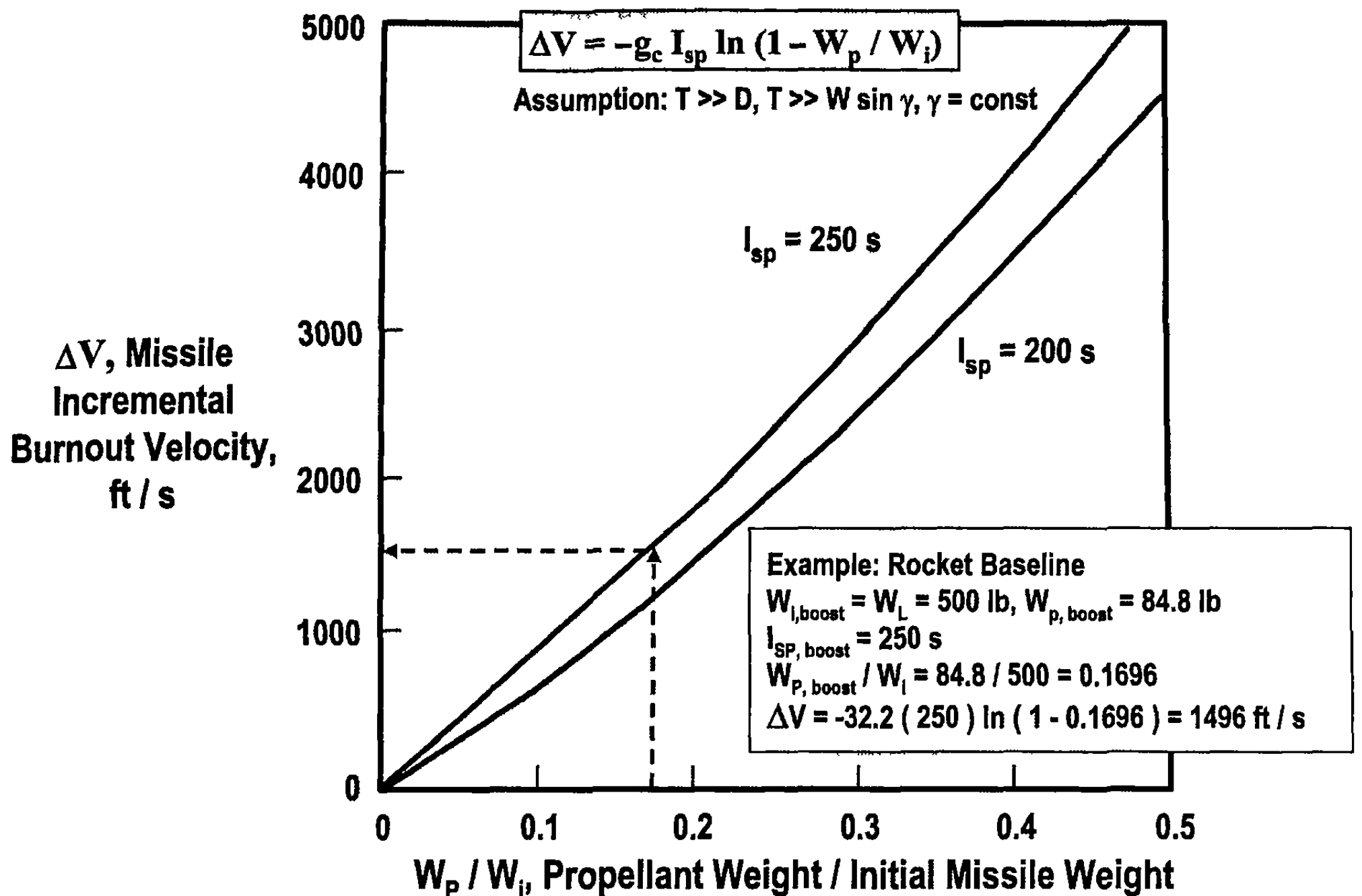


Fig. 3.33 High propellant fraction increases burnout velocity.

For a typical launch condition of Mach 0.8/20,000 ft altitude, the launch velocity is 820 ft/s, resulting in a theoretical end-of-boost velocity of

$$V_{EB} = 820 + 1496 = 2316 \text{ ft/s} \quad (3.85)$$

The actual end-of-boost velocity would be lower because of drag.

Specific impulse and thrust are primary considerations in the motor sizing process. Assuming isentropic flow, the equations are

$$I_{SP} = c_d \left\{ \left[\frac{2\gamma^2}{(\gamma-1)} \right] \left[\frac{2}{(\gamma+1)} \right]^{(\gamma-1)/(\gamma+1)} \left[1 - (p_e/p_c)^{(\gamma-1)/\gamma} \right] \right\}^{1/2} + (p_e/p_c)\epsilon - (p_0/p_c)\epsilon \} c^* / g_c \quad (3.86)$$

$$T = \dot{W}_p I_{SP} = (g_c / c^*) p_c A_t I_{SP}$$

$$\epsilon = \left\{ \left[\frac{2}{(\gamma+1)} \right]^{1/(\gamma-1)} \left[\frac{(\gamma-1)}{(\gamma+1)} \right]^{1/2} \right\} / \left\{ (p_e/p_c)^{1/\gamma} \left[1 - (p_e/p_c)^{(\gamma-1)/\gamma} \right]^{1/2} \right\} \quad (3.87)$$

Specific impulse of the rocket baseline is shown in Fig. 3.34 as a function of nozzle expansion ratio and chamber pressure. These results are based on typical values of specific heat ratio ($\gamma = 1.18$), discharge coefficient ($c_d = 0.96$), propellant characteristic velocity ($c^* = 5200 \text{ ft/s}$), and $h = 20,000 \text{ ft}$ altitude ($p_0 = 6.75 \text{ psi}$). The assumed characteristic velocity of 5200 ft/s is a representative value for a propellant. For a low-performance, minimum smoke propellant, c^* may be less than 4800 ft/s. It may be greater than 5600 ft/s for a high-performance, high-smoke propellant. Referring again

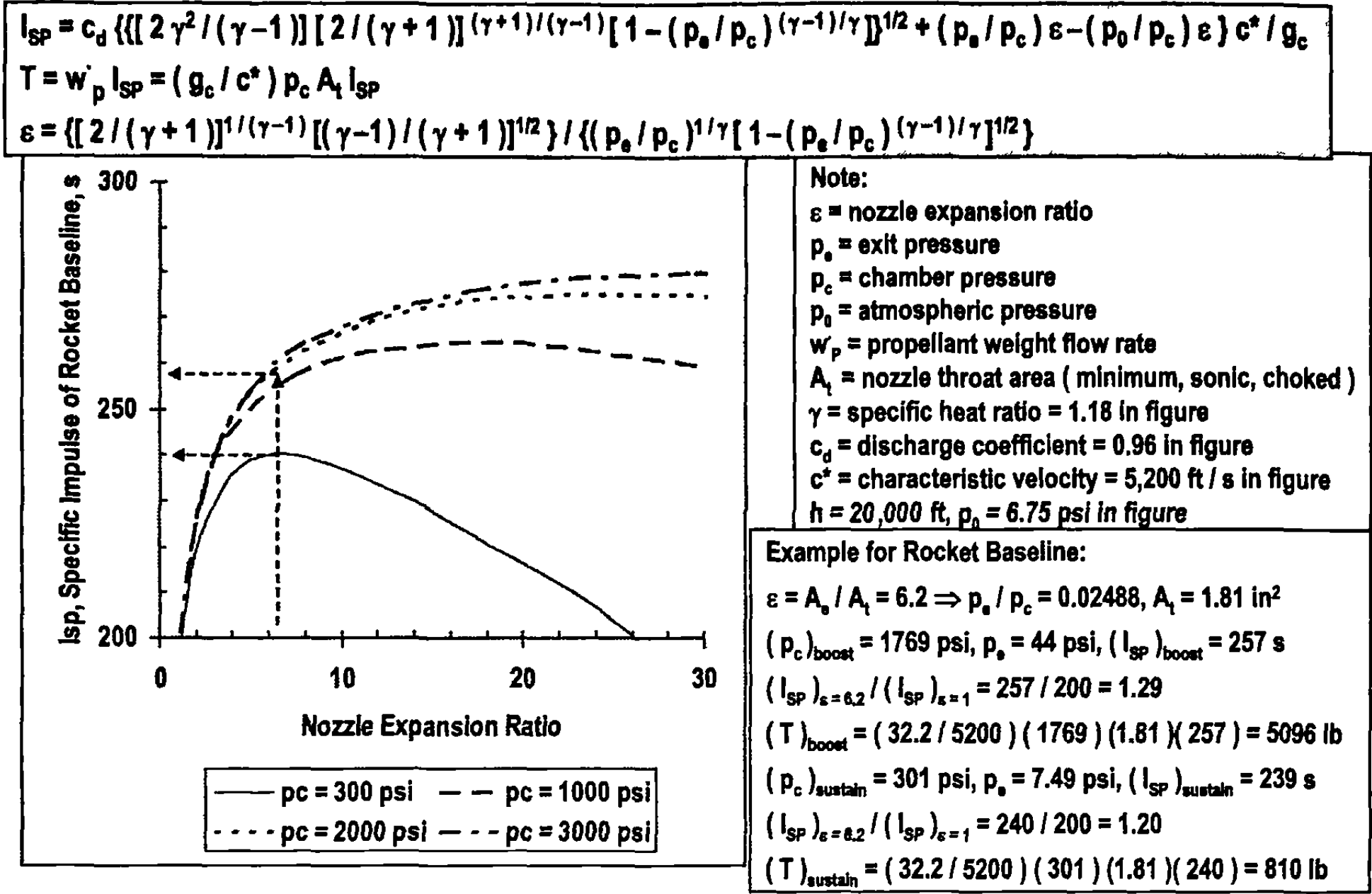


Fig. 3.34 High specific impulse requires high chamber pressure and optimum nozzle expansion.

to the figure, note that a high chamber pressure requires a high nozzle expansion ratio for maximum specific impulse. Also note that for a low chamber pressure, a relatively small value of nozzle expansion is desired. At high-altitude flight conditions and high chamber pressure, a large expansion ratio is desirable for the nozzle, within the limit of the missile diameter and length constraints. For a length-limited missile, a tradeoff is a large expansion nozzle with higher I_{SP} versus a lower expansion nozzle and shorter nozzle length. Although the lower expansion nozzle has lower I_{SP} , the shorter nozzle length allows more length for propellant for a length-limited missile. Another possible design approach for a length-limited missile would be to use two or more adjacent nozzles instead of a single nozzle, such as the TOW missile. Multiple nozzles can be designed for optimum expansion and higher I_{SP} in a shorter length.

As an example calculation of specific impulse and thrust, the rocket baseline has a nozzle exit-area-to-throat-area expansion ratio of $\varepsilon = 6.2$ and a nozzle throat area (minimum flow area with sonic, choked flow) $A_t = 1.81 \text{ in}^2$. At an altitude of 20,000 ft, the specific impulse and thrust during boost ($p_c = 1,769 \text{ psi}$) are computed to be $I_{SP} = 257 \text{ s}$ and $T = 5096 \text{ lb}$ respectively. During the lower chamber pressure sustain ($p_c = 301 \text{ psi}$), the computed specific impulse and thrust are $I_{SP} = 239 \text{ s}$ and $T = 810 \text{ lb}$. Note from the figure that for the rocket baseline during sustain flight, its expansion ratio provides maximum I_{SP} .

A purpose of using nozzle expansion is to provide increased specific impulse and thrust. For the rocket baseline, nozzle expansion increases the specific impulse by 29% during boost (257 s for $\varepsilon = 6.2$ vs 200 s for $\varepsilon = 1$). During

sustain, the nozzle expansion increases the specific impulse by 20% (239 s with the nozzle vs 200 s if there were no nozzle expansion).

The actual specific impulse of the rocket baseline is about 3% lower than prediction (250 vs 257 s during boost and 230 vs 239 s during sustain). The actual thrust of the rocket baseline is about 20% higher than prediction (5750 vs 5096 lb during boost and 1018 vs 810 lb during sustain). The predicted burn time is about 20% longer than the actual burn time. A possible contributor to the differences between prediction and data is the assumed value of specific heat ratio γ of the nozzle gases. Although the gases experience a large variation in temperature, the prediction is based on an assumption of a constant $\gamma = 1.18$ during the expansion of the gases through the nozzle.

Shown in Fig. 3.35 are the chamber pressure and the area of the nozzle throat that are necessary to provide a required propellant weight flow rate. The equation is

$$A_t = c^* w_p / (g_c p_c) \tag{3.88}$$

Note from the figure that the necessary throat area A_t and chamber pressure p_c increase with increased values of the design propellant burn rate w_p and characteristic velocity c^* . Also note that the necessary throat area decreases with an increased value of the design chamber pressure. The range of values shown are the figure are typical values of A_t , w_p , c^* , and p_c for tactical missiles.

As an example, the rocket baseline has a propellant characteristic velocity of $c^* = 5200$ ft/s. At the boost condition the motor chamber pressure is $p_c = 1769$

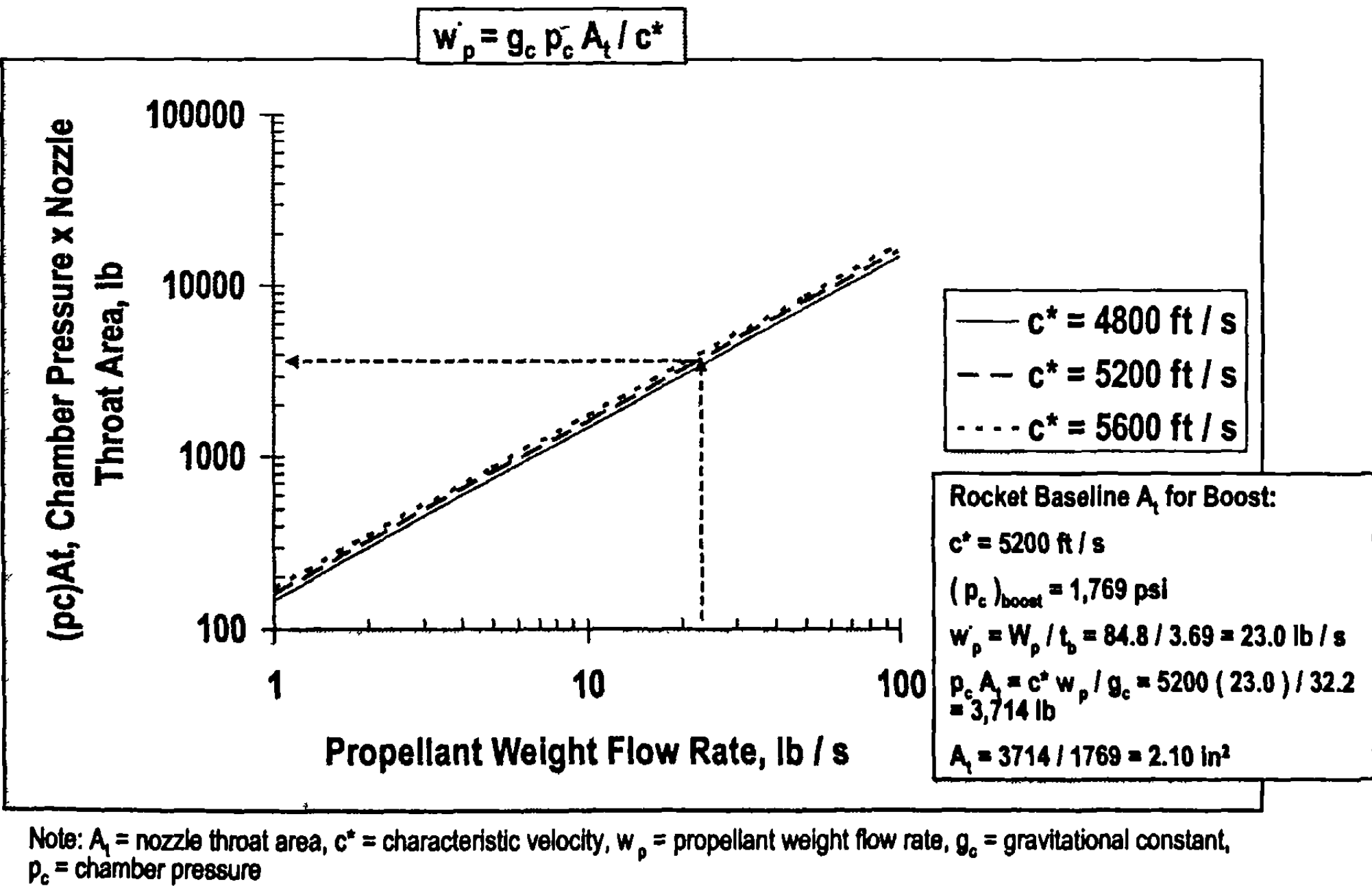


Fig. 3.35 High propellant weight flow rate requires a high chamber pressure and a large nozzle throat

psi and the propellant weight flow rate is $w_p = 23.0$ lb/s. The necessary nozzle throat area is computed to be

$$A_t = 5200(23.0)/[32.2(1769)] = 2.10 \text{ in}^2 \tag{3.89}$$

The computed nozzle throat area is 16% higher than the actual nozzle throat area of the rocket baseline (2.10 vs 1.81 in²).

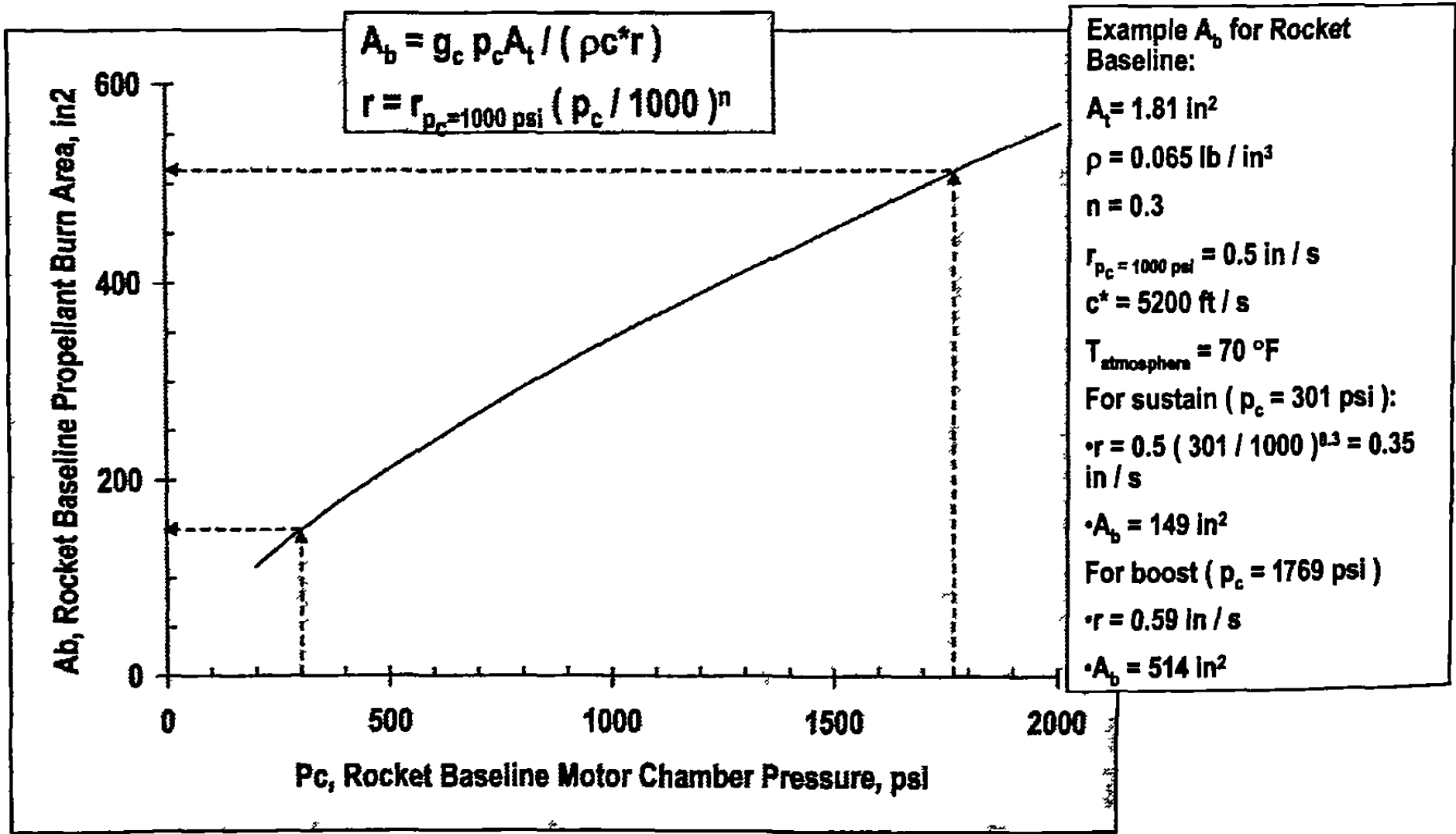
Nozzle exit area and nozzle throat area define the nozzle expansion ratio ($\epsilon = A_e/A_t$). It may not be possible to use the value of expansion ratio that gives the maximum value of specific impulse and thrust. The expansion ratio that provides $(I_{SP})_{max}$ and T_{max} may not allow sufficient volume for packaging subsystems around the nozzle, it may result in an excessively long nozzle, or it may exceed the diameter of the missile. A lower value of the design ϵ may be required, reducing the value of the achievable specific impulse and thrust.

Figure 3.36 shows a relationship of the propellant burn area to the chamber pressure of the rocket baseline. The equations are

$$A_b = g_c p_c A_t / (\rho c^* r) \tag{3.90}$$

$$r = r_{p_c=1000 \text{ psi}} (p_c / 1000)^n \tag{3.91}$$

Note that the required burn area A_b is a function of the chamber pressure, throat area, propellant density, characteristic velocity, and the propellant burn rate. To achieve a high chamber pressure, a large propellant burn area and a small nozzle throat are required. Burn rate r is a function of the propellant



Note: A_b = propellant burn area, g_c = gravitation constant, A_t = nozzle throat area, ρ = density of propellant, c^* = characteristic velocity, r = propellant burn rate, $r_{p_c=1000 \text{ psi}}$ = propellant burn rate at $p_c = 1,000$ psi, p_c = chamber pressure, n = burn rate exponent

Fig. 3.36 High chamber pressure requires large propellant burn area and a small nozzle throat.

reference burn rate at a reference chamber pressure of 1000 psi ($r_{p_c=1000 \text{ psi}}$), chamber pressure p_c , and burn rate exponent n . For statically stable combustion, $n < 1$. For a boost-sustain motor, it may be desirable for the value of n to approach 1. This allows a large change in burn rate, mass flow rate, and thrust with a small change in chamber pressure and burn area.

For a solid rocket, the oxidizer and fuel are all already mixed in the propellant. As a result, combustion occurs nearly instantaneously at the burning surface. For air-breathing or liquid-propellant propulsion, combustion occurs downstream of the mixing of the fuel and oxidizer. As a result, air-breathing propulsion and liquid-propellant rocket propulsion require additional length and volume to complete combustion compared to a solid rocket.

The rocket baseline has a throat area $A_t = 1.81 \text{ in}^2$, a propellant density $\rho = 0.065 \text{ lb/in}^3$, and a characteristic velocity $c^* = 5200 \text{ ft/s}$. The burn rate exponent is $n = 0.3$, referenced to a nominal chamber pressure $p_c = 1000 \text{ psi}$ and a nominal atmospheric temperature $T = 70^\circ \text{ F}$. The nominal burn rate at standard conditions is $r = 0.5 \text{ in/s}$. During sustain, with a chamber pressure of $p_c = 301 \text{ psi}$, the burn rate is $r = 0.5(301/1000)^{0.3} = 0.35 \text{ in/s}$. The required burn area for sustain is

$$A_b = 32.2(301)(1.81)/[0.065(5200)(0.35)] = 149 \text{ in}^2 \quad (3.92)$$

For the 8-in diameter rocket baseline, the body cross-sectional area is 50 in^2 . An end-burning grain for the rocket baseline, constrained by its nozzle geometry, propellant burn rate exponent, and propellant burn rate, would not provide the required chamber pressure. Therefore a center cavity grain, with three radial slots, is used for the rocket baseline to obtain the required burn surface area for boost and sustain thrust. For the rocket baseline boost chamber pressure $p_c = 1769 \text{ psi}$, the burn rate during boost is $r = 0.5(1769/1000)^{0.3} = 0.59 \text{ in/s}$. The required burn area for boost is larger,

$$A_b = 32.2(1769)(1.81)/[0.065(5200)(0.59)] = 514 \text{ in}^2 \quad (3.93)$$

Figure 3.37 summarizes the rocket motor sizing process for conceptual design. It is a highly iterative process that harmonizes a large number of design parameters and constraints. The process begins by 1) specifying an altitude and assuming a desired thrust-time profile. In the next step, 2) assumptions are made for the type of propellant, the desired chamber pressure, the burn area(s), and the nozzle geometry. These are used in step 3) to calculate an initial estimate of the thrust and specific impulse. If the thrust and specific impulse are acceptable, the process continues to the next step. However, if the initial estimate of thrust and specific impulse are not acceptable, we go back to step 2 and try again with a different combination of propellant, chamber pressure, burn area, and nozzle geometry. If we still cannot get an acceptable specific impulse and thrust after iterating in step 2, we may have to go back to step 1 and change the desired thrust-time profile. Assuming that the desired specific impulse and thrust are achievable, we next go to step 4) and calculate of the propellant weight flow rate and the weight of the propellant used. If this is acceptable, we

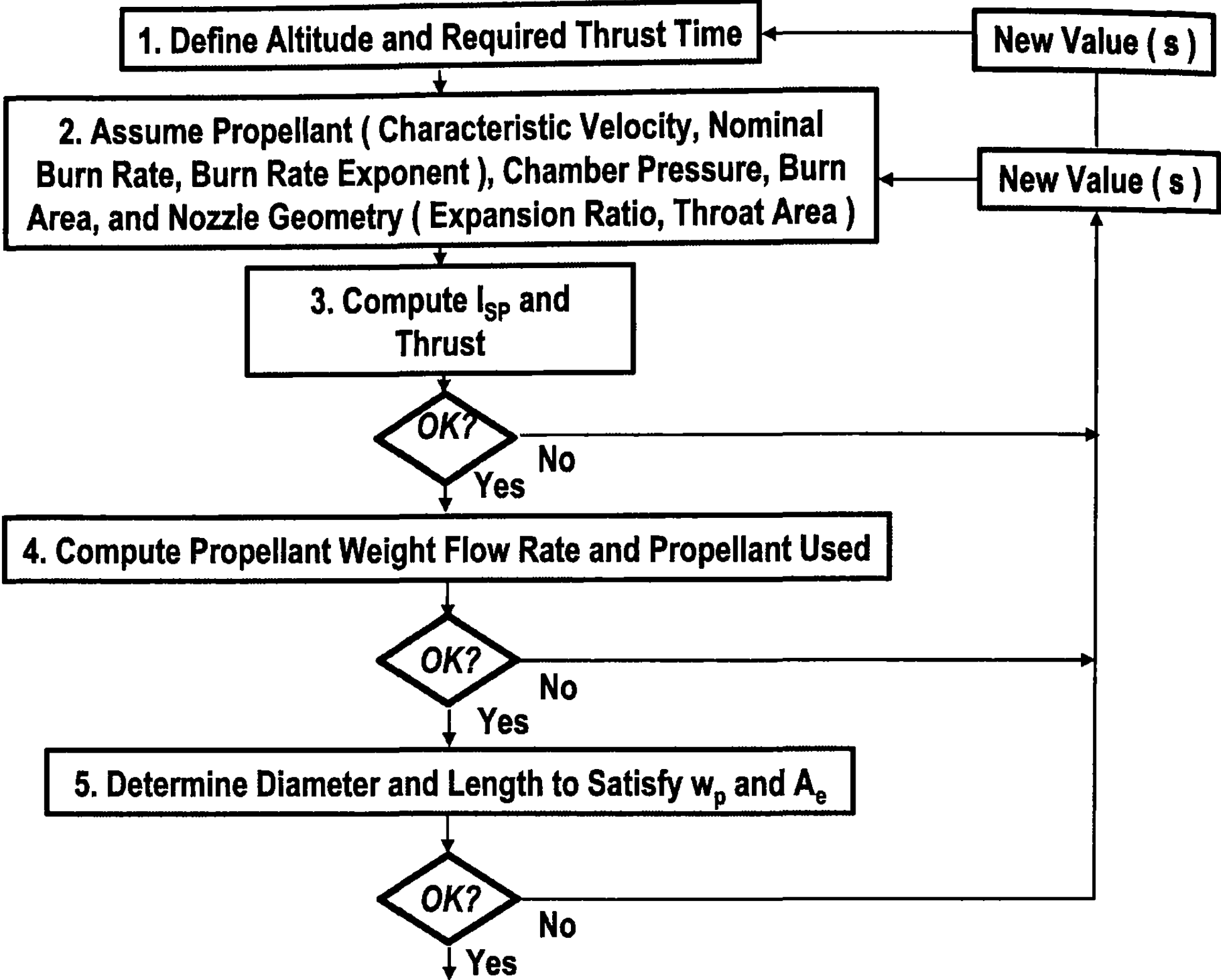


Fig. 3.37 Conceptual design sizing process for a rocket motor.

go to the next step. If not, we go back and iterate step 2 until we reach a convergence. Again, if there is no convergence, we may have to go back to step 1 and change the desired thrust-time history. Assuming that there is a convergence, we next go to step 5), determining the diameter and length to satisfy the propellant weight and nozzle exit area. If this is acceptable, we have completed one of the possible design solutions. If it is not acceptable, we have to go back and iterate until there is a solution.

3.17 Solid Propellant Grain Alternatives

To define a varying thrust-time profile, a cavity through the center of the propellant grain can be used to provide the required burning surface area time profile. As shown in Fig. 3.38, a variety of shapes (e.g., wagon wheel, star, dendrite, end-burner, radial slotted tube) provide larger or smaller burn area and thus more or less hot gas, providing more or less thrust. The change in the burn area as a function of time provides the desired thrust-time profile. If the thrust increases with burn time, it is referred to as progressive burning. If the grain design produces a decrease in thrust with burn time, the rocket motor is said to have regressive burning characteristics. Shown in the figure are typical volumetric loadings for grain design. The volumetric efficiency of the propellant

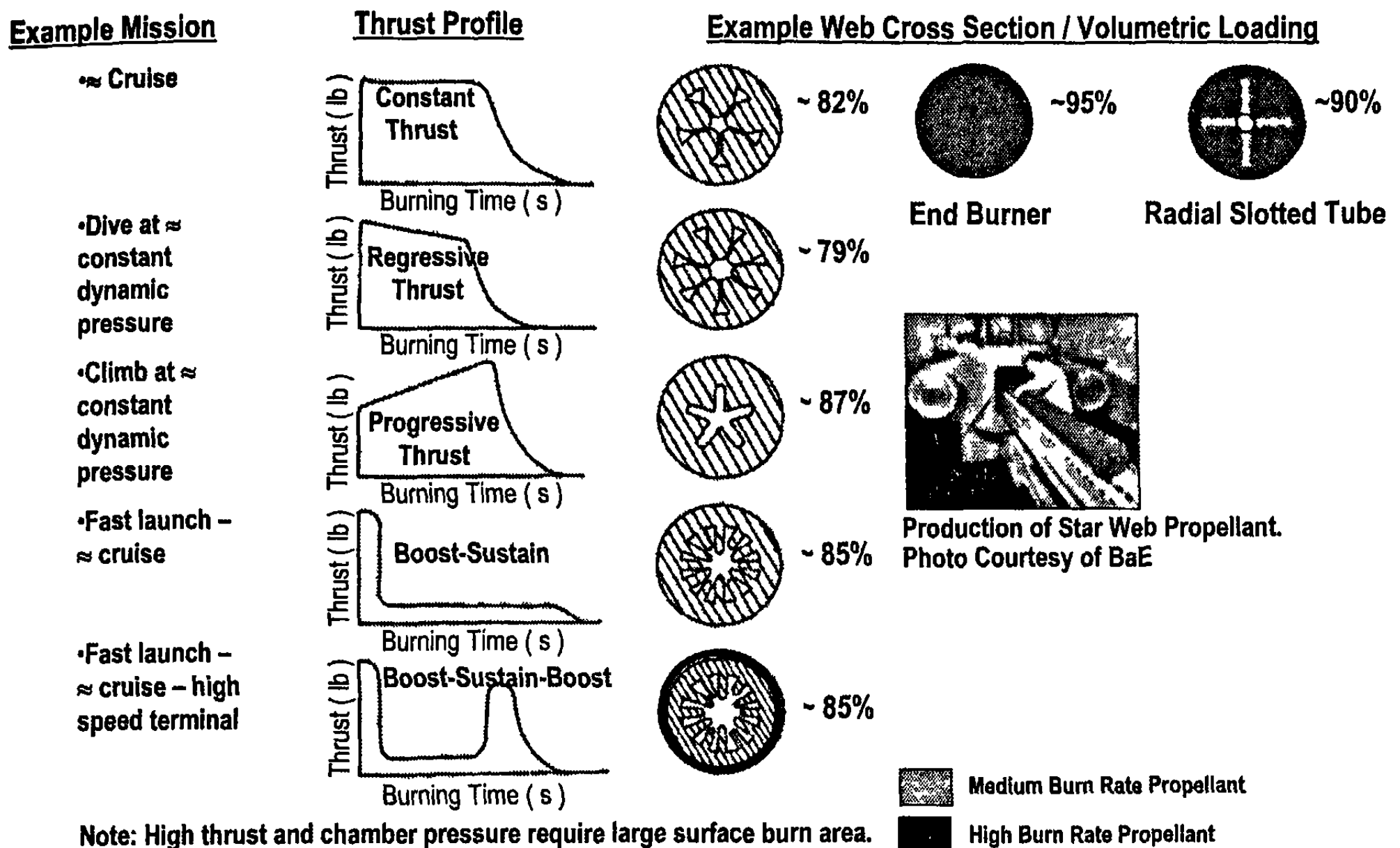


Fig. 3.38 Conventional solid rocket thrust-time design alternatives are a function of the propellant cross section.

grain cavity loading for operational missiles varies from about 79 to 87%. High volumetric efficiency is desirable to maximize the amount of propellant and to minimize the weight of the motor case. A disadvantage of high volumetric loading of the propellant grain cavity is that there may be problems of localized stress and cracks in the the grain.

For a propellant grain with radial burn, the length of the radial-burn motor affects the thrust. A longer motor has more burn area, which provides higher thrust.

Shown in the top section of the figure is an end-burner. An end-burner has no cavity, only the end of the grain burns. An end-burning grain has nearly constant burning area, and if the throat area is constant, provides nearly constant thrust. An advantage of an end-burner is high volumetric loading. A potential disadvantage is relatively low thrust.

Another popular type of cross section is a radial slotted tube, shown in the top right section of the figure. Compared to a web cross section, a radial slotted motor has an advantage of higher volumetric loading. A radial slotted tube has broad application; it can be designed to provide nearly constant, regressive, progressive, and boost-sustain thrust profiles. The most common type of slotted tube provides a boost-sustain thrust profile. For boost-sustain, high burn rate propellant in the aft section of the motor provides an initial high thrust while a different, slower burn rate propellant in the forward section provides the follow-on sustain thrust.

Shown in the middle of the figure is a five-point star cross section. It provides progressive thrust. Volumetric loading is fairly high, about 87%, for this cross section. The middle section of the figure is a photograph of a three-point star cross section extrusion of a double-base propellant. Following extrusion, the propellant can be loaded into the motor case as a free-standing cartridge. A second

bottom concept in the figure is another pulse motor concept based on a radial grain for boost (with high burn rate propellant), followed by second radial grain for either sustain (with lower burn rate propellant) or a second boost (with a high burn rate propellant).

A design concern of layering a high burn rate propellant on top of a low burn rate propellant is the difference in the thermal expansion of the two different propellants. Localized stress and cracks can develop because of thermal expansion.

Potential advantages of a pulse motor include longer range, longer flight duration, and reduced aerodynamic heating (from lower velocity at burnout). Pulse motors are more expensive than conventional single-burn motors. Each pulse of a pulse motor increases the motor production cost by about 40%. Examples of missiles with a pulse motor are the AGM-69 SRAM two-pulse rocket motor and the SM-3 Standard Missile two-pulse TSRM.

Approaches to rocket propulsion energy management for tactical missiles include pulsed, pintle, and gel rocket motors. Figure 3.40 shows these alternatives for thrust magnitude control.

As mentioned previously, the solid pulse motor uses thermal or mechanical barriers to provide separate motor pulses. It is usually limited to two or three pulses. Because the motor can be designed for an optimum combustion pressure, a pulse motor can be designed for maximum specific impulse. Another advantage is that the time delay between pulses can be controlled to optimize the flight trajectory profile. As a result, a boost-coast-boost-coast pulse motor can have longer range and reduced aerodynamic heating compared to conventional single-burn boost-coast and boost-sustain-coast motors.

Solid Pulse Motor

☺ High I_{sp}

⊗ Limited Pulses

Solid Pintle Motor

☺ Continuously Select Up to 40:1 Variation in Thrust

☺ Reduce MEOP on Hot Day

☺ Good I_{sp} Only If Burn Rate Exponent $n \rightarrow 1$

Bi-propellant Gel Motor

☺ High I_{sp}

☺ Duty Cycle Thrust

☺ Insensitive Munition

⊗ Lower Max Thrust

⊗ Toxicity

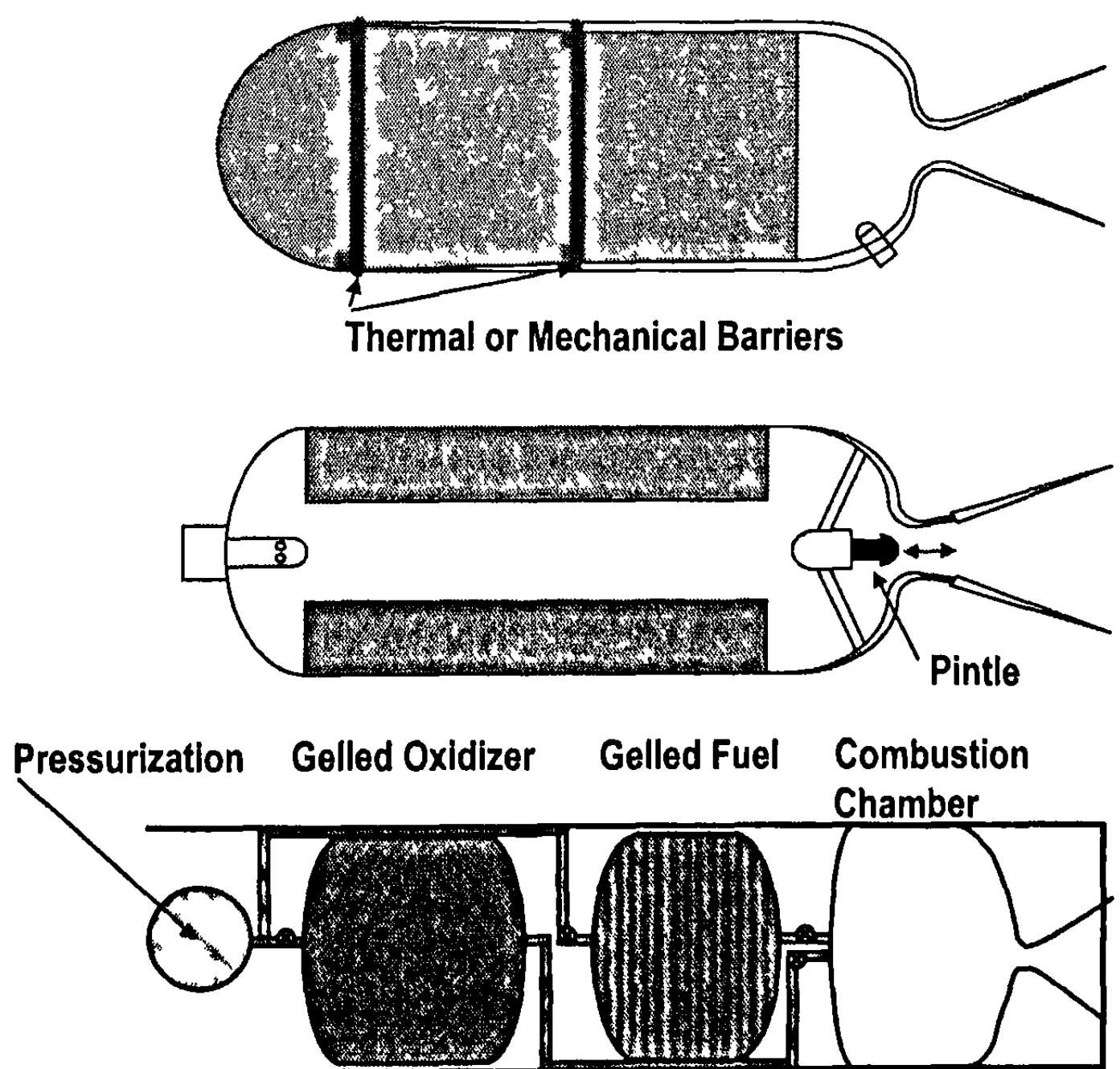


Fig. 3.40 Tactical rocket motor thrust magnitude control alternatives.

The second approach to thrust magnitude control, a solid pintle motor, has a pintle plug that is moved in and out of the throat area. The pintle is constructed from a material with a very high temperature capability, such as tungsten or carbon carbon. Moving the pintle into the throat area increases the chamber pressure, providing higher burn rate and higher thrust. Moving the pintle out of the throat area decreases the chamber pressure, reducing the burn rate and thrust.

One advantage of a pintle motor is that it can be used to reduce the maximum effective operating pressure (MEOP) on a hot day. This capability allows a smaller factor of safety and a lighter weight motor case.

For a typical value of propellant burn rate exponent (e.g., $n = 0.3$), there is a large penalty in specific impulse for the motor to operate over a wide range in chamber pressure. It is desirable for a pintle motor to use a propellant with a high burn rate exponent (e.g., $n \rightarrow 1$), to allow a large change in burn rate/mass flow rate/thrust with only a small change in chamber pressure. Pintle motors have demonstrated maximum-to-minimum thrust ratios of greater than to 40:1. Combustion can be sustained down to a chamber pressure that is the minimum for a sonic throat, allowing an “idle” mode similar to that of a pulse motor.

A technology opportunity is the development of propellants with burn rate exponents that approach the value of one, allowing a large change in burn rate (and thrust) with a relatively small change in chamber pressure. Propellants with $n \rightarrow 1$ require high bandwidth and tight control of the position of the pintle to avoid the hazard of combustion instability if $n > 1$.

A third alternative for thrust magnitude control of a tactical rocket is a gel propellant rocket using a low-toxicity gelled fuel such as alcohol and a low-toxicity gelled oxidizer such as hydrogen peroxide. The gelled propellant is nearly solid at atmospheric pressure but flows like a liquid when it is pumped at high pressure. The thrust is controlled by the rate at which the fuel and oxidizer are pumped into the combustion chamber. The specific impulse of a low-toxicity gel-propellant rocket can be higher than that of a solid-propellant rocket. The specific impulse of a higher toxicity gel propellant could be comparable to that of a liquid-propellant rocket. A disadvantage of a gel-propellant rocket compared to a solid-propellant motor is lower maximum thrust. A second disadvantage is concern for toxicity, particularly for naval platforms. Although a gel-propellant rocket is much less toxic than a liquid-propellant rocket, it is probably unacceptable in a closed environment storage with humans, such as below deck of a ship. In regard to insensitive munitions, a gel-propellant rocket is superior to a solid propellant motor. The gelled fuel and gelled oxidizer are stored in separate tanks, reducing their sensitivity. The U.S. Army has been a leader in the development of gel-propellant rockets. For example, gel propellant is used in the THAAD divert and attitude control system (DACS).

3.19 Solid Propellant Alternatives

Figure 3.41 is a comparison of solid-propellant alternatives based on tradeoffs of rocket motor visual observable versus performance and safety. In high smoke and reduced smoke motors, the solid propellant is a composite consisting of a fuel and an oxidizer blended together and mixed into a binder, or polymer. The mixture is cast (poured) into the motor case and cured. The highest performance

Type	I_{sp} , Specific Impulse, s	ρ , Density, lb / in ³	Burn Rate @ 1000 psi, in / s	Safety	Observables
• Min Smoke. No Al fuel or AP oxidizer. Either Composite with Nitramine Oxidizer (CL-20, ADN, HMX, RDX) or Double Base. Very low contrail (H ₂ O).	— 220–255	— 0.055–0.062	○ 0.25–2.0	—	●
• Reduced Smoke. No Al (binder fuel). AP oxidizer. Low contrail (HCl).	○ 250–260	○ 0.062	● 0.1–1.5	○	○
• High Smoke. Al fuel. AP oxidizer. High smoke (Al ₂ O ₃).	● 260–265	● 0.065	● 0.1–3.0	●	—

● Superior ● Above Average ○ Average — Below Average

Fig. 3.41 Solid rocket propellant alternatives.

solid propellants are composite propellants based on binding a metal fuel with an oxidizer such as ammonium perchlorate. Powdered aluminum is the metallic fuel most widely used in solid rocket propellants. Benefits of a metal fuel include 1) increased combustion temperature, resulting in an increase in specific impulse; 2) increased density of the propellant, resulting in improved volumetric performance; and 3) improved combustion stability. Binders have been based on both natural and man-made materials. The organic prepolymers have rubbery properties following cure and form a strong binding within the inorganic fuels and chemical oxidizers. The most commonly used liquid polymers cure much like rubber. Prepolymers used as binders in composite propellants are initially liquid, so that the fuel and oxidizer can be blended more easily prior to the start of the polymerization cure. Unfortunately, metal fuels such as aluminum have high observable smoke particles (e.g., Al₂O₃).

An initial approach that was used to reduce the plume observables was reduced smoke motors. Reduced smoke motors replace the metal fuel with a binder fuel, such as HTPB binder. The performance and insensitive munition capability of reduced smoke motors are somewhat lower than that of high smoke motors. Reduced smoke propellants can still have visual observables from a hydrogen chloride contrail, particularly at low atmospheric temperature.

A third type of propellant is minimum smoke propellant. Minimum smoke propellants eliminate the HCl contrail by eliminating ammonium perchlorate as an oxidizer, resulting in lower visual observables. However, the performance and safety of the current minimum smoke propellants are not as good as that of high smoke propellants. Current minimum smoke propellants are either composite propellants with a nitramine oxidizer or cross-linked double base (XLDB) propellants. In the older minimum smoke double-base propellants, the propellant

consists generally of cotton (cellulose) combined with nitric acid to form nitrocellulose (guncotton), which in turn is combined with nitroglycerin, another fuel oxidizer. In double-base propellant, the nitrocellulose serves as the binder, and the nitroglycerin causes it to solidify. A nitramine oxidizer contains the radical group NHNO_2 . Examples of the nitramine oxidizer of current minimum smoke composite propellants are HMX (cyclotetramethylene tetranitramine, $\text{C}_4\text{H}_8\text{N}_8\text{O}_8$) and RDX (cyclotrimethylene trinitramine, $\text{C}_3\text{H}_6\text{N}_6\text{O}_6$). An example of a new minimum smoke propellant is the U.S. Navy China Lake CL-20 propellant. CL-20 is a cyclic polynitramine with a unique caged structure that provides higher crystal density, higher heat of formation, and higher oxidizer-to-fuel ratio. CL-20 propellant has 10–20% higher performance than HMX and RDX. CL-20 also has reduced shock sensitivity (Class 1.3 vs 1.1) and milder cookoff reaction than either HMX or RDX. A disadvantage of CL-20 propellant is high cost (currently more than \$400 per lb). The relatively high performance of CL-20 is primarily from its high flame temperature. As a first-order estimate, to illustrate the benefit of high flame temperature, specific impulse can be calculated from the equation

$$I_{\text{SP}} \approx 25(T_f/MW)^{1/2} \quad (3.94)$$

In the above equation T_f is the flame temperature in Kelvin and MW is the molecular weight.

Another example of a relatively new minimum smoke propellant developed by Russia is ammonium dinitramine (ADN). ADN performance is superior to HMX and RDX and somewhat less than that of CL-20. ADN is lower density and has a lower flame temperature than CL-20.

3.20 Motor Case Material Alternatives

Primary design considerations in selecting the materials for the motor case are shown in Fig. 3.42. These are high maximum temperature, high volumetric efficiency, light weight, ease of attachment (e.g., bolt, weld, bond) to the airframe or launcher, and low cost. Other considerations include manufacturing, handling, storage, and the environment (e.g., sea salt corrosion in a naval environment).

The most common types of motor cases are conventional quench and temper steel (hardened by heating then plunging into liquid to cool suddenly) and nickel precipitation-hardening alloy steel. Other types include aluminum alloys, strip metal/epoxy laminate (e.g., steel/epoxy laminate), titanium alloys, and composite (e.g., fiberglass-reinforced plastics). If a combination of high performance and low cost is required, steel is often the best material. Although steel has high strength, it is susceptible to localized buckling and may require a thicker wall to avoid localized buckling. If the motor has relatively low heating (e.g., a small rocket motor usually has low heating), aluminum may be the best material. Because the maximum allowable temperature of aluminum is lower than steel, it requires more insulation. A strip metal/epoxy laminate may be the best choice of moderate performance and cost. Another good feature of a strip metal/laminate motor case is it is often easier to qualify as an insensitive

Type	Temperature	Volumetric Efficiency	Weight	Airframe and Launcher Attachment	Cost
♦ Steel	●	●	○	●	●
♦ Aluminum	—	—	○ ○	○	○ ●
♦ Strip Steel / Epoxy Laminate	○	●	○	—	○
♦ Composite	○	—	●	—	—
♦ Titanium	○	○	○	—	—

● Superior ○ Above Average ○ Average — Below Average

Fig. 3.42 Steel is the most common motor case material.

munition. Composite and titanium cases have had relatively little application to date because of their high cost. However, the cost of a composite motor case is expected to decrease in the future, resulting in increased application of composite motor cases. Current applications of composite cases include LOSAT, PAC-3, and the SM-3 third stage rocket motor. The fifth candidate material, titanium, is rarely used in motor cases because of its higher cost. Molten titanium requires an expensive vacuum manufacturing process to avoid oxidation.

For a volume-limited missile, such as a missile packaged inside a Vertical Launch System (VLS) canister that has a diameter limit, the volumetric efficiency of the material may be as important as the weight efficiency. As an example, a composite motor case may have the lightest weight but also a greater thickness. The thickness of the composite case may limit the available volume for propellant, resulting in a reduction in flight range.

Internal insulation protects the motor case from the high flame temperature (e.g., 5200° F) of combustion. For a typical case-bonded propellant, the internal insulator also provides the bonding of the propellant to the motor case and must have sufficient strength to prevent the propellant from debonding. Most of the internal motor case insulators are rubber-based materials such as hydroxyl terminated polybutadiene or ethylene propylene dimethyl monomer (EPDM). Reinforcing erosion-resistant fibers such as silica are usually added. Upon exposure to the combustion flame temperature, the insulator chars. A design consideration is the required insulation thickness to compensate for the recession and the change in thermal conductivity as a result of char. Another design consideration is the potential debonding from the motor case because of outgassing.

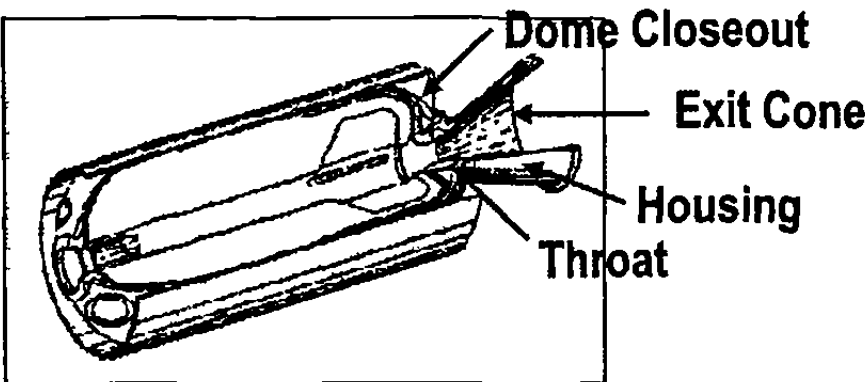
3.21 Rocket Nozzle Material Alternatives

The rocket nozzle is an attachment to the end of the motor case. It is shaped to control the flow of gases and to convert the chemical energy released from combustion into kinetic energy. Figure 3.43 shows material alternatives for the rocket nozzle components (i.e., housing, dome closeout, blast tube, throat, and exit cone).

The housing is the overall structure of the nozzle. It may be made of high temperature steel for a high heating motor case or lower temperature materials such as cellulose/phenolic or aluminum for a low heating motor. The inside of the housing usually must be insulated to maintain structural integrity.

The dome closeout is a convergent section of the nozzle that is downstream of the motor case and upstream of the throat. In the dome closeout the flow area is reduced, and the speed of the subsonic exhaust gas is increased as it is channeled to the throat. For a high heating motor with a high chamber pressure and long burn duration, an insulation insert is required in the dome closeout housing. The candidate dome closeout insulation materials include silica/phenolic, graphite/phenolic, and silicone elastomer. For a relatively low heating motor, low-cost glass/phenolic is a candidate insulation material for the dome closeout. For a very low heating motor, it may not be necessary to have an insulation insert in the housing of the dome closeout.

A blast tube may be downstream of the dome closeout if it is necessary to package subsystems, such as flight control actuators, around the nozzle. Ideally, the blast tube should have a small diameter to allow maximum packaging volume for the surrounding subsystems. However, as the blast tube diameter



Rocket Nozzle Element	High Heating (High Chamber Pressure or Long Burn) ⇒ High Cost / Heavy Nozzle	Low Heating (Low Chamber Pressure or Short Burn) ⇒ Low Cost / Light Weight Nozzle
♦ Housing Material Alternatives	♦ Steel	♦ Cellulose / Phenolic ♦ Aluminum
♦ Throat Material Alternatives	♦ Tungsten Insert ♦ Rhenium Insert ♦ Molybdenum Insert	♦ Cellulose / Phenolic Insert ♦ Silica / Phenolic Insert ♦ Graphite Insert ♦ Carbon – Carbon Insert
♦ Exit Cone, Dome Closeout, and Blast Tube Material Alternatives	♦ Silica / Phenolic Insert ♦ Graphite / Phenolic Insert ♦ Silicone Elastomer Insert	♦ No Insert ♦ Glass / Phenolic Insert

Fig. 3.43 Heat transfer drives rocket nozzle materials, weight, and cost.

approaches the throat diameter, the local gas Mach number approaches Mach 1. Because of severe erosion from the hot gas at Mach 1, the blast tube diameter is usually significantly larger than the throat diameter, such that the local Mach number is less than Mach 0.8.

The nozzle throat is the smallest internal diameter of the nozzle assembly. At the nozzle throat the exhaust gas passes through the minimum flow area and reaches the speed of sound. Because of high heat transfer at the throat, it may be necessary to have a refractory metal insert at the throat of a high-performance motor to prevent erosion. Examples of low-eroding refractory materials are tungsten, rhenium (used in SM-3 third-stage rocket motor and the Harpoon booster), and molybdenum. For a low heating motor, a lighter weight insert such as cellulose/phenolic, silica/phenolic, graphite, or carbon/carbon may be used.

Downstream of the throat the gases expand in the exit cone to supersonic speed. The exit cone may be contoured or conical. A contoured nozzle provides higher specific impulse, whereas a conical nozzle is shorter length, lighter weight, and lower cost. Most tactical missiles use conical nozzles. Finally, the exit plane is at the end of the exit cone, where the static pressure of the gas is ideally nearly equal to the atmospheric pressure. As shown in the figure, the material selection for the exit cone is driven by heating and cost considerations.

3.22 Summary

Chapter 3 provided conceptual design methods, discussed design trades, and identified high-payoff technologies for tactical missile propulsion. Conceptual design methods were provided for predicting the thrust and specific impulse of turbojet, ramjet, and rocket propulsion. Design trades for turbine-based propulsion include turbine material, compressor pressure ratio, and the propulsion cycle. Design trades for ramjets include combustor material alternatives, ramjet engine/booster integration alternatives, ramjet inlet options, and ramjet fuel alternatives. Desired trades for rockets include rocket propellant burn area required to provide the desired chamber pressure, nozzle throat area required to provide the desired mass flow rate, nozzle expansion ratio required for maximum specific impulse and maximum thrust, rocket motor grain alternatives, thrust magnitude control alternatives, solid-propellant alternatives, motor case material alternatives, and nozzle material alternatives. New propulsion technologies that have high payoff for tactical missiles were identified. These include hypersonic turbojets, ramjets, ducted rockets, scramjets, combined cycle propulsion, high-temperature turbine materials, high-temperature combustor, oblique shock airframe compression, mixed compression inlet, low-drag inlet, high-density fuel/propellant, ramjet throttle fuel control, endothermic fuel, solid rocket thrust magnitude control, high burn exponent propellant, and low observable fuel/propellant.

Problems

- 3.1. An advantage of turbojets compared to ramjets is s_____thrust.
- 3.2. The specific impulse of a turbojet is often limited by the maximum allowable temperature of the t_____.